

How Cells Read the Genome: From DNA to Protein

CHAPTER 6

Since the structure of DNA was discovered in the early 1950s, progress in cell and molecular biology has been astounding. We now know the complete genome sequences for thousands of different organisms, revealing fascinating details of their biochemistry as well as important clues as to how these organisms evolved. Complete genome sequences have also been obtained for hundreds of thousands of individual humans, as well as for a few of our now-extinct relatives, such as the Neanderthals. Knowing the maximum amount of information that is required to produce a complex organism like ourselves puts constraints on the biochemical and structural features of cells and makes it clear that biology is not infinitely complex.

As discussed in Chapter 1, most of the genetic information carried by DNA specifies the sequence of amino acids in proteins. But this DNA does not direct the synthesis of proteins directly, instead producing RNA as an intermediary. When the cell needs a particular protein, the nucleotide sequence of the appropriate portion of the DNA molecule in a chromosome is first copied into RNA (a process called *transcription*). These RNA copies of segments of the DNA sequence are then used to direct the synthesis of the protein (a process called *translation*). The genetic information in cells thereby flows from DNA to RNA to protein. All cells, from bacteria to humans, express their genetic information in this way—a principle so fundamental that it is termed the *central dogma* of molecular biology (Figure 6-1).

Despite this universality, there are important variations between organisms in the way information flows from DNA to protein. Most notable, the RNA transcripts in eukaryotic cells are subject to a series of processing steps in the nucleus, including *RNA splicing*, before they are permitted to exit from the nucleus and be translated into protein. As we discuss in this chapter, these processing steps can critically change the “meaning” of an RNA molecule, and they are therefore crucial for understanding how eukaryotic cells read their genome.

Although we shall focus in this chapter on the production of the proteins encoded by the genome, for some genes RNA is the final product. Like proteins, some of these RNAs fold into precise three-dimensional structures that have structural and catalytic roles in the cell. Although the functions of many non-coding RNAs are not yet known, some have been studied in great detail and are discussed in this and the following chapter.

One might have expected that the information present in genomes would be arranged in an orderly fashion, resembling a dictionary or a telephone directory. But it turns out that the genomes of most multicellular organisms are surprisingly disorderly, reflecting their chaotic evolutionary histories. The genes in these organisms largely consist of a long string of alternating short exons and long introns, as discussed in Chapter 4 (see Figure 4-15D). Moreover, small bits of DNA sequence that code for protein are interspersed with large blocks of seemingly meaningless DNA. Some sections of the genome contain many genes and others lack genes altogether. Proteins that work closely with one another in the cell usually have their genes located on different chromosomes, and adjacent genes typically encode proteins that have little to do with each other in the cell. Decoding genomes is therefore no simple matter. Even with the aid of powerful

IN THIS CHAPTER

From DNA to RNA

From RNA to Protein

The RNA World and the Origins of Life

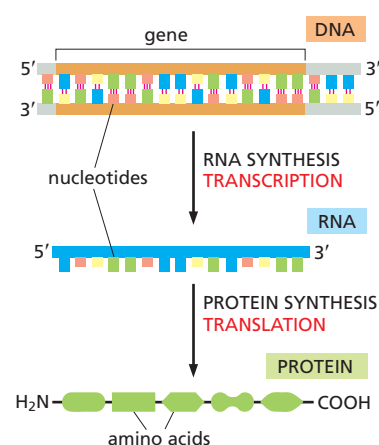
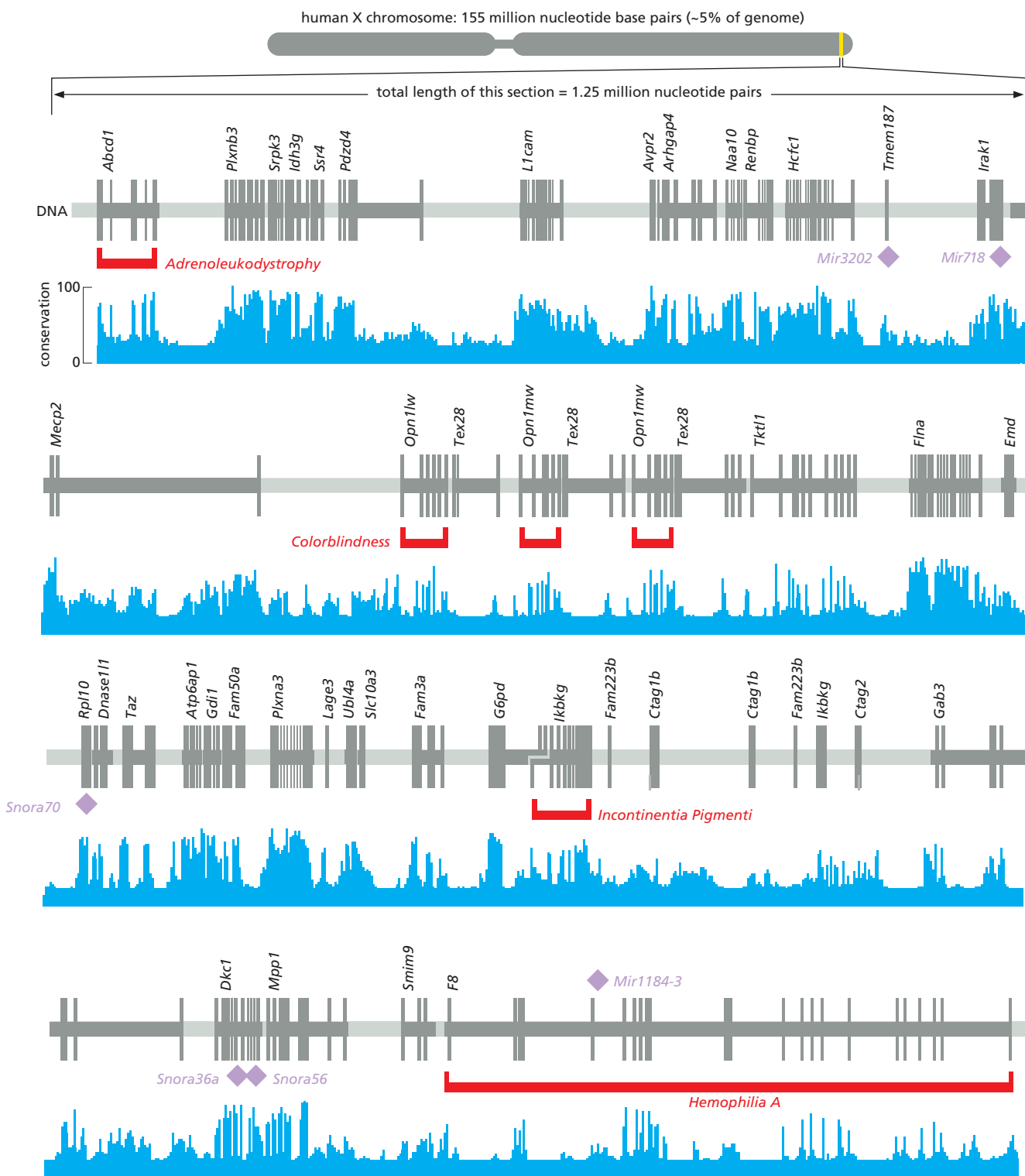
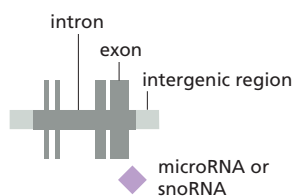


Figure 6-1 Genetic information directs the synthesis of proteins. The flow of genetic information from DNA to RNA (transcription) and from RNA to protein (translation) occurs in all living cells. As we saw in Chapter 5, DNA can also be copied—or replicated—to produce new DNA molecules.



KEY:



Incontinentia Pigmenti

disease phenotype caused by
nucleotide changes in the
indicated gene

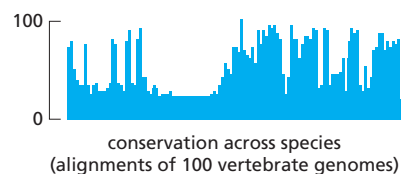


Figure 6–2 Schematic depiction of a small portion of the human X chromosome. As summarized in the key, the known protein-coding genes (starting with *Abcd1* and ending with *F8*) are marked by a *dark gray* central line, with their coding regions (exons) indicated by *dark gray* bars that extend above and below this line. Noncoding RNAs with known functions are indicated by *purple diamonds*. The *blue* histogram indicates the extent to which portions of the human genome are conserved with other vertebrate species. It is likely that additional genes, currently unrecognized, also lie within this portion of the human genome.

Genes whose mutation causes an inherited human condition are indicated by *red brackets*. The *Abcd1* gene codes for a protein that imports fatty acids into the peroxisome; mutations in the gene cause demyelination of nerves, which can result in cognition and movement disorders. *Incontinentia pigmenti* is a disease of the skin, hair, nails, teeth, and eyes. *Hemophilia A* is a bleeding disorder caused by mutations in the Factor VIII gene, which codes for a blood-clotting protein (see Figure 6–25B). Because males have only a single copy of the X chromosome, most of the conditions shown here affect only males; females that inherit one of these defective genes are often asymptomatic because a functional protein is made from their other X chromosome. (Courtesy of Alex Williams, data obtained from the University of California, Genome Browser, <http://genome.ucsc.edu>.)

computers, it is difficult for researchers, in the absence of direct experimental evidence, to locate definitively the beginning and end of genes, much less to decipher when and where each gene is expressed in the life of the organism. Yet the cells in our body do this automatically, thousands of times a second.

The problems that cells face in decoding genomes can be appreciated by considering a tiny portion of the human genome (**Figure 6–2**). The region illustrated represents less than 1/2000 of our genome and includes at least 48 genes that encode proteins plus 6 genes for noncoding RNAs. When we consider the entire human genome, we can only marvel at the capacity of our cells to rapidly and accurately handle such large amounts of information.

In this chapter, we explain how cells decode and use the information in their genomes. Much has been learned about how the genetic instructions written in an alphabet of just four “letters”—the four different nucleotides in DNA—direct the formation of a bacterium, a fruit fly, or a human. Nevertheless, we still have a great deal to discover about how the information stored in an organism’s genome produces even the simplest unicellular bacterium with about 500 genes, let alone how it directs the development of a human with approximately 25,000 genes. An enormous amount of ignorance remains; many fascinating challenges therefore await the next generation of cell biologists.

FROM DNA TO RNA

Transcription and translation are the means by which cells read out, or express, the genetic instructions in their genes. Because many identical RNA copies can be made from the same gene, and each RNA molecule can direct the synthesis of many identical protein molecules, cells can synthesize a large amount of protein from a single gene when necessary. Importantly, genes can be transcribed and translated with different efficiencies, allowing the cell to make vast quantities of some proteins and tiny amounts of others (**Figure 6–3**). Moreover, as we see in the next chapter, a cell can change (or regulate) the expression of each of

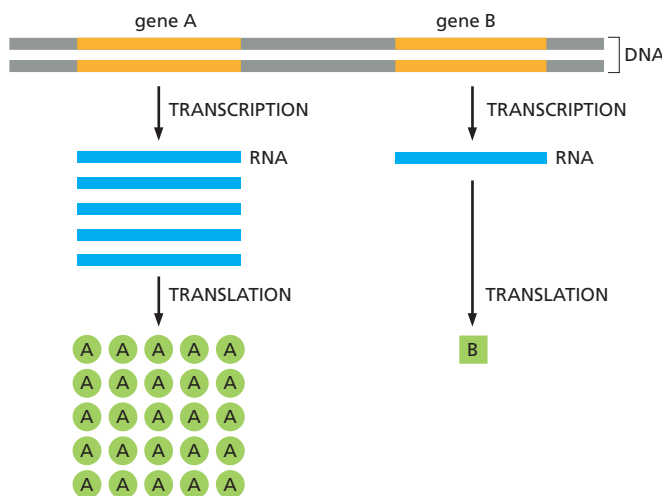


Figure 6–3 Genes can be expressed with different efficiencies. In this example, gene A is transcribed much more efficiently than gene B, and each RNA molecule that it produces is also translated more frequently. This causes the amount of protein A in the cell to be much greater than that of protein B. In this and later figures, the portions of DNA that are transcribed are shown in *orange*.

its genes according to its needs—commonly by controlling the production of its RNA—and many genes will not be expressed at all in some cells.

One of the central problems in producing proteins from the information carried in genomes is that most steps depend on conventional nucleic acid base-pairing, which on its own has only modest specificity. In many contexts, a correct base pair is only 10–100 times more thermodynamically stable than an incorrect base pair, so that most steps of gene expression rely on mechanisms that both improve the specificity of the base-pairing and correct the many mistakes that arise. A central theme of this chapter, therefore, is the way cells deal with the fundamentally inaccurate base-pairing process that lies at the heart of the mechanisms that they use to read their genome.

RNA Molecules Are Single-Stranded

The first step a cell takes in reading out its genetic instructions is to copy a particular portion of its DNA nucleotide sequence—a gene—into an RNA nucleotide sequence (Figure 6-4). The information in RNA, although copied into another chemical form, is still written in essentially the same language as it is in DNA—the language of a nucleotide sequence. Hence the name given to producing RNA molecules on DNA is *transcription*.

Like DNA, RNA is a linear polymer made of four different types of nucleotide subunits linked together by phosphodiester bonds (see Figure 6-4). It differs from DNA chemically in two respects: (1) the nucleotides in RNA are *ribonucleotides*—that is, they contain the sugar ribose (hence the name *ribonucleic acid*) rather than deoxyribose; (2) although, like DNA, RNA contains the bases adenine (A), guanine (G), and cytosine (C), it contains the base uracil (U) instead of the thymine (T) in DNA (Figure 6-5). Because U, like T, can base-pair by hydrogen-bonding with A (Figure 6-6), the complementary base-pairing properties described for DNA in Chapters 4 and 5 apply also to RNA (in RNA, G pairs with C, and A pairs with U).

Although these chemical differences are slight, DNA and RNA differ quite dramatically in overall structure. Whereas DNA always occurs in cells as a double-strand helix, RNA is single-stranded. An RNA chain can therefore fold up into a particular shape, just as a polypeptide chain folds up to form the final shape of a protein (Figure 6-7). As we see later in this chapter, the ability to fold into complex three-dimensional shapes allows some RNA molecules to have precise structural and catalytic functions.

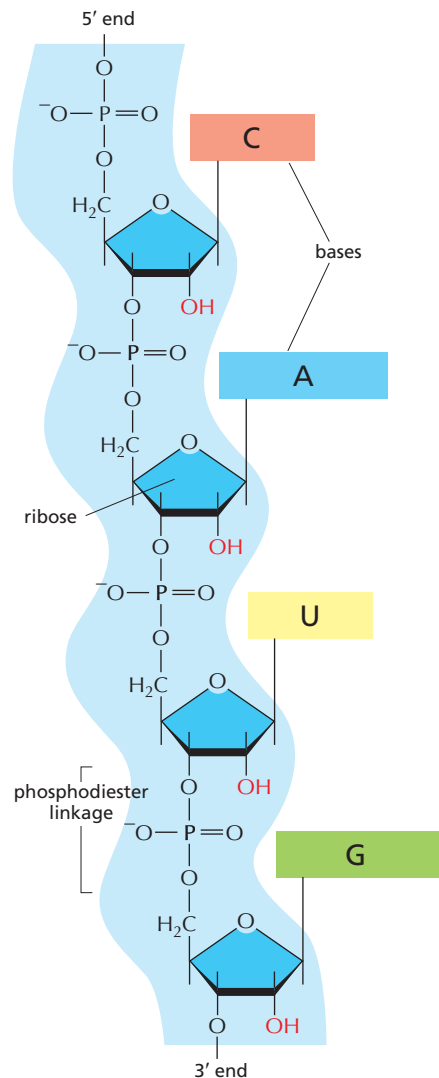


Figure 6-4 A short length of RNA. The phosphodiester chemical linkage between nucleotides in RNA is the same as that in DNA.

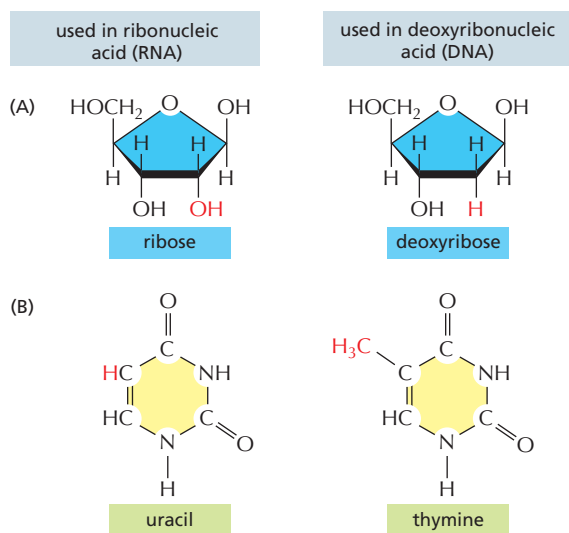


Figure 6-5 The chemical structure of RNA. (A) RNA contains the sugar ribose, which differs from deoxyribose, the sugar used in DNA, by the presence of an additional -OH group. (B) RNA contains the base uracil, which differs from thymine, the equivalent base in DNA, by the absence of a -CH_3 group.

Transcription Produces RNA Complementary to One Strand of DNA

The RNA in a cell is made by **DNA transcription**, a process that has certain similarities to the process of DNA replication discussed in Chapter 5. Transcription begins with the opening and unwinding of a small portion of the DNA double helix to expose the bases on each DNA strand. One of the two strands of the DNA double helix then acts as a template for the synthesis of an RNA molecule. As in DNA replication, the nucleotide sequence of the RNA chain is determined by the complementary base-pairing between incoming nucleotides and the DNA template. When a good match is made (A with T, U with A, G with C, and C with G), the incoming ribonucleotide is covalently linked to the growing RNA chain in an enzymatically catalyzed reaction. The RNA chain produced by transcription—the *transcript*—is therefore elongated one nucleotide at a time, and it has a nucleotide sequence that is exactly complementary to the strand of DNA used as the template (Figure 6–8).

Transcription, however, differs from DNA replication in several crucial ways. Unlike a newly formed DNA strand, the RNA strand does not remain hydrogen-bonded to the DNA template strand. Instead, just behind the region where the ribonucleotides are being added, the RNA chain is displaced and the DNA helix re-forms. Thus, the RNA molecules produced by transcription are released from the DNA template as single strands. In addition, because they are copied from only a limited region of the DNA, RNA molecules are much shorter than DNA molecules. A DNA molecule in a human chromosome can be up to 250 million nucleotide-pairs long; in contrast, most RNAs are no more than a few thousand nucleotides long, and many are considerably shorter.

RNA Polymerases Carry Out DNA Transcription

The enzymes that perform transcription are called **RNA polymerases**. Like the DNA polymerase that catalyzes DNA replication (discussed in Chapter 5), RNA polymerases catalyze the formation of the phosphodiester bonds that link the nucleotides together to form a linear chain. The RNA polymerase moves stepwise

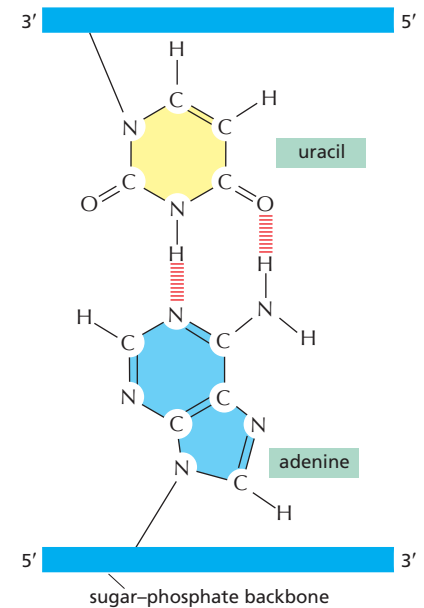


Figure 6–6 Uracil base pairs with adenine. The absence of a methyl group in U has no effect on base-pairing; thus, U–A base pairs closely resemble T–A base pairs (see Figure 4–5).

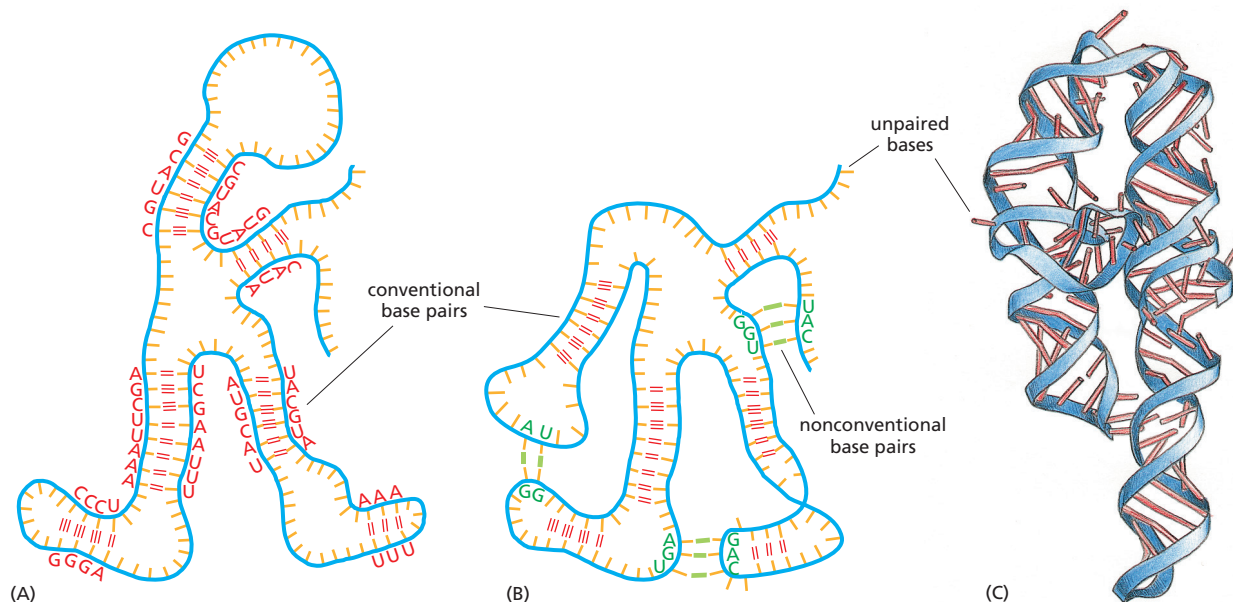


Figure 6–7 RNA can fold into specific structures. RNA is largely single-stranded, but it often contains short stretches of nucleotides that can form conventional base pairs with complementary sequences found elsewhere on the same molecule. These interactions, along with additional “nonconventional” base-pair interactions (for example, A–G), allow an RNA molecule to fold into a three-dimensional structure that is determined by its sequence of nucleotides (Movie 6.1). (A) Diagram of a folded RNA structure showing only conventional (G–C and A–U) base-pair interactions (red). (B) Formation of nonconventional (green) base-pair interactions folds the hypothetical structure shown in A even further. (C) Structure of an actual RNA molecule, in this case one that catalyzes its own splicing (see pp. 347–348). Each conventional base-pair interaction is indicated by a “rung” in the double helix. Bases in other configurations are indicated by broken rungs.

along the DNA, unwinding the DNA helix just ahead of its active site for polymerization to expose a new region of the template strand for complementary base-pairing. In this way, the growing RNA chain is extended by one nucleotide at a time in the 5'-to-3' direction (Figure 6-9). The substrates are ribonucleoside triphosphates (ATP, CTP, UTP, and GTP); as in DNA replication, the hydrolysis of high-energy bonds provides the energy needed to drive the reaction forward (see Figure 5-4 and Movie 6.2).

The almost immediate separation of the RNA strand from the DNA as it is synthesized means that many RNA copies can be made from the same gene in a relatively short time, with the synthesis of additional RNA molecules being started before the previous RNA molecules are completed (Figure 6-10). When RNA polymerase molecules follow hard on each other's heels in this way, each moving at speeds up to 50 nucleotides per second, more than a thousand transcripts can be synthesized in an hour from a single gene.

Although RNA polymerase catalyzes essentially the same chemical reaction as DNA polymerase, there are some important differences between the activities of the two enzymes. First, and most obviously, RNA polymerase catalyzes the linkage of ribonucleotides, not deoxyribonucleotides. Second, unlike the DNA polymerases involved in DNA replication (see pp. 259-260), RNA polymerases can start an RNA chain without a primer. This difference is thought possible because transcription need not be as accurate as DNA replication (see Table 5-1, p. 260). RNA polymerases make about one mistake for every 10^4 nucleotides copied into RNA (compared with an error rate for direct copying and proofreading by DNA polymerase of about one in 10^7 nucleotides), and the consequences of an error in RNA transcription are much less significant as RNA does not permanently store genetic information in cells. Finally, unlike DNA polymerases, which make their products in segments that are later stitched together, RNA polymerases are processive; that is, the same RNA polymerase that begins an RNA molecule must finish it without dissociating from the DNA template.

Although not nearly as accurate as the DNA polymerases that replicate DNA, RNA polymerases nonetheless have a modest proofreading mechanism. If an incorrect ribonucleotide is added to the growing RNA chain, the polymerase can back up, and the active site of the enzyme can perform an excision reaction that resembles the reverse of the polymerization reaction, except that a water molecule replaces the pyrophosphate and a nucleoside monophosphate is released.

Given that DNA and RNA polymerases both carry out template-dependent nucleotide polymerization, it might be expected that the two types of enzymes would be structurally related. However, x-ray crystallographic studies reveal that, other than containing a critical Mg^{2+} ion at the catalytic site, the two enzymes are quite different. Template-dependent nucleotide-polymerizing enzymes seem to have arisen at least twice during the early evolution of cells. One lineage led to the

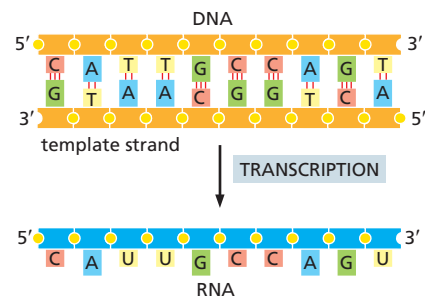


Figure 6-8 DNA transcription produces a single-strand RNA molecule that is complementary to one strand of the DNA double helix. Note that the sequence of bases in the RNA molecule produced is the same as the sequence of bases in the non-template DNA strand, except that a U replaces every T base in the DNA.

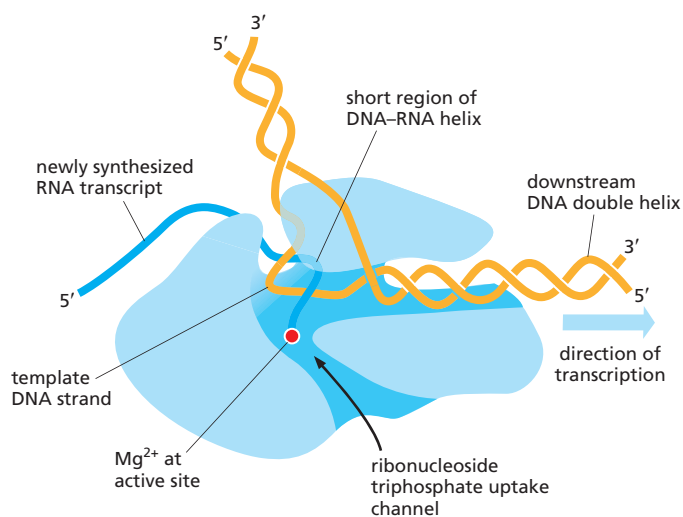


Figure 6-9 DNA is transcribed by the enzyme RNA polymerase. The RNA polymerase (pale blue) moves stepwise along the DNA, unwinding the DNA helix at its active site indicated by the Mg^{2+} (red), which is required for catalysis. As it progresses, the polymerase adds nucleotides one by one to the RNA chain at the polymerization site, using an exposed DNA strand as a template. The RNA transcript is thus a complementary copy of one of the two DNA strands. A short region of DNA-RNA helix (approximately nine nucleotide pairs in length) is formed only transiently, and a “window” of DNA-RNA helix therefore moves along the DNA with the polymerase as the DNA double helix re-forms behind it. The incoming nucleotides are in the form of ribonucleoside triphosphates (ATP, UTP, CTP, and GTP), and the energy stored in their phosphate-phosphate bonds provides the driving force for the polymerization reaction. The figure, based on an x-ray crystallographic structure, shows a cutaway view of the polymerase: the part facing the viewer has been sliced away to reveal the interior (Movie 6.3). (Adapted from P. Cramer et al., *Science* 288:640-649, 2000. PDB code: 1HQM.)

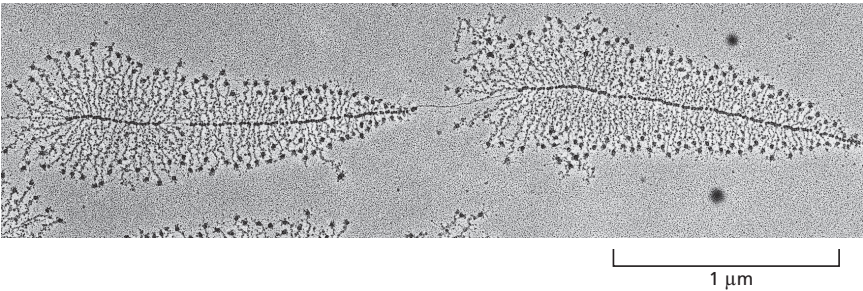


Figure 6–10 Transcription of two genes as observed under the electron microscope. The micrograph shows many molecules of RNA polymerase simultaneously transcribing each of two adjacent genes. Molecules of RNA polymerase are visible as a series of dots along the DNA with the newly synthesized transcripts (fine threads) attached to them. The RNA molecules (ribosomal RNAs; rRNAs) shown in this example are not translated into protein but are instead used directly as components of ribosomes, the machines on which translation takes place. The particles at the 5' end (the free end) of each rRNA transcript are believed to reflect the beginnings of ribosome assembly. From the relative lengths of the newly synthesized transcripts, it can be deduced that the RNA polymerase molecules are transcribing from right to left. (Courtesy of Ulrich Scheer.)

modern DNA polymerases and reverse transcriptases discussed in Chapter 5, as well as to a few RNA polymerases from viruses. The other lineage formed all of the RNA polymerases that we discuss in this chapter.

Cells Produce Different Categories of RNA Molecules

The majority of genes carried in a cell’s DNA specify the amino acid sequence of proteins, and the RNAs that are copied from these genes (which ultimately direct the synthesis of proteins) are called **messenger RNA (mRNA)** molecules. The final product of other genes is the RNA molecule itself. These RNAs are known as **non-coding RNAs**, because they do not code for protein. In a well-studied, single-celled eukaryote, the yeast *Saccharomyces cerevisiae*, more than 1200 genes (about 15% of the total) produce RNA as their final product. Humans produce about 5000 different noncoding RNAs. These RNAs, like proteins, serve as enzymatic, structural, and regulatory components for a wide variety of processes in the cell. In Chapter 5, we encountered one of them as the template RNA carried by the enzyme telomerase. We shall see in this chapter that *ribosomal RNA (rRNA)* molecules form the core of ribosomes, that *transfer RNA (tRNA)* molecules serve as the adaptors that select amino acids and hold them in place on a ribosome for incorporation into protein, and that *small nuclear RNA (snRNA)* molecules direct the splicing of pre-mRNA to form mRNA. In Chapter 7, we shall see that *microRNA (miRNA)* molecules and *small interfering RNA (siRNA)* molecules serve as key regulators of eukaryotic gene expression, and that *piwi-interacting RNAs (piRNAs)* protect animal germ lines from transposons; we also discuss the *long noncoding RNAs (lncRNAs)*, a diverse set of RNAs, many of whose functions are just being discovered (**Table 6–1**).

TABLE 6–1 Principal Types of RNAs Produced in Cells	
Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as the adaptors between mRNA and amino acids
Telomerase RNA	Serves as the template for the telomerase enzyme that extends the ends of chromosomes
snRNAs	Small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	Small nucleolar RNAs, help to process and chemically modify rRNAs
lncRNAs	Long noncoding RNAs, not all of which appear to have a function; some serve as scaffolds and regulate diverse cell processes, including X-chromosome inactivation
miRNAs	MicroRNAs, regulate gene expression by blocking translation of specific mRNAs and causing their degradation
siRNAs	Small interfering RNAs, turn off gene expression by directing the degradation of selective mRNAs and helping to establish repressive chromatin structures
piRNAs	Piwi-interacting RNAs, bind to piwi proteins and protect the germ line from transposable elements

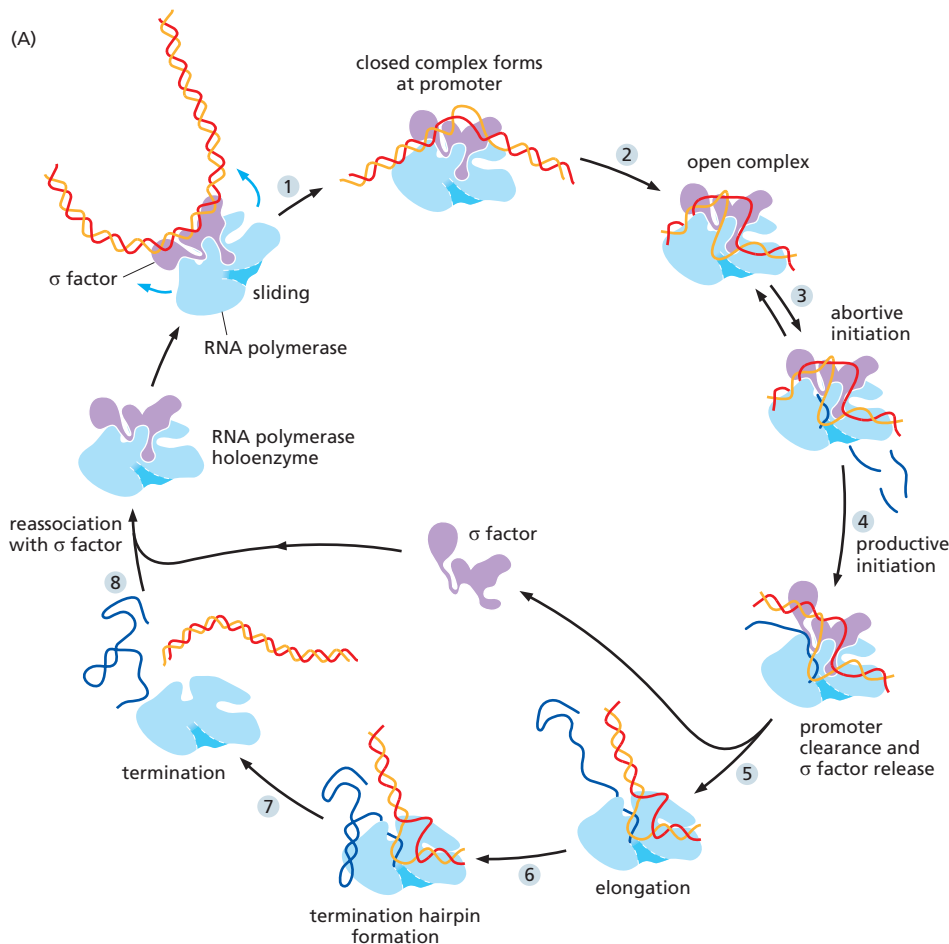


Figure 6-11 The transcription cycle of bacterial RNA polymerase. (A) In step 1, the RNA polymerase holoenzyme (polymerase core enzyme plus σ factor) assembles and then, by sliding, locates a promoter DNA sequence (see Figure 6-12). The polymerase opens (unwinds) the DNA at the position at which transcription is to begin (step 2) and begins transcribing (step 3). This initial RNA synthesis (abortive initiation) is relatively inefficient as short, unproductive transcripts are often released. However, once RNA polymerase has managed to synthesize about 10 nucleotides of RNA, it breaks its interactions with the promoter DNA (step 4) and eventually releases σ factor—as the polymerase tightens around the DNA and shifts to the elongation mode of RNA synthesis, moving along the DNA (step 5). During the elongation mode, transcription is highly processive, with the polymerase leaving the DNA template and releasing the newly transcribed RNA only when it encounters a termination signal (steps 6 and 7). Termination signals are typically encoded in DNA, and many function by forming an RNA hairpin-like structure that destabilizes the polymerase's hold on the RNA.

In bacteria, all RNA molecules are synthesized by a single type of RNA polymerase, and the cycle depicted in the figure therefore applies to the production of mRNAs as well as structural and catalytic RNAs. (B) Two-dimensional image of an elongating bacterial RNA polymerase, as determined by atomic force microscopy. (C) Interpretation of the image in B. (Adapted from K.M. Herbert et al., *Annu. Rev. Biochem.* 77:149–176, 2008. With permission from Annual Reviews.)

Each transcribed segment of DNA is called a *transcription unit*. In eukaryotes, a transcription unit typically carries the information of just one gene, and therefore codes for either a single RNA molecule or a single protein (or group of related proteins if the initial RNA transcript is spliced in more than one way to produce different mRNAs). In bacteria, a set of adjacent genes is often transcribed as a unit; the resulting mRNA molecule therefore carries the information for producing several distinct proteins.

Overall, RNA makes up a few percent of a cell's dry weight, whereas proteins compose about 50%. Most of the RNA in cells is rRNA; mRNA composes only 3–5% of the total RNA in a typical mammalian cell. The mRNA population is made up of tens of thousands of different species, and there are on average only 10–15 molecules of each species of mRNA present in each cell.

Signals Encoded in DNA Tell RNA Polymerase Where to Start and Stop

To transcribe a gene accurately, RNA polymerase must recognize where on the genome to start and where to finish. The way in which RNA polymerases perform these tasks differs somewhat between bacteria and eukaryotes. Because the processes in bacteria are simpler, we discuss them first.

The initiation of transcription is an especially important step in gene expression because it is the main point at which the cell regulates which proteins are to be produced and at what rate. The bacterial RNA polymerase core enzyme is a multisubunit complex that synthesizes RNA using the DNA template as a guide. An additional subunit called *sigma* (σ) *factor* associates with the core enzyme and assists it in reading the signals in the DNA that tell it where to begin transcribing (Figure 6-11). Together, σ factor and the core enzyme are known as the

RNA polymerase holoenzyme; this complex adheres only weakly to DNA when the two collide, and a holoenzyme typically slides rapidly along the long bacterial DNA molecule and then dissociates. However, when the polymerase holoenzyme slides into a special sequence of nucleotides indicating the starting point for RNA synthesis called a **promoter**, the polymerase binds tightly because its σ factor makes specific contacts with the edges of bases exposed on the outside of the DNA double helix (step 1 in Figure 6-11A).

The tightly bound RNA polymerase holoenzyme at a promoter opens up the double helix to expose a short stretch of nucleotides on each strand (step 2 in Figure 6-11A). The region of unpaired DNA (about 10 nucleotides) is called the *transcription bubble*, and it is stabilized by the binding of σ factor to the unpaired bases on one of the exposed strands. The other exposed DNA strand then acts as a template for complementary base-pairing with incoming ribonucleotides, two of which are joined together by the polymerase to begin an RNA chain (step 3 in Figure 6-11A).

The first 10 or so nucleotides of RNA are synthesized using a “scrunching” mechanism, in which RNA polymerase remains bound to the promoter and pulls the upstream DNA into its active site, thereby expanding the transcription bubble. This process creates considerable stress, and the short RNAs are often released, thereby relieving the stress and forcing the polymerase, which remains in place, to begin synthesis over again. Eventually this process of *abortive initiation* is overcome, and the stress generated by scrunching helps the core enzyme to break free of its interactions with the promoter DNA (step 4 in Figure 6-11A) and discard the σ factor (step 5 in Figure 6-11A).

At this point, the polymerase begins to move down the DNA, synthesizing RNA in a stepwise fashion: the polymerase moves forward one base pair for every nucleotide added. During this process, the transcription bubble continually expands at the front of the polymerase and contracts at its rear. Chain elongation continues (at a speed of approximately 50 nucleotides per second for bacterial RNA polymerases) until the enzyme encounters a second signal, the terminator (step 6 in Figure 6-11A), where the polymerase halts and releases both the newly made RNA molecule and the DNA template (step 7 in Figure 6-11A). The free polymerase core enzyme then reassociates with a free σ factor to form a holoenzyme that can begin the process of transcription again (step 8 in Figure 6-11A).

The process of transcription initiation is complicated and requires that the RNA polymerase holoenzyme and the DNA undergo a series of conformational changes, first opening the DNA double helix at promoters and subsequently tightening the enzyme around the DNA and RNA so that it does not dissociate before it has finished transcribing a gene. If an RNA polymerase does dissociate prematurely, it must start over again at the promoter.

How do the termination signals in the DNA stop the elongating polymerase? For most bacterial genes, a termination signal consists of a string of A-T nucleotide pairs preceded by a twofold symmetric DNA sequence, which, when transcribed into RNA, folds into a “hairpin” structure through Watson-Crick base-pairing (see Figure 6-92). As the polymerase transcribes across a terminator, the formation of the hairpin helps release the RNA transcript, which is held in place by relatively weak A-T and U-A base pairs (step 7 in Figure 6-11A). As we shall see, the folding of RNA into specific structures affects many steps in decoding the genome.

Bacterial Transcription Start and Stop Signals Are Heterogeneous in Nucleotide Sequence

As we have just seen, the processes of transcription initiation and termination involve a complicated series of structural transitions in protein, DNA, and RNA molecules. The signals encoded in DNA that specify these transitions are often difficult for researchers to recognize. Indeed, a comparison of many different bacterial promoters reveals a surprising degree of variation. Nevertheless, they all contain related sequences, reflecting aspects of the DNA that are recognized

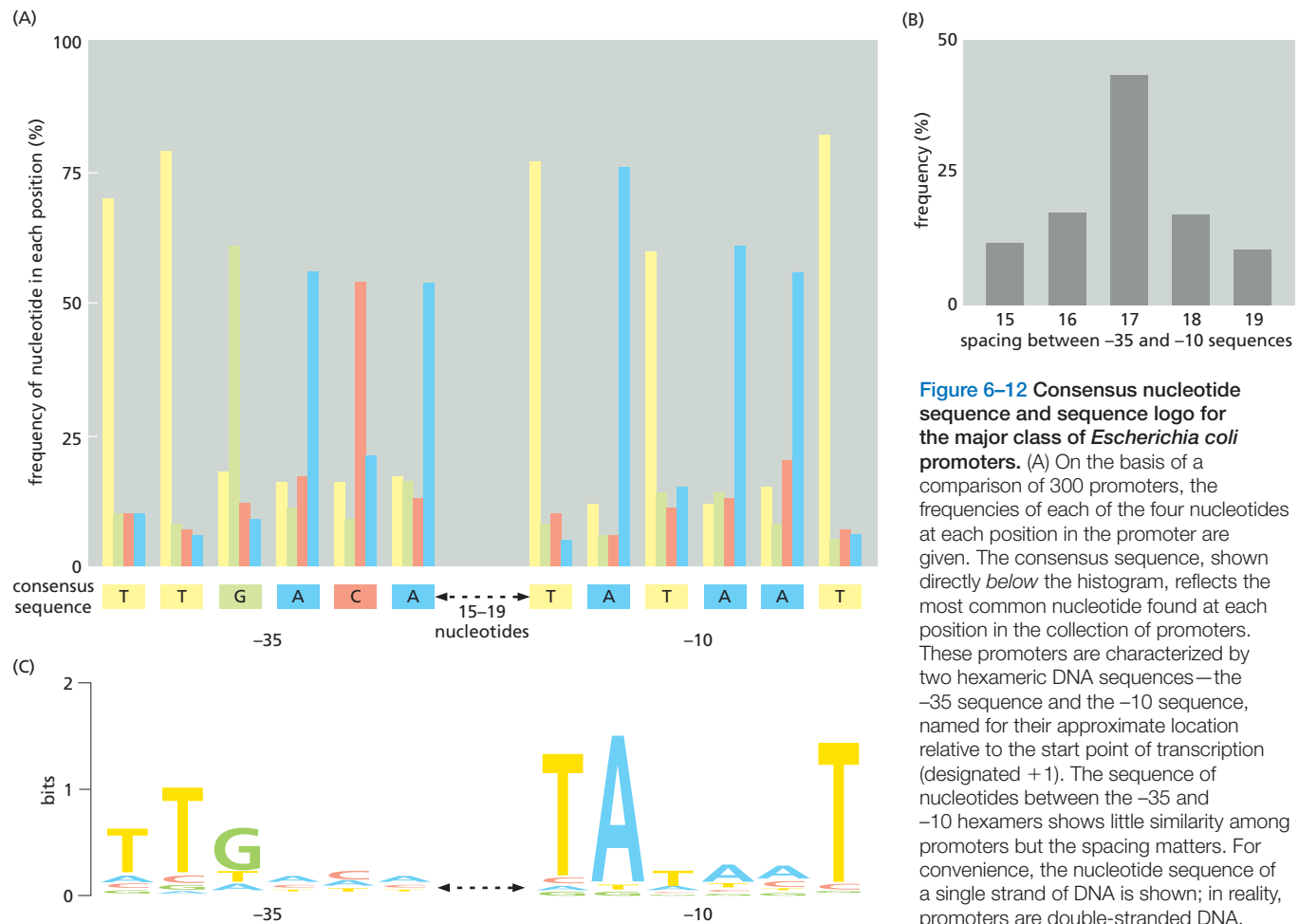


Figure 6-12 Consensus nucleotide sequence and sequence logo for the major class of *Escherichia coli* promoters. (A) On the basis of a comparison of 300 promoters, the frequencies of each of the four nucleotides at each position in the promoter are given. The consensus sequence, shown directly below the histogram, reflects the most common nucleotide found at each position in the collection of promoters. These promoters are characterized by two hexameric DNA sequences—the -35 sequence and the -10 sequence, named for their approximate location relative to the start point of transcription (designated +1). The sequence of nucleotides between the -35 and -10 hexamers shows little similarity among promoters but the spacing matters. For convenience, the nucleotide sequence of a single strand of DNA is shown; in reality, promoters are double-stranded DNA. The nucleotides shown in the figure are recognized by σ factor, a subunit of the RNA polymerase holoenzyme. (B) The distribution of spacing between the -35 and -10 hexamers found in *E. coli* promoters. (C) A sequence logo displaying the same information as in panel A. Here, the height of each letter is proportional to the frequency at which that base occurs at that position across a wide variety of promoter sequences. The total height of all the letters at each position is proportional to the information content (expressed in bits) at that position. For example, the total information content of a position that can tolerate several different bases is small (see the last three bases of the -35 sequences) but statistically greater than random.

directly by the σ factor. These common features are often summarized in the form of a *consensus sequence* (Figure 6-12). A **consensus nucleotide sequence** is derived by comparing many sequences with the same basic function and tallying up the most common nucleotides found at each position. It therefore serves as a summary or “average” of a large number of individual nucleotide sequences. A more accurate way of displaying the range of DNA sequences recognized by a protein is through the use of a *sequence logo*, which reveals the relative frequencies of each nucleotide at each position (Figure 6-12C).

The DNA sequences of individual bacterial promoters differ in ways that determine their strength, that is, the number of initiation events per unit time. Evolutionary processes have fine-tuned each to initiate as often as necessary and have thereby created a wide spectrum of promoter strengths. Promoters for genes that code for abundant proteins are much stronger than those associated with genes that encode rare proteins, and the nucleotide sequences of their promoters are responsible for these differences.

Like bacterial promoters, transcription terminators also have a wide range of sequences, with the potential to form a simple hairpin RNA structure being the most important common feature. Because an almost unlimited number of nucleotide sequences have this potential, terminator sequences are even more heterogeneous than promoter sequences.

We have discussed bacterial promoters and terminators in some detail to illustrate an important point regarding the analysis of genome sequences. Although we know a great deal about bacterial promoters and terminators and can construct “average” sequences that summarize their most salient features, their variation in nucleotide sequence makes it difficult to definitively locate them simply by

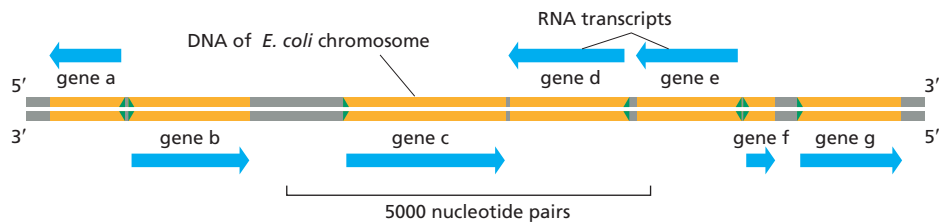


Figure 6–13 Directions of transcription along a short portion of a bacterial chromosome. Some genes are transcribed using one DNA strand as a template, while others are transcribed using the other DNA strand. The direction of transcription is determined by the orientation of the promoter at the beginning of each gene (green arrowheads). This diagram shows approximately 0.2% (9000 base pairs) of the *E. coli* chromosome. The genes transcribed from *left to right* use the bottom DNA strand as the template; those transcribed from *right to left* use the top strand as the template. DNA that is not transcribed is indicated in gray.

analysis of the nucleotide sequence of a genome. It is even more difficult to locate analogous sequences in eukaryotic genomes, due in part to the excess DNA carried in these genomes. Often we need additional information, some of it from direct experimentation, to locate and accurately interpret the short DNA signals in genomes.

As shown in Figure 6–12, promoter sequences are asymmetric, ensuring that RNA polymerase can bind in only one orientation. Because the polymerase can synthesize RNA only in the 5′-to-3′ direction, the promoter orientation specifies the strand to be used as a template. Genome sequences reveal that the DNA strand that is used as the template for RNA synthesis varies from gene to gene, depending on the orientation of the promoter (Figure 6–13).

Having considered transcription in bacteria, we now turn to the situation in eukaryotes, where the synthesis of RNA molecules is a much more elaborate affair.

Transcription Initiation in Eukaryotes Requires Many Proteins

In contrast to bacteria, which contain a single type of RNA polymerase, eukaryotic nuclei have three: *RNA polymerase I*, *RNA polymerase II*, and *RNA polymerase III*. The three polymerases are structurally similar to one another and share some common subunits, but they transcribe different categories of genes (Table 6–2). RNA polymerases I and III transcribe the genes encoding transfer RNA, ribosomal RNA, and various small RNAs. RNA polymerase II transcribes most genes, including all those that encode proteins, and our subsequent discussion therefore focuses on this enzyme.

Eukaryotic RNA polymerase II has many structural similarities to bacterial RNA polymerase (Figure 6–14). But there are several important differences in the way in which the bacterial and eukaryotic enzymes function, two of which concern us immediately.

1. While bacterial RNA polymerase requires only a single transcription-initiation factor (σ) to begin transcription, eukaryotic RNA polymerases require many such factors, collectively called the *general transcription factors*.
2. Eukaryotic transcription initiation must take place on DNA that is packaged into nucleosomes and higher-order forms of chromatin structure (described in Chapter 4), features that are absent from bacterial chromosomes.

TABLE 6–2 The Three RNA Polymerases in Eukaryotic Cells	
Type of polymerase	Genes transcribed
RNA polymerase I	5.8S, 18S, and 28S rRNA genes
RNA polymerase II	All protein-coding genes, plus snoRNA genes, miRNA genes, siRNA genes, lncRNA genes, and most snRNA genes
RNA polymerase III	tRNA genes, 5S rRNA genes, some snRNA genes, and genes for other small RNAs
The rRNAs were named according to their “S” values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.	

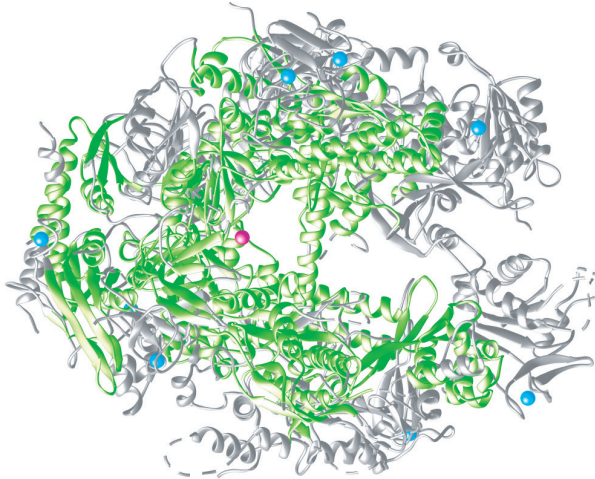


Figure 6-14 Structural similarity between a bacterial RNA polymerase and a eukaryotic RNA polymerase II. Regions of the two RNA polymerases that have similar structures are indicated in green. The eukaryotic polymerase is larger than the bacterial enzyme (12 subunits instead of 5), and some of the additional regions are shown in gray. The red sphere represents the Mg atom present at the active site, where polymerization takes place, while the blue spheres denote Zn atoms that serve as structural components. The RNA polymerases in all modern-day cells (bacteria, archaea, and eukaryotes) are closely related, indicating that the basic features of the enzyme were in place before the divergence of the three major branches of life. (Courtesy of P. Cramer and R. Kornberg.)

To Initiate Transcription, RNA Polymerase II Requires a Set of General Transcription Factors

The **general transcription factors** help to position eukaryotic RNA polymerase correctly at the promoter, aid in pulling apart the two strands of DNA to allow transcription to begin, and release RNA polymerase from the promoter to start its elongation mode. The proteins are “general” because they are needed at nearly all promoters used by RNA polymerase II. They consist of a set of interacting proteins denoted arbitrarily as TFIIA, TFIIB, and so on (TFII standing for transcription factor for polymerase II). In a broad sense, the eukaryotic general transcription factors carry out functions that are equivalent to those of the σ factor in bacteria.

Figure 6-15 illustrates how the general transcription factors assemble at promoters used by RNA polymerase II, and **Table 6-3** summarizes their activities. The assembly process begins when TFIID binds to a short double-helical DNA sequence primarily composed of T and A nucleotides. For this reason, this sequence is known as the TATA sequence, or **TATA box**, and the subunit of TFIID that recognizes it is called TBP (for TATA-binding protein). The TATA box is typically located about 30 nucleotides upstream from the transcription start site. It is not the only DNA sequence that signals the start of transcription (**Figure 6-16**), but for many polymerase II promoters it is the most important. The binding of

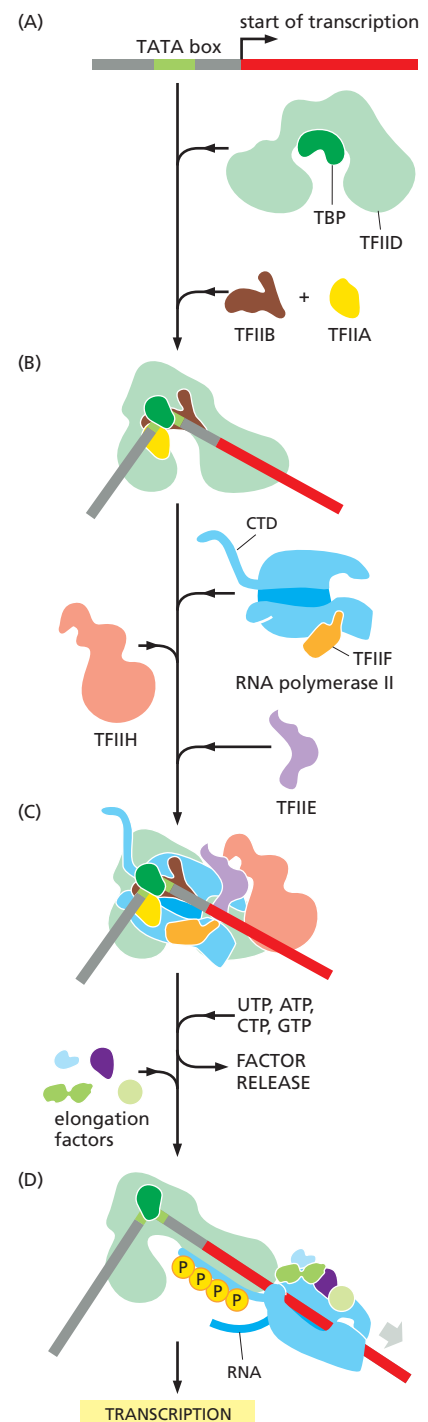


Figure 6-15 Initiation of transcription of a eukaryotic gene by RNA polymerase II. To begin transcription, RNA polymerase requires several general transcription factors. (A) Many promoters contain a DNA sequence called the TATA box, which, in humans, is located about 30 nucleotides away from the site at which transcription is initiated. (B) Through its subunit TBP, TFIID recognizes and binds the TATA box, which then enables the adjacent binding of TFIIB and TFIIA. (C) The RNA polymerase and the rest of the general transcription factors assemble at the promoter. (D) TFIIF then uses energy from ATP hydrolysis to pry apart the DNA double helix at the transcription start point, locally exposing the template strand. TFIIF also phosphorylates the long C-terminal polypeptide tail of RNA polymerase II, also called the C-terminal domain (CTD). This causes the polymerase to be released from the general factors and begin the elongation phase of transcription. For most genes, TFIID remains bound at the promoter whereas most of the other general transcription factors are released when the polymerase begins transcribing.

TABLE 6–3 The General Transcription Factors Needed for Transcription Initiation by Eukaryotic RNA Polymerase II

Name	Number of subunits	Roles in transition initiation
TFIID	12	Recognizes TATA box and other DNA sequences near the transcription start point
TFIIB	1	Recognizes BRE element in promoters; accurately positions RNA polymerase at the start site of transcription
TFIIA	2	Not required in all promoters; stabilizes binding of TFIID
TFIIF	3	Stabilizes RNA polymerase interaction with TFIIB; helps attract TFIIIE and TFIIH
TFIIIE	2	Attracts and regulates TFIIH
TFIIH	10	Unwinds DNA at the transcription start point, phosphorylates Ser5 of the RNA polymerase C-terminal domain (CTD); releases RNA polymerase from the promoter

TFIID is composed of TBP and 11 additional subunits called TAFs (TBP-associated factors).

TFIID causes a large distortion in the DNA of the TATA box (**Figure 6–17**). This distortion is thought to serve as a physical landmark for the location of an active promoter in the midst of a very large genome, and it brings DNA sequences on both sides of the distortion closer together to allow for subsequent protein assembly steps. The additional general transcription factors then assemble, along with RNA polymerase II, to form a complete *transcription initiation complex* (see **Figure 6–15**). The most complicated of the general transcription factors is TFIIH (shown in *pink*). Consisting of 10 subunits, it is nearly as large as RNA polymerase II itself and, as we shall see shortly, performs several enzymatic steps needed for the initiation of transcription.

After forming a transcription initiation complex on the promoter DNA, RNA polymerase II must gain access to the template strand at the transcription start point. TFIIH makes this step possible by hydrolyzing ATP and pulling apart the DNA strands at the start site, thereby exposing the template strand. Next, RNA polymerase II, like the bacterial polymerase, remains at the promoter synthesizing short lengths of RNA until it undergoes a series of conformational changes that allow it to move away from the promoter and enter the elongation phase of transcription. A key step in this transition is the addition of phosphate groups to the “tail” of the RNA polymerase (known as the CTD, or C-terminal domain). In

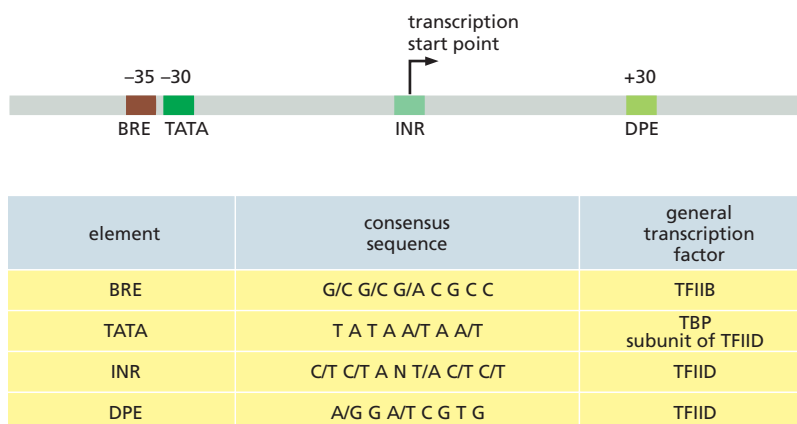


Figure 6–16 Consensus sequences found in the vicinity of eukaryotic RNA polymerase II start points. The name given to each consensus sequence (first column) and the general transcription factor that recognizes it (last column) are indicated. N indicates any nucleotide, and two nucleotides separated by a slash indicate an equal probability of either nucleotide at the indicated position. In reality, each consensus sequence is a shorthand representation of a histogram similar to that of **Figure 6–12**.

For most RNA polymerase II transcription start points, only two or three of the four sequences are present. For example, many polymerase II promoters have a TATA box sequence, but those that do not typically have a “strong” INR sequence. Although most of the DNA sequences that influence transcription initiation are located upstream of the transcription start point, a few, such as the DPE shown in the figure, are located within the transcribed region.

humans, the CTD consists of 52 tandem repeats of a seven-amino-acid sequence, which extend from the RNA polymerase core structure. During transcription initiation, the serines located at the fifth position in each repeat sequence (Ser5) are phosphorylated by TFIIH, which contains a protein kinase in one of its subunits. Triggered by these phosphorylations, the polymerase disengages from the cluster of general transcription factors (see Figure 6-15D). During this process, it undergoes a series of conformational changes that tighten its interaction with DNA, and it acquires new proteins that allow it to transcribe for long distances, in some cases for many hours, without dissociating from DNA.

Once the polymerase II has begun elongating the RNA transcript, most of the general transcription factors are released from the DNA so that they are available to initiate another round of transcription with a new RNA polymerase molecule. As we see shortly, the phosphorylation of the tail of RNA polymerase II has an additional function: it causes components of the RNA-processing machinery to load onto the polymerase and thereby be positioned to modify the newly transcribed RNA as it emerges from the polymerase.

In Eukaryotes, Transcription Initiation Also Requires Activator, Mediator, and Chromatin-modifying Proteins

The model for transcription initiation just described is based on experiments performed *in vitro* using purified proteins and DNA. However, as discussed in Chapter 4, DNA in eukaryotic cells is packaged into nucleosomes, which are further arranged in higher-order chromatin structures. As a result, transcription initiation in a eukaryotic cell is more complex and requires more proteins than it does on purified DNA. First, regulatory proteins known as *transcriptional activators* must bind to specific sequences in DNA (called *enhancers*) to help attract the general transcription factors and RNA polymerase II to the start point of transcription (Figure 6-18). We discuss the role of these activators in Chapter 7, because they are one of the main ways in which cells regulate expression of their genes. Here we simply note that their presence on DNA is required for transcription initiation in a eukaryotic cell. Second, eukaryotic transcription initiation *in vivo* requires the presence of a large protein complex known as *Mediator*, which

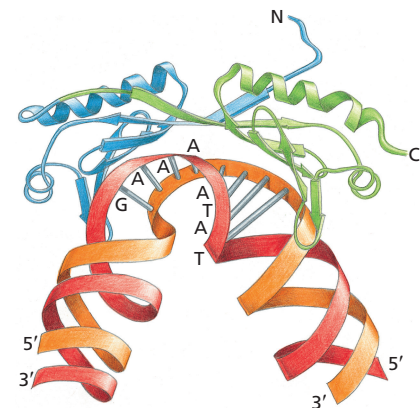
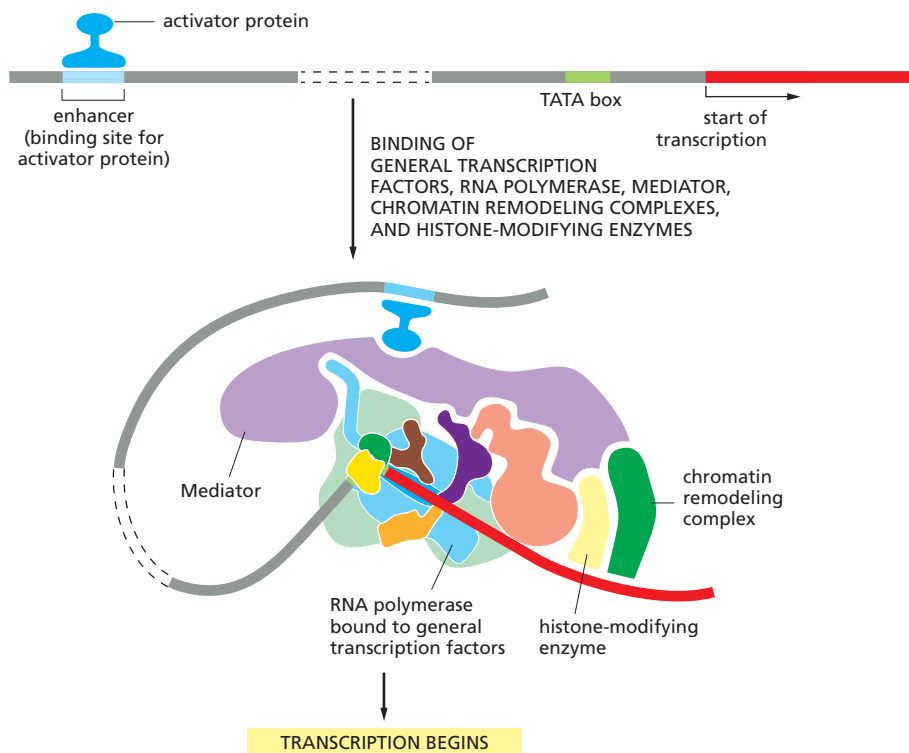


Figure 6-17 Three-dimensional structure of TBP (TATA-binding protein) bound to DNA. The TBP is the subunit of the general transcription factor TFIID that is responsible for recognizing and binding the TATA box sequence in the DNA. The unique DNA bending caused by TBP—kinks in the double helix separated by partly unwound DNA—is thought to serve as a landmark that helps to attract the other general transcription factors (Movie 6.4). TBP is a single polypeptide chain that is folded into two very similar domains (blue and green). (Adapted from J.L. Kim et al., *Nature* 365:520–527, 1993.)

Figure 6-18 Transcription initiation by RNA polymerase II in a eukaryotic cell.

Transcription initiation *in vivo* requires the presence of transcription activator proteins. As described in Chapter 7, these proteins bind to short, specific sequences in DNA that are located in regulatory regions called enhancers. Although only one activator is shown here (in blue), a typical eukaryotic gene utilizes many DNA-bound transcription activator proteins, which act together to determine that gene's rate and pattern of transcription across different cell types. Often acting from a distance of many thousand nucleotide pairs along DNA (indicated by the dashes), these proteins help RNA polymerase, the general transcription factors, and Mediator all to assemble at the promoter. In addition, ATP-dependent chromatin remodeling complexes and histone-modifying enzymes are needed at most genes. One of the main roles of Mediator is to coordinate the assembly of all these proteins at the promoter so that transcription can begin. As discussed in Chapter 4, the "default" state of eukaryotic DNA is to be packaged into nucleosomes and higher-order chromatin structures; for simplicity, these are not shown in this figure.

allows the activator proteins to communicate properly with the polymerase II and with the general transcription factors. Mediator also correctly positions TFIIF near the tail of RNA polymerase, facilitating phosphorylation of the tail and the consequent release of the polymerase from the promoter to begin synthesizing RNA. Finally, transcription initiation in a eukaryotic cell typically requires the recruitment of chromatin-modifying enzymes, including chromatin remodeling complexes and histone-modifying enzymes. As discussed in Chapter 4, both types of enzymes can increase access to the DNA in chromatin, and by doing so they facilitate the assembly of the transcription initiation machinery onto DNA.

To summarize, as illustrated schematically in Figure 6–18, many proteins (well over 100 individual subunits) must assemble at the start point of transcription to initiate transcription in a eukaryotic cell. We shall return to some of these proteins—especially transcription activator proteins, chromatin remodeling complexes, and histone-modifying enzymes—in the following chapter, where we discuss how eukaryotic cells regulate the process of transcription initiation.

Transcription Elongation in Eukaryotes Requires Accessory Proteins

Once RNA polymerase has initiated transcription, it moves jerkily, pausing at some DNA sequences and rapidly transcribing through others. Elongating RNA polymerases, both bacterial and eukaryotic, are associated with a series of *elongation factors*, proteins that decrease the likelihood that RNA polymerase will dissociate before it reaches the end of a gene. These factors typically associate with RNA polymerase shortly after initiation, and they help the polymerase move both through nucleosomes (Figure 6–19) and through the wide variety of different DNA sequences that are found in genes.

We will see in the next chapter that, like the process of transcription initiation, transcription elongation can be regulated by the cell; more specifically, we will see that at many human genes, RNA polymerase pauses shortly after it initiates transcription. This pause can last from several seconds to many hours, and the cell controls the duration of this pause as part of gene regulatory processes.

As RNA polymerase II moves along a gene, some of the enzymes bound to it modify the histones, leaving behind a record of where the polymerase has been. Although it is not clear exactly how the cell uses this information, it may aid in transcribing a gene over and over again once it has become active for the first time.

Transcription Creates Superhelical Tension

Nucleosomes are not the only impediment to elongating RNA polymerases, and in this section, we describe an entirely different type of barrier, one that applies to both bacterial and eukaryotic polymerases. To introduce this issue, we need first to consider a subtle property inherent in the DNA double helix called DNA supercoiling. **DNA supercoiling** is the name given to a conformation that DNA

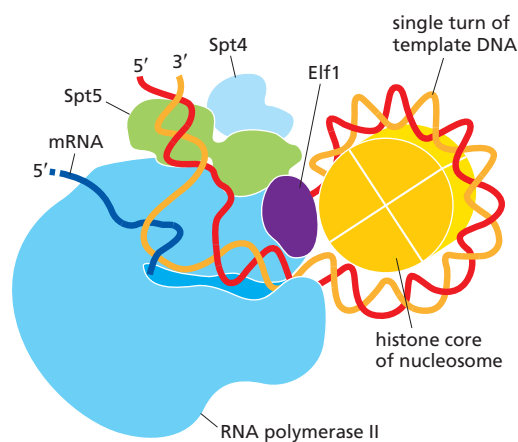


Figure 6–19 Structure of an RNA polymerase II transcribing through a nucleosome. In the structure diagrammed here, which was determined by cryo-electron microscopy, the polymerase has moved about halfway through the DNA of the nucleosome, leaving only one of the two loops of duplex DNA still bound to the histone core. The polymerase is shown in blue, associated with three elongation factors (Spt4, Spt5, and Elf1) that help the polymerase transcribe through nucleosomes. These factors act in several ways: they form a wedge to pry the DNA away from the histone core as the polymerase moves forward; they directly destabilize histone–DNA interactions by pushing a positively charged surface ahead of the RNA polymerase; and they reduce the intrinsic “stickiness” of RNA polymerase for nucleosomes. In addition to these factors, eukaryotic transcription is typically aided by ATP-dependent chromatin remodeling complexes that seek out and rescue the occasional stalled polymerase, as well as by histone chaperones that can partially disassemble nucleosomes in front of a moving RNA polymerase and reassemble them behind. (Based on PDB code 6IR9.)

can adopt in response to superhelical tension. Alternatively, the creation of loops or coils in a double-helical DNA molecule will produce such tension.

Figure 6–20 illustrates why. There are approximately 10 nucleotide pairs for every helical turn in a DNA double helix. Imagine a helix whose two ends are fixed with respect to each other (as they are in a DNA circle, such as a bacterial chromosome, or in a tightly clamped loop, as can exist in eukaryotic chromosomes). In this case, one large DNA supercoil will form to compensate for each 10 nucleotide pairs that are opened (unwound). The formation of this supercoil is energetically favorable because it restores a normal helical twist to the base-paired regions that remain, which would otherwise become overwound because of the fixed ends.

RNA polymerase creates superhelical tension as it moves along a stretch of DNA that is anchored at its ends. As illustrated in Figure 6–20C, if the polymerase is not free to rotate rapidly (and such rotation is unlikely given the size of RNA polymerases and their attached transcripts), a moving polymerase will generate positive superhelical tension in the DNA in front of it and negative helical tension

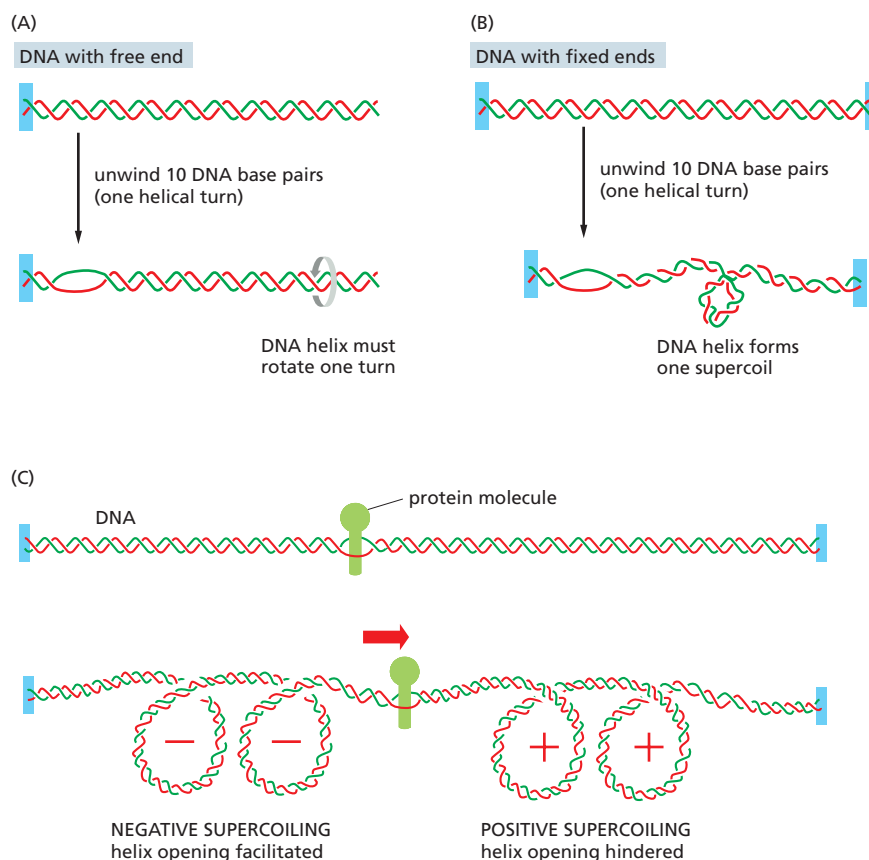
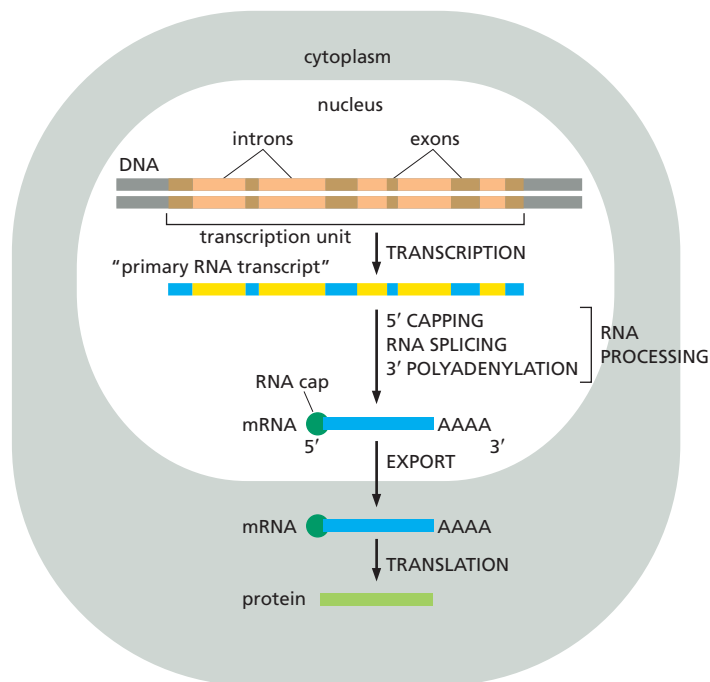


Figure 6–20 Superhelical tension in DNA causes DNA supercoiling. (A) For a DNA molecule with one free end (or a break in one strand that serves as a swivel), the DNA double helix rotates by one turn for every 10 nucleotide pairs opened. (B) If rotation is prevented, superhelical tension is introduced into the DNA by helix opening. In the example shown, the DNA helix contains 10 helical turns, one of which is opened. One way of accommodating the tension created would be to increase the helical twist from 10 to 11 nucleotide pairs per turn in the double helix that remains. The DNA helix, however, resists such a deformation in a springlike fashion, preferring to relieve the superhelical tension by bending into supercoiled loops. As a result, one DNA supercoil forms in the DNA double helix for every 10 nucleotide pairs opened. The supercoil formed in this case is a positive supercoil. (C) Supercoiling of DNA is induced by a protein tracking through the DNA double helix. The two ends of the DNA shown here are unable to rotate freely relative to each other, and the protein molecule is assumed also to be prevented from rotating freely as it moves. Under these conditions, the movement of the protein causes an excess of helical turns to accumulate in the DNA helix ahead of the protein (inducing positive supercoils) and a deficit of helical turns to arise in the DNA behind the protein (inducing negative supercoils), as shown. Because locally pulling apart the two strands of the double helix relieves the tension from negative supercoils, it is easier to do behind the moving protein than ahead of it.

(A) EUKARYOTES



(B) BACTERIA

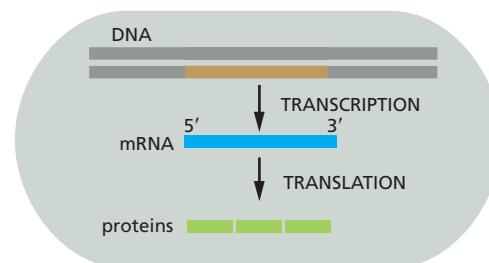


Figure 6–21 Comparison of the steps leading from gene to protein in eukaryotes and bacteria. The final amount of a protein in the cell depends on the efficiency of each step and on the rates of degradation of the RNA and protein molecules. (A) In eukaryotic cells, the mRNA molecule resulting from transcription contains both coding (exon) and noncoding (intron) sequences. Before it can be translated into protein, the two ends of the RNA are modified, the introns are removed by an enzymatically catalyzed RNA splicing reaction, and the resulting mRNA is transported from the nucleus to the cytoplasm. For convenience, the steps in this figure are depicted as occurring one at a time; in reality, many occur concurrently. For example, the RNA cap is added and splicing begins before transcription has been completed. Because of the coupling between transcription and RNA processing, intact primary transcripts—the full-length RNAs that would, in theory, be produced if no processing had occurred—are found only rarely. (B) In bacteria, the production of mRNA is much simpler. The 5' end of an mRNA molecule is produced by the initiation of transcription, and the 3' end is produced by the termination of transcription. Because bacteria lack a nucleus, transcription and translation take place in a common compartment, and the translation of a bacterial mRNA often begins before its synthesis has been completed. As indicated, a single bacterial mRNA typically produces several different proteins, another feature that distinguishes eukaryotes from bacteria.

behind it. If this tension is not relieved, the polymerase will grind to a halt because further unwinding requires more energy than the transcription process can provide. For eukaryotes, the mild buildup of tension is thought to provide a bonus: the positive superhelical tension ahead of the polymerase facilitates the partial unwrapping of the DNA in nucleosomes, inasmuch as the release of DNA from the histone core helps to relax this tension. The tension can also be relieved by DNA topoisomerase enzymes, as we saw in the previous chapter for the similar kind of tension generated by DNA polymerases during DNA replication (see Figure 5–21).

In bacteria (but not eukaryotes), a specialized topoisomerase called *DNA gyrase* uses the energy of ATP hydrolysis to pump supercoils continually into the DNA, thereby maintaining the DNA under constant tension. These are *negative supercoils*, having the opposite handedness from the *positive supercoils* that form when a region of DNA helix opens (see Figure 6–20B). Whenever a region of helix opens, it removes these negative supercoils from bacterial DNA, reducing the superhelical tension. DNA gyrase therefore makes the opening of the DNA helix in bacteria energetically favorable compared with helix opening in DNA that is not supercoiled. For this reason, it facilitates those genetic processes in bacteria, such as the initiation of transcription by bacterial RNA polymerase, that require helix opening (see Figure 6–11).

Transcription Elongation in Eukaryotes Is Tightly Coupled to RNA Processing

We saw earlier that bacterial mRNAs are synthesized by the RNA polymerase starting and stopping at specific spots on the genome. The situation in eukaryotes is substantially different. In particular, transcription is only the first of several steps needed to produce a mature mRNA molecule. Other critical steps are the covalent modification of the ends of the RNA and the removal of *intron sequences* that are discarded from the middle of the RNA transcript by the process of *RNA splicing* (Figure 6–21). As we shall see, RNA splicing not only joins together different portions of an RNA transcript to eliminate the intron sequences; it also provides eukaryotes with the ability to synthesize several related but different proteins from the same gene.

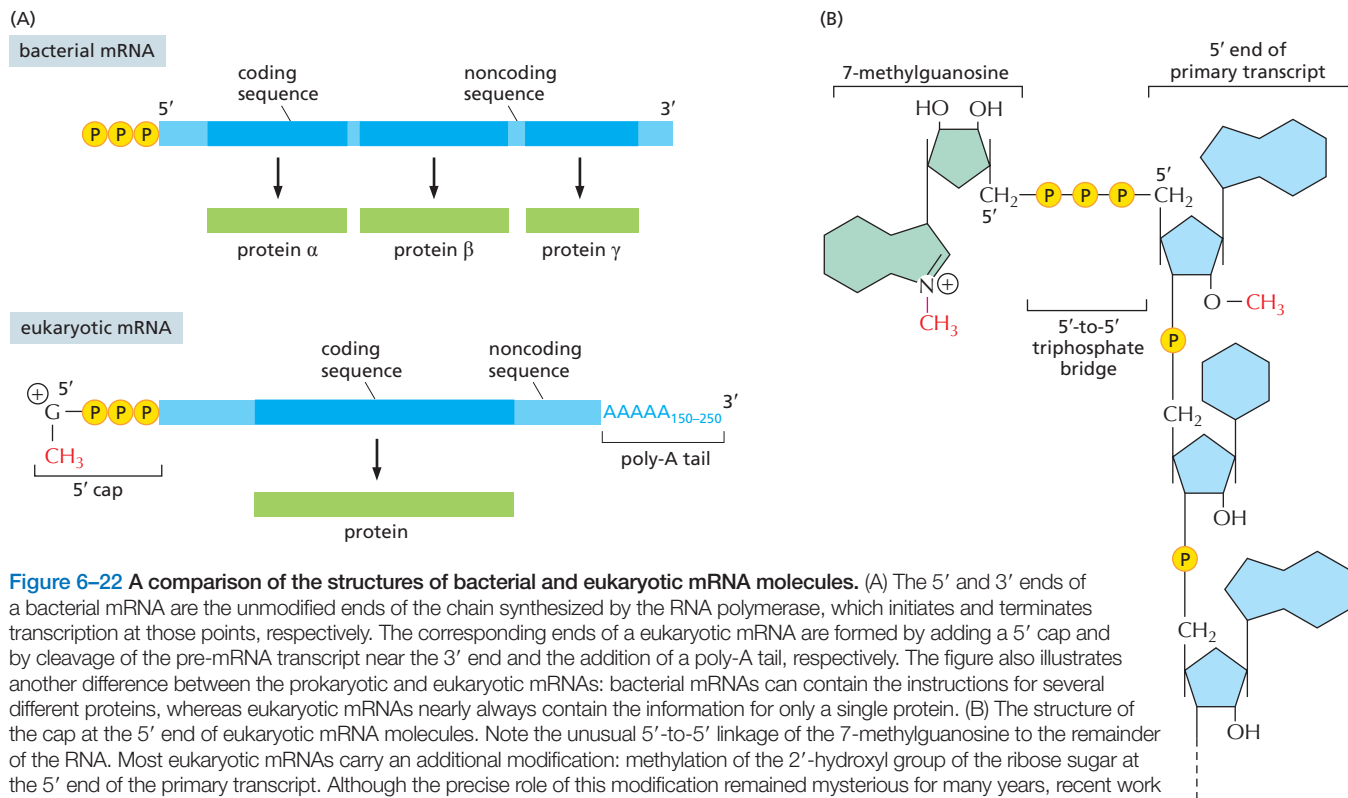


Figure 6-22 A comparison of the structures of bacterial and eukaryotic mRNA molecules. (A) The 5' and 3' ends of a bacterial mRNA are the unmodified ends of the chain synthesized by the RNA polymerase, which initiates and terminates transcription at those points, respectively. The corresponding ends of a eukaryotic mRNA are formed by adding a 5' cap and by cleavage of the pre-mRNA transcript near the 3' end and the addition of a poly-A tail, respectively. The figure also illustrates another difference between the prokaryotic and eukaryotic mRNAs: bacterial mRNAs can contain the instructions for several different proteins, whereas eukaryotic mRNAs nearly always contain the information for only a single protein. (B) The structure of the cap at the 5' end of eukaryotic mRNA molecules. Note the unusual 5'-to-5' linkage of the 7-methylguanosine to the remainder of the RNA. Most eukaryotic mRNAs carry an additional modification: methylation of the 2'-hydroxyl group of the ribose sugar at the 5' end of the primary transcript. Although the precise role of this modification remained mysterious for many years, recent work indicates that it aids cells in distinguishing their own mRNAs from those of invading viruses, which typically lack this modification. On the basis of this difference, cells can block translation of viral RNAs, thereby defending themselves from viral attack.

Both ends of eukaryotic mRNAs are modified: by *capping* on the 5' end and by *polyadenylation* of the 3' end (Figure 6-22). These special ends allow the cell to assess whether both ends of an mRNA molecule are present (and if the message is therefore intact) before it exports the RNA from the nucleus and translates it into protein.

A simple mechanism has evolved to couple all of the above RNA-processing steps to transcription elongation. As discussed previously, a key step in transcription initiation by RNA polymerase II is the phosphorylation of the RNA polymerase II tail, also called the CTD (C-terminal domain). This phosphorylation, which proceeds gradually as the RNA polymerase initiates transcription and moves along the DNA, not only helps to dissociate the RNA polymerase II from other proteins present at the start point of transcription but also allows a new set of proteins to associate with the RNA polymerase tail, which function in transcription elongation and RNA processing, as we discuss in the following sections. Some of the processing proteins are thought to “hop” from the polymerase tail onto the nascent RNA molecule to begin their processing reactions as soon as this RNA emerges from the RNA polymerase. Thus, we can view RNA polymerase II in its elongation mode as an RNA factory that not only moves along the DNA synthesizing an RNA molecule but also processes the RNA that it produces (Figure 6-23). Fully extended, the CTD is nearly 10 times longer than the remainder of the RNA polymerase. As a flexible protein domain, it serves as a scaffold or tether, holding a variety of proteins close by so that they can rapidly act when needed. This scaffolding strategy, which greatly speeds up the overall rate of a series of consecutive reactions, is one that is commonly utilized in the cell (see Figure 3-76).

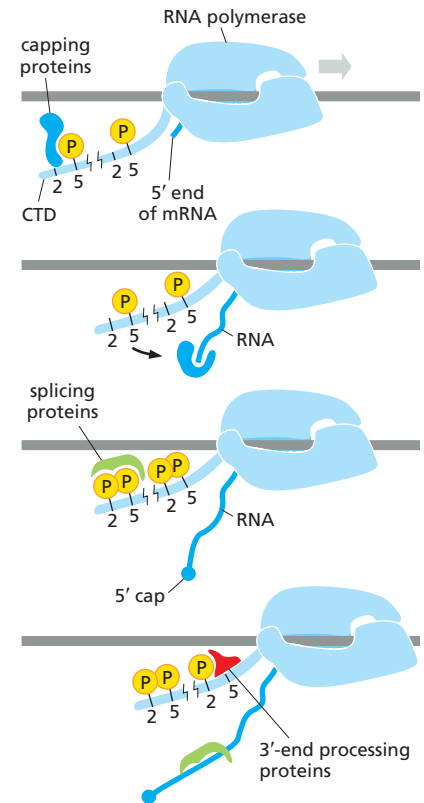
RNA Capping Is the First Modification of Eukaryotic Pre-mRNAs

As soon as RNA polymerase II has produced about 20 nucleotides of RNA, the 5' end of the new RNA molecule is modified by addition of a cap that consists of a modified guanine nucleotide (see Figure 6-22B). Three enzymes, acting in

Figure 6–23 Eukaryotic RNA polymerase II as an RNA synthesis and processing machine.

As the polymerase transcribes DNA into RNA, it carries RNA-processing proteins on its tail that are transferred to the nascent RNA at the appropriate time. The tail contains 52 tandem repeats of a seven-amino-acid sequence, and there are two serines in each repeat. The capping proteins first bind to the RNA polymerase tail when it is phosphorylated on Ser5 of the heptad repeats late in the process of transcription initiation (see Figure 6–15). This strategy ensures that the RNA molecule is efficiently capped as soon as its 5' end emerges from the RNA polymerase. As the polymerase continues transcribing, its tail is extensively phosphorylated on the Ser2 positions by a kinase associated with the elongating polymerase and is eventually dephosphorylated at Ser5 positions. These further modifications attract splicing and 3'-end processing proteins to the moving polymerase, positioning them to act on the newly synthesized RNA as it emerges from the RNA polymerase. There are many RNA-processing enzymes, and not all travel with the polymerase. In RNA splicing, for example, the tail carries only a few critical components; once bound to an emerging RNA molecule, they serve as a nucleation site for the remaining components.

When RNA polymerase II finishes transcribing a gene, it is released from DNA, and protein phosphatases remove the phosphates on its tail so that it can reinitiate transcription. Only the fully dephosphorylated form of RNA polymerase II is competent to begin RNA synthesis at a promoter.



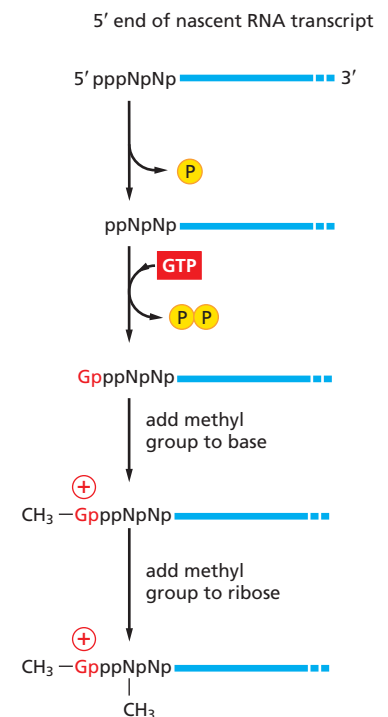
succession, perform the capping reaction: a phosphatase removes a phosphate from the triphosphate left at the 5' end of the nascent RNA molecule, a guanyl transferase adds a GMP in a reverse linkage (5'-to-5' instead of 5'-to-3') to the 5' diphosphate just produced, and a methyl transferase adds a methyl group to the guanosine (Figure 6–24). Because all three enzymes bind to the RNA polymerase tail phosphorylated at the Ser5 position—the modification added by TFIIF during transcription initiation—they are poised to modify the 5' end of the nascent transcript as soon as it emerges from the polymerase.

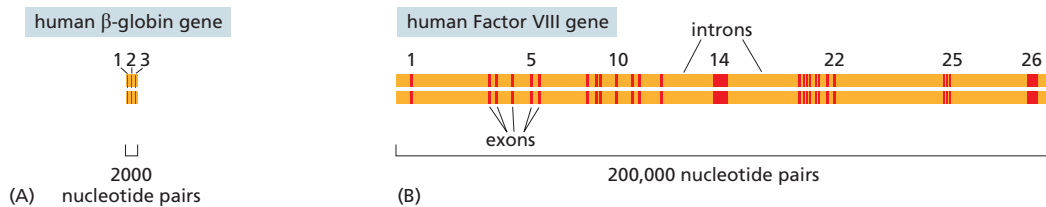
The 7-methylguanosine cap signifies the 5' end of eukaryotic mRNAs, and this landmark helps the cell to distinguish mRNAs from the other types of RNA molecules present in the cell. For example, RNA polymerases I and III produce uncapped RNAs during transcription, in part because these polymerases lack a CTD. In the nucleus, the cap binds a protein complex called CBC (cap-binding complex), which, as we discuss in subsequent sections, helps a future mRNA to be further processed and exported. The 5' cap also has an important role in the translation of mRNAs in the cytosol, as we discuss later in the chapter.

Although the vast majority of eukaryotic mRNAs possess a 7-methylguanosine cap at their 5' ends, an alternative type of cap is found on some mRNAs, specifically a nicotinamide adenine dinucleotide phosphate (NADP⁺). We saw earlier in Chapter 2 that NADP⁺ is an important cofactor for many biochemical reactions, and it is added to certain mRNAs by RNA polymerase itself as the first nucleotide when a new mRNA molecule chain is begun. The role of this particular type of mRNA cap is not known with certainty, but one hypothesis holds that it provides a way for the cell to link the expression of some of its genes to its overall metabolic “health.”

RNA Splicing Removes Intron Sequences from Newly Transcribed Pre-mRNAs

As discussed in Chapter 4, the protein-coding sequences of eukaryotic genes are typically interrupted by noncoding intervening sequences (introns). Discovered in 1977, this feature of eukaryotic genes came as a surprise to scientists, who had been, until that time, familiar only with bacterial genes, which consist of a continuous stretch of coding DNA that is directly transcribed into mRNA. In marked contrast, eukaryotic genes were found to be broken up into small pieces of coding sequence (*expressed sequences*, or **exons**) interspersed with much longer

Figure 6–24 The reactions that cap the 5' end of each RNA molecule synthesized by RNA polymerase II. The final cap contains a novel 5'-to-5' linkage between the positively charged 7-methylguanosine residue and the 5' end of the RNA transcript (see Figure 6–22B). The letter N represents any one of the four ribonucleotides, although the nucleotide that starts an RNA chain is usually a purine (an A or a G). (After A.J. Shatkin, *Bioessays* 7:275–277, 1987. With permission from John Wiley & Sons.)



intervening sequences, or **introns**; thus, the coding portion of a eukaryotic gene is often only a small fraction of the length of the gene (**Figure 6-25**).

Both intron and exon sequences are transcribed into RNA. The intron sequences are removed from the newly synthesized RNA through the process of **RNA splicing**. The vast majority of RNA splicing that takes place in cells functions in the production of mRNA, and our discussion of splicing focuses on this so-called precursor-mRNA (or pre-mRNA) splicing. Only after 5'- and 3'-end processing and splicing have taken place is an RNA transcript called an mRNA molecule.

Each splicing event removes one intron, proceeding through two sequential phosphoryl-transfer reactions known as transesterifications; these join two exons together while removing the intron between them as a “lariat” (**Figure 6-26**). The machinery that catalyzes pre-mRNA splicing is complex, consisting of five additional RNA molecules and several hundred proteins, and it hydrolyzes many ATP molecules per splicing event. This complexity ensures that splicing is accurate, while at the same time being flexible enough to deal with the enormous variety of introns found in a typical eukaryotic cell. On average, there are 11 introns in each of the approximately 20,000 human protein-coding genes, so the cell devotes considerable resources to this step in gene expression.

Although it seems wasteful to first produce and then remove large numbers of introns from an RNA transcript, this process provides an advantage to the cell. In many organisms, the transcript of a given gene can be spliced in more than one way, and this allows the same gene to produce a set of different but related proteins (**Figure 6-27**). It has been proposed that 95% of human gene transcripts are spliced in more than one way, but this is almost certainly an overestimate: many

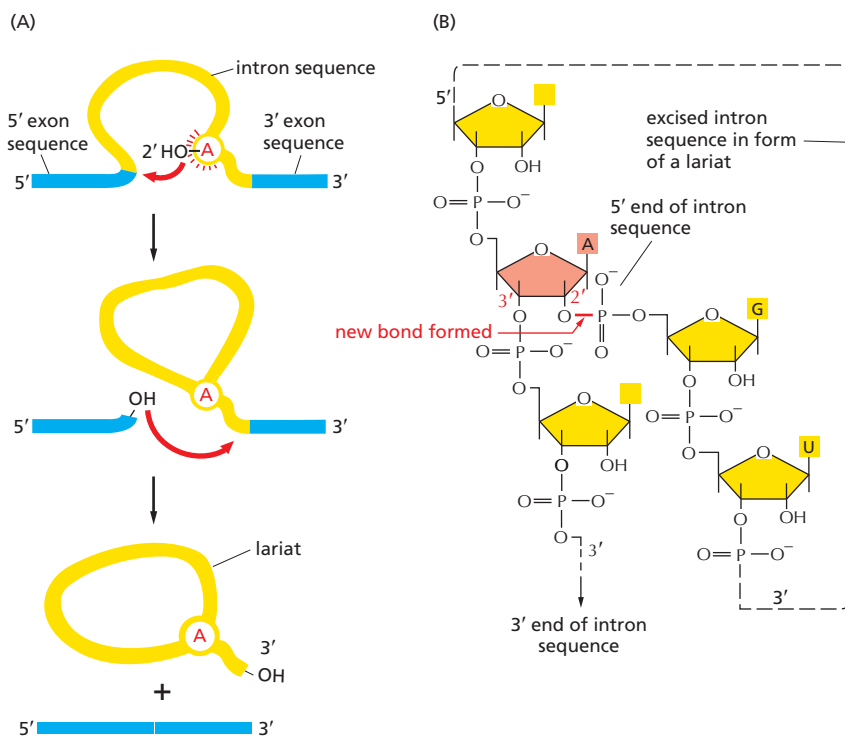


Figure 6–25 Structure of two human genes showing the arrangement of exons and introns. (A) The relatively small β -globin gene, which encodes a subunit of the oxygen-carrying protein hemoglobin, contains 3 exons (see also Figure 4–9). (B) The much larger Factor VIII gene contains 26 exons; it codes for a protein (Factor VIII) that functions in the blood-clotting pathway. The most prevalent form of hemophilia results from mutations in this gene.

Figure 6–26 The pre-mRNA splicing reaction. (A) In the first step, a specific adenine nucleotide in the intron sequence (indicated in red) attacks the 5' splice site and cuts the sugar–phosphate backbone of the RNA at this point. The cut 5' end of the intron becomes covalently linked to the adenine nucleotide, as shown in detail in (B), thereby creating a loop in the RNA molecule. The released free 3'-OH end of the exon sequence then reacts with the start of the next exon sequence, joining the two exons together and releasing the intron sequence in the shape of a *lariat*. The two exon sequences thereby become joined into a continuous coding sequence. The released intron sequence is discarded, eventually being broken down into single nucleotides, which are recycled.

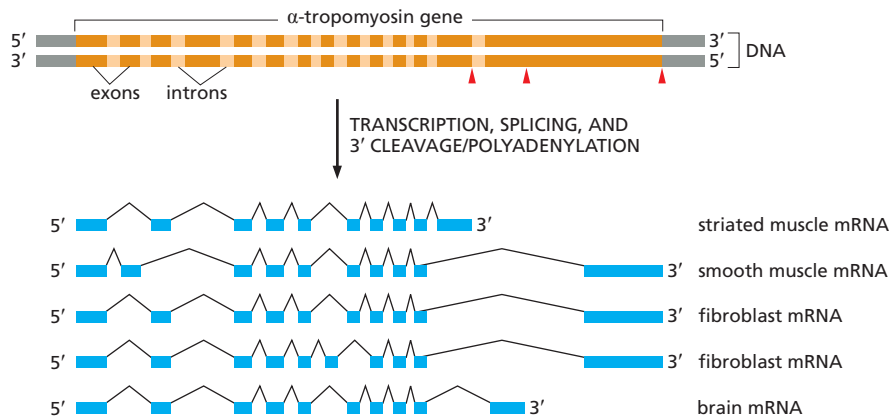


Figure 6-27 Alternative splicing of the α -tropomyosin gene from rat.

α -Tropomyosin is a coiled-coil protein (see Figure 3-8) involved in the regulation of contraction in muscle cells. Its initial RNA transcript can be spliced in different ways, as indicated in the figure, to produce distinct mRNAs, which then give rise to variant proteins. Some of the splicing patterns are specific for certain types of cells. For example, the α -tropomyosin made in striated muscle is different from that made from the same gene in smooth muscle. The arrowheads in the top part of the figure mark the sites where cleavage and poly-A addition form the 3' ends of the mature mRNAs.

of the splicing products that can be detected are the results of splicing errors and do not produce functional proteins. Nonetheless, *alternative splicing* is a key feature of gene expression in many organisms, and we shall return to this subject later, after we describe the cellular machinery that performs the basic reaction.

Nucleotide Sequences Signal Where Splicing Occurs

The mechanism of pre-mRNA splicing shown in Figure 6-26 requires that the splicing machinery recognize three portions of the precursor RNA molecule: the 5' splice site, the 3' splice site, and the branch point in the intron sequence that forms the base of the excised lariat. Not surprisingly, each site has a consensus nucleotide sequence that is similar from intron to intron and provides the cell with cues for where splicing is to take place (**Figure 6-28**). These consensus sequences are relatively short and can accommodate extensive sequence variability, and as we shall see shortly, the cell uses additional types of information to ultimately choose exactly where, on each RNA molecule, splicing is to take place.

RNA Splicing Is Performed by the Spliceosome

Unlike the other steps in mRNA production we have discussed, key steps in RNA splicing are performed by RNA molecules rather than proteins. Specialized RNA molecules recognize the nucleotide sequences that specify where splicing is to occur and also form the active site that catalyzes the chemistry of splicing. These RNA molecules are relatively short (less than 200 nucleotides each), and there are five of them, U1, U2, U4, U5, and U6. Known as **snRNAs (small nuclear RNAs)**, each is complexed with at least seven protein subunits to form an **snRNP (small nuclear ribonucleoprotein)**. These snRNPs form the core of the **spliceosome**, the large assembly of RNA and protein molecules that performs pre-mRNA splicing in the cell. Recognition of the 5' splice junction, the branch-point site, and the 3' splice junction is performed through base-pairing between the snRNAs and RNA sequences in the pre-mRNA substrate.

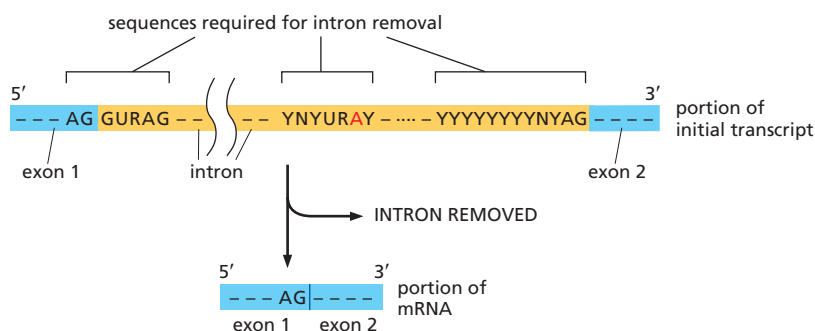
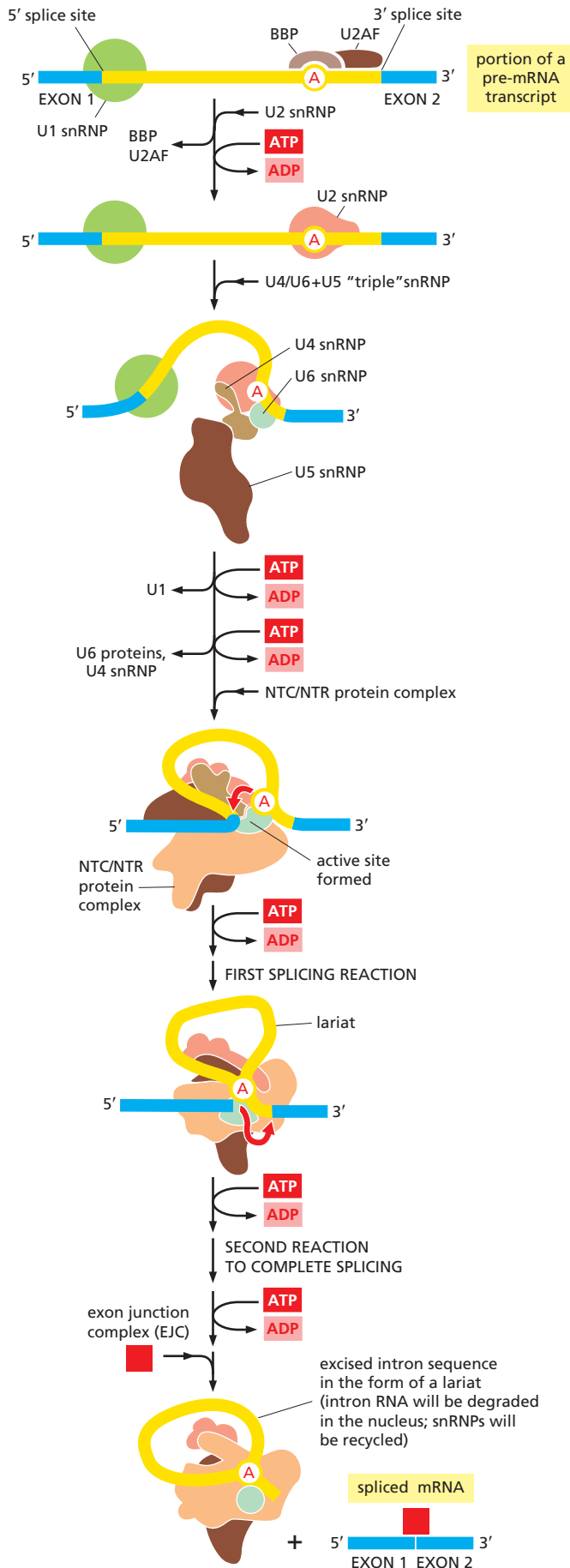


Figure 6-28 The consensus nucleotide sequences in an RNA molecule that signal the beginning and the end of most introns in humans. The three blocks of nucleotide sequences shown are required to remove an intron sequence. Here A, G, U, and C are the standard RNA nucleotides; R stands for a purine (A or G); and Y stands for a pyrimidine (C or U). The A highlighted in red forms the branch point of the lariat produced by splicing (see Figure 6-26). Only the GU at the start of the intron and the AG at its end are invariant nucleotides in the splicing consensus sequences. Several different nucleotides can occupy the remaining positions, although the indicated nucleotides are preferred. Although the distances along the RNA between the three splicing consensus sequences are highly variable, the distance between the branch point and 3' splice junction is typically much shorter than the distance between the 5' splice junction and the branch point.



The U1 snRNP forms base pairs with the 5' splice junction. BBP (branch-point binding protein) recognizes the branch-point site and binds cooperatively with U2AF, which recognizes the polypyrimidine tract and the 3' splice junction (see Figure 6–28).

The U2 snRNP displaces BBP and U2AF and forms base pairs with the branch-point site consensus sequence.

The U4/U6+U5 "triple" snRNP enters the reaction. In this triple snRNP, the U4 and U6 snRNAs are held firmly together by base-pair interactions. Subsequent rearrangements break apart the U4/U6 base pairs, allowing U6 to displace U1 at the 5' splice junction and ejecting the U4 snRNP and some of the proteins of the U6 snRNP.

Addition of the NTC/NTR protein complex positions the snRNPs to form the active site of the spliceosome and brings the branch point in proximity to the 5' splice site.

Lariat formed. Additional rearrangements bring the two exon segments together and place them in the active site.

Spliced RNA is released. Spliceosome components recycled using ATP hydrolysis to "reset" them.

Figure 6–29 The pre-mRNA splicing mechanism. RNA splicing is catalyzed by an assembly of snRNPs (shown as *colored shapes*) plus other proteins (most of which are not shown), which together constitute the spliceosome. The spliceosome recognizes the splicing signals on a pre-mRNA molecule, brings the two ends of the intron together, and forms the active site that catalyzes the two covalent bond-making and -breaking steps required (see Figure 6–26A and **Movie 6.5**). As shown, nearly every step is accompanied by hydrolysis of a molecule of ATP, which prevents the reaction from stalling or moving backwards and, as discussed below, increases the accuracy of splicing. Although only six molecules of ATP are shown in this simplified diagram, some steps require more than one, and a total of eight ATPs are consumed in each splicing reaction. As indicated in the last step, a set of proteins called the exon junction complex (EJC) is retained on the spliced mRNA molecule; its subsequent role will be discussed shortly.

The spliceosome is a complex and dynamic machine. When studied *in vitro*, a few components of the spliceosome assemble on pre-mRNA and, as the splicing reaction proceeds, new components enter and those that have already performed their tasks are jettisoned (**Figure 6–29**). However, many scientists believe that, inside the cell, the spliceosome is a preexisting, loose assembly of all the components—capturing, splicing, and releasing RNA as a coordinated unit, and undergoing extensive rearrangements each time a splice is made.

The Spliceosome Uses ATP Hydrolysis to Produce a Complex Series of RNA–RNA Rearrangements

There are many unusual features of the splicing reaction compared to the typical catalytic processes in the cell introduced in Chapter 2. First, splicing seems grossly inefficient, requiring more than a hundred proteins, five RNA molecules, and the hydrolysis of eight molecules of ATP to produce a single splice. Many genes require multiple splicing events to produce a single functional mRNA molecule (more than 20 in the example in Figure 6–25B), and the process seems inordinately complex. Second, each splicing reaction requires that the catalytic site for the reaction be assembled *de novo* on each pre-mRNA molecule through a complex, multistep process. Third, as mentioned earlier, the catalytic site of the spliceosome (which catalyzes both steps of the splicing reaction) is formed by RNA molecules (the snRNAs) rather than by proteins (**Figure 6–30**), with the proteins required to correctly position these RNAs. In the final section of this chapter, we describe the structure and chemical properties of RNA molecules that allow them to act as catalysts, as well as the proposal that the RNA-based reaction mechanisms that exist today are “leftovers” from ancient, RNA-only biological systems.

How might we rationalize the unusual biochemical complexity of splicing compared to other steps in gene expression? As we discuss shortly, pre-mRNA splicing has evolved from a much simpler, purely RNA-based process, and its complexity may in part be an example of what some evolutionary biologists term “runaway bureaucracies,” accruing more and more parts over time that are now required for the process without necessarily making it any better. We do know, however, that some of the complexity of pre-mRNA splicing provides advantages. Even though ATP hydrolysis is not required for the splicing reaction *per se*, the numerous ATP hydrolysis steps keep the reaction from stalling or running backwards and, as we will see shortly, they also increase the accuracy of pre-mRNA splicing.

Most of the spliceosome proteins that hydrolyze ATP use the released energy to break existing RNA–RNA interactions to allow the formation of new ones. These RNA–RNA rearrangements allow the splicing signals on the pre-mRNA to be examined several times during the course of splicing. For example, the U1 snRNP initially recognizes the 5′ splice site through conventional base-pairing; as splicing proceeds, these base pairs are broken (using the energy of ATP hydrolysis) and U1 is replaced by U6 (see Figure 6–30A and B). Likewise, the branch point is “examined” twice, the first time by the branch-point binding protein and the second time by the U2 snRNP (see Figure 6–29). In this way, the spliceosome checks and rechecks the splicing signals before the active site for the two transesterification reactions forms (see Figure 6–30C), thereby increasing overall accuracy.

Note that both catalytic steps of the splicing reaction occur in the same active site (the second reaction is almost the reverse of the first). ATP-mediated

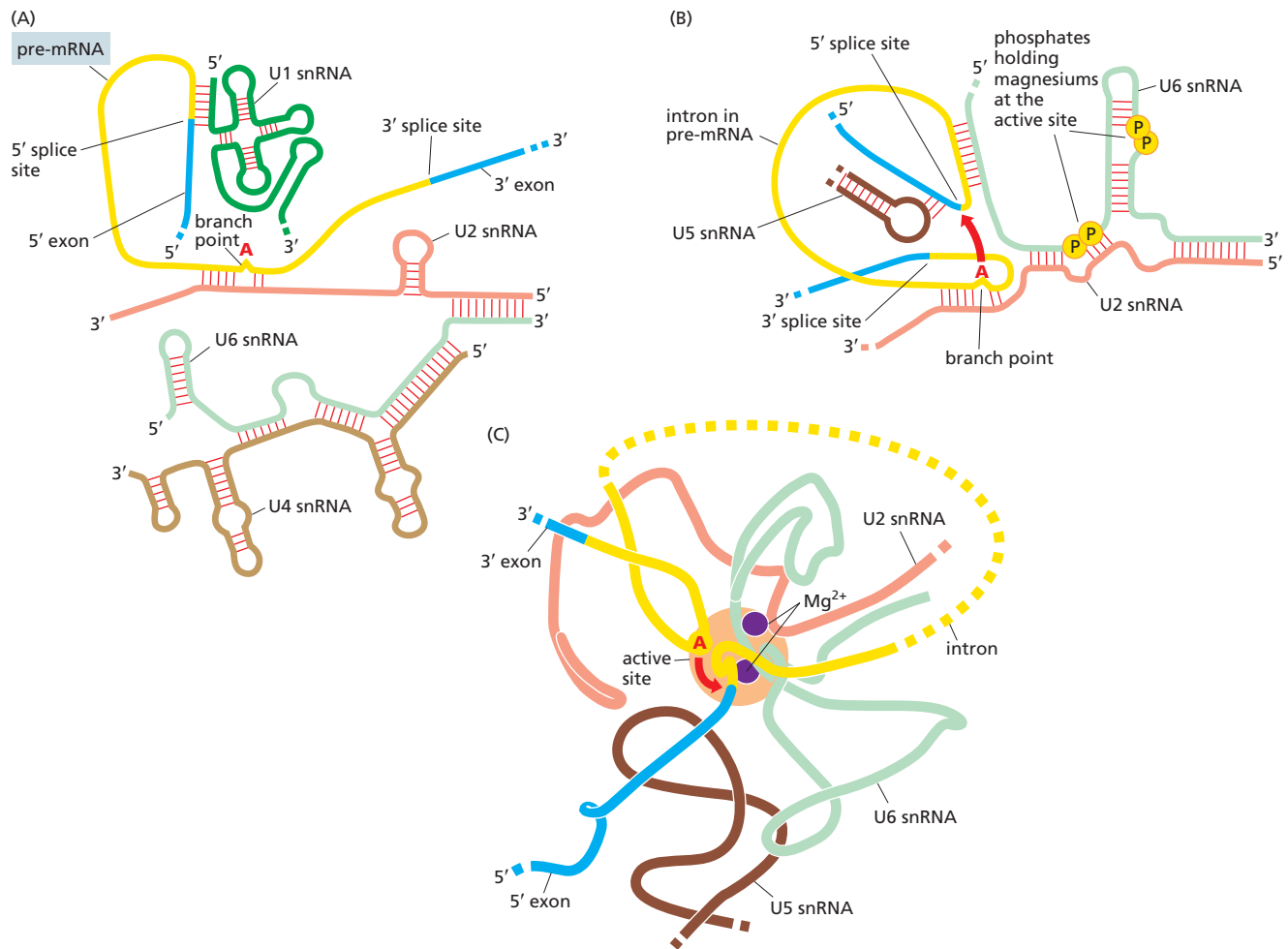


Figure 6–30 An example of an ATP hydrolysis–driven RNA–RNA rearrangement that occurs during splicing. Schematic diagram of the arrangement of RNA molecules before (A) and after (B) the active site has been formed for the two phosphoryl-transfer reactions. This rearrangement also brings the branch point and the 5' splice site together in preparation for the first reaction. The active site (which grasps the two magnesium ions needed for the chemistry of the reactions) is formed by U2 and U6. Several sequential ATP-dependent rearrangement steps are needed to convert the configuration in A to that in B. (C) The actual structure, determined by cryo-electron microscopy, of the RNA-based catalytic core of the spliceosome that was schematically illustrated in B. For simplicity, the proteins that surround the RNAs are not shown, even though their conformations are known. High-resolution structures have also been determined for most of the other spliceosome intermediates. (B, adapted from M.E. Wilkinson, C. Charenton, and K. Nagai, *Annu. Rev. Biochem.* 89:359–398, 2020. With permission from Annual Reviews.)

rearrangements are required between the two reactions to properly reposition the pre-mRNA for the second reaction, helping to ensure that splicing accidents occur only rarely.

A general process associated with ATP hydrolysis, called *kinetic proofreading*, further increases spliceosome accuracy. Because an incorrect, “off-target” base-pairing interaction will be weaker than the correct one, incorrect interactions will dissociate more rapidly than correct ones. Each ATP-mediated rearrangement of the spliceosome takes a finite amount of time, and this time delay will favor the correct choice, because off-target interactions will often dissociate during this time window, giving the correct interaction multiple chances to form. Such kinetic proofreading is used throughout biology. For example, we saw in Chapter 5 that the initial selection of the correct nucleotides by DNA polymerase during DNA replication takes advantage of this principle. We will discuss it in more detail later in the chapter when we describe translation by the ribosome.

Once the splicing chemistry is completed, the snRNPs remain bound to the excised lariat. The disassembly of these snRNPs from the lariat (and from

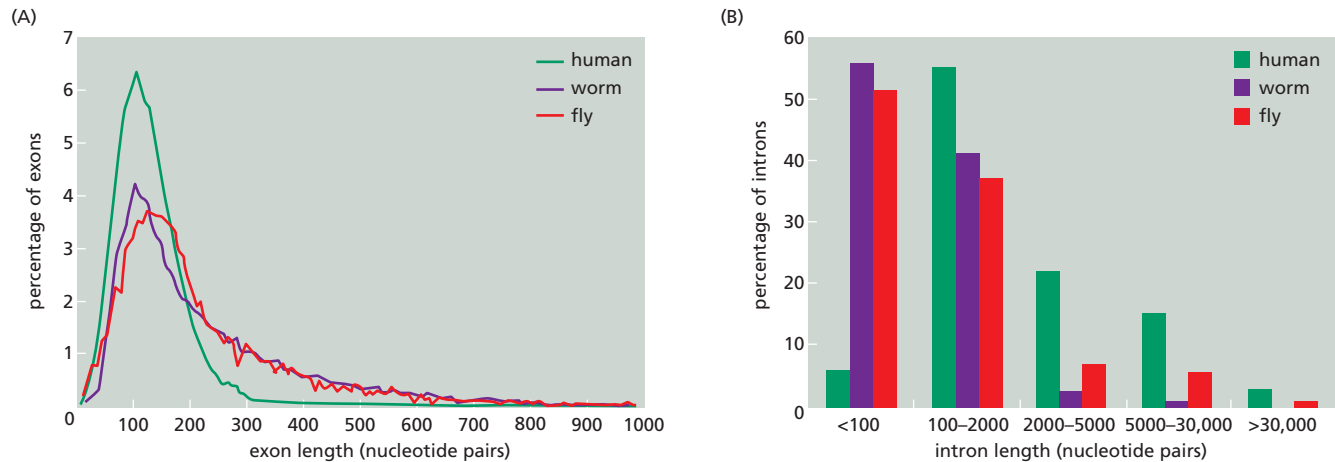


Figure 6-31 Variation in intron and exon lengths in the human, worm, and fly genomes. (A) Size distribution of exons. (B) Size distribution of introns. Note that exon length is much more uniform than intron length. (Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, published 2001 by Macmillan Magazines Ltd. Reproduced with permission of SCSNC.)

each other) requires another series of ATP-driven RNA–RNA rearrangements, thereby returning the snRNAs to their original configuration to be used again in a new reaction.

As splicing is completed, the spliceosome directs a set of proteins to bind to the mRNA near the position formerly occupied by the intron. Called the *exon junction complex (EJC)*, these proteins mark the site of a successful splicing event, and, as we shall see later in this chapter, they influence the subsequent fate of the mRNA.

Other Properties of Pre-mRNA and Its Synthesis Help to Explain the Choice of Proper Splice Sites

As shown in **Figure 6-31**, intron sequences vary enormously in size; some are more than 100,000 nucleotides long. One might therefore expect frequent splicing mistakes—including exon skipping and the mistaken use of “cryptic” splice sites (**Figure 6-32**). To minimize these problems the fidelity mechanisms built into the spliceosome are supplemented by two additional strategies that further increase the accuracy of splicing. The first is a simple consequence of splicing being coupled to transcription (**Figure 6-33**). As transcription proceeds, the phosphorylated tail of RNA polymerase carries several components that stimulate formation of the spliceosome (see **Figure 6-23**), and these components are transferred directly from the polymerase to the RNA as the RNA emerges from the polymerase. This strategy helps the cell to keep track of introns and exons; for example, the snRNPs that assemble at a 5′ splice site are initially presented only with the single 3′ splice site that emerges next from the polymerase, inasmuch as the potential sites further downstream have not yet been synthesized. The coordination of transcription with splicing is thus important for preventing inappropriate exon skipping.

Another mechanism, called *exon definition*, also helps cells choose the appropriate splice sites. Exon size tends to be much more uniform than intron size,

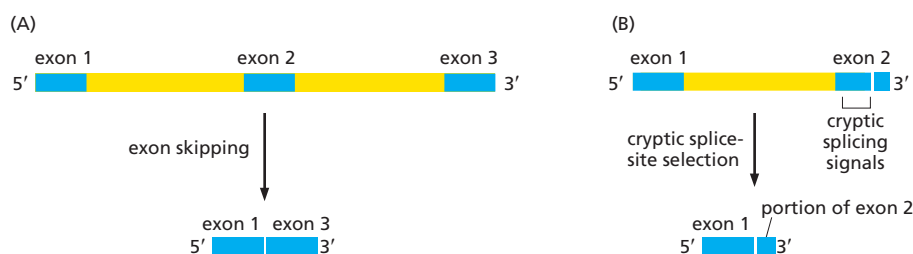


Figure 6-32 Two types of potential splicing errors. (A) Exon skipping. (B) Cryptic splice-site selection. Cryptic splicing signals are nucleotide sequences of RNA that closely resemble true splicing signals and are sometimes mistakenly used by the spliceosome.

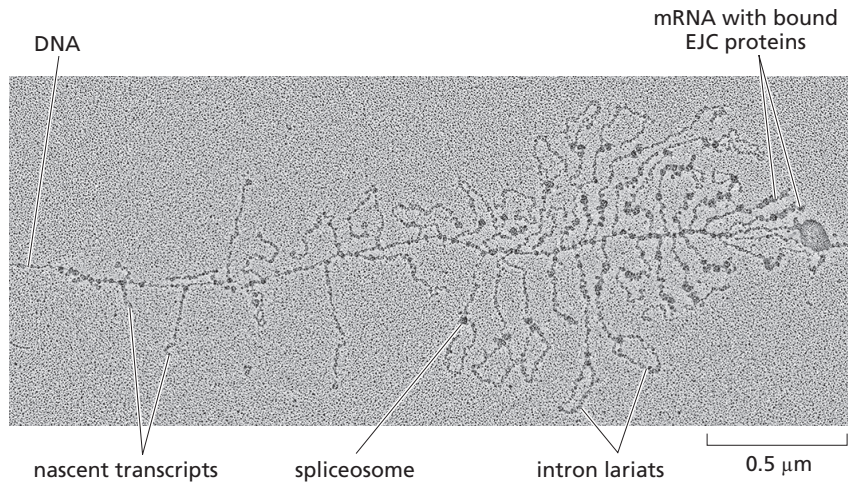


Figure 6–33 Electron micrograph of a heavily transcribed gene containing multiple introns in an early *Drosophila* embryo, illustrating the co-transcriptional splicing of its RNA transcripts. (Courtesy of Victoria Foe, Friday Harbor Laboratories, Friday Harbor WA.)

averaging about 150 nucleotide pairs across a wide variety of eukaryotic organisms (see Figure 6–31). Through exon definition, the splicing machinery seeks out the relatively homogeneously sized exon sequences rather than the intron sequences. It does this in the following way: as RNA synthesis proceeds, a group of additional components (most notably SR proteins, so-named because each contains a domain rich in serines and arginines) assemble on exon sequences and help to mark off each 3' and 5' splice site, starting at the 5' end of the RNA (Figure 6–34). These proteins, in turn, recruit the U1 snRNP, which marks the one exon boundary, and U2AF, which, along with BBP, specifies the other. By marking out the exons in this way and thereby taking advantage of the relatively uniform size of exons, the cell increases the accuracy with which it deposits the initial splicing components on the nascent RNA and thereby avoids “near miss” splice sites. To further aid the cell in marking off exons and distinguishing them from introns, some SR proteins bind tightly to specific RNA sequences, often termed *splicing enhancers*, that are preferentially found in exons. Because several different codons specify most amino acids, it is possible for a splicing enhancer to evolve without a change in the amino acid sequence coded by the exon.

Both the marking of exon and intron boundaries and the assembly of the spliceosome begin on an RNA molecule while it is still being elongated by RNA polymerase (see Figure 6–33). However, because the actual chemistry of splicing can be delayed, intron sequences are not necessarily removed from a pre-mRNA molecule in the order in which they occur along the RNA chain.

RNA Splicing Has Remarkable Plasticity

We have seen that the choice of splice sites depends on such features of the pre-mRNA transcript as the strength of the three signals on the RNA recognized by

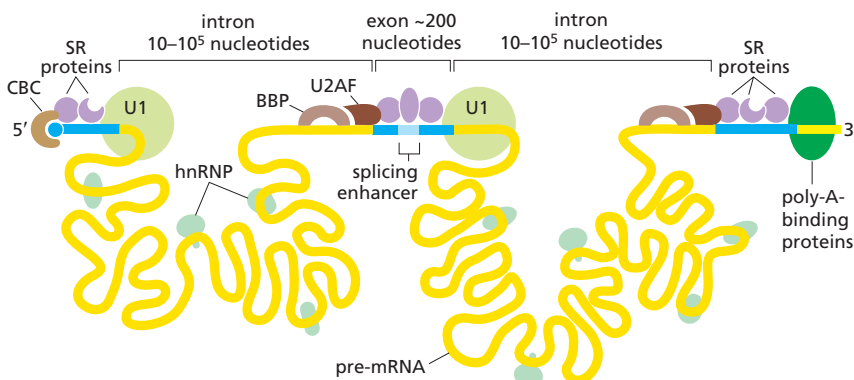


Figure 6–34 The exon definition hypothesis. SR proteins bind to each exon sequence in the pre-mRNA and thereby help to guide the snRNPs to the proper intron–exon boundaries. This demarcation of exons by the SR proteins occurs co-transcriptionally, beginning at the CBC (cap-binding complex) at the 5' end. It has also been proposed that a group of proteins known as the heterogeneous nuclear ribonucleoproteins (hnRNPs) preferentially associates with intron sequences, further helping the spliceosome distinguish introns from exons. (Adapted from R. Reed, *Curr. Opin. Cell Biol.* 12:340–345, 2000. With permission from Elsevier.)

the splicing machinery (the 5' and 3' splice junctions and the branch point), the co-transcriptional assembly of the spliceosome, and the “bookkeeping” that underlies exon definition. We do not know exactly how accurate splicing normally is because, as we see later, there are several quality-control systems that rapidly destroy mRNAs whose splicing goes awry. However, we do know that, compared with other steps in gene expression, splicing is unusually flexible.

Thus, for example, a mutation in a nucleotide sequence critical for splicing of a particular intron does not necessarily prevent splicing of that intron altogether. Instead, the mutation typically creates a new pattern of splicing (**Figure 6-35**). Most common, an exon is simply skipped (**Figure 6-35B**). In other cases, the mutation causes a cryptic splice junction to become the default choice (**Figure 6-35C**). Apparently, the splicing machinery has evolved to pick out the best possible pattern of splice junctions, and if the optimal one is damaged by mutation, it will seek out the next best pattern, and so on. This inherent plasticity in the process of RNA splicing suggests that changes in splicing patterns caused by random mutations have been important in the evolution of genes and organisms. It also means that mutations that affect splicing can be severely detrimental to the organism: in addition to the β -thalassemia example presented in **Figure 6-35**, aberrant splicing plays important roles in the development of cystic fibrosis, frontotemporal dementia, Parkinson's disease, retinitis pigmentosa, spinal muscular atrophy, myotonic dystrophy, premature aging, and cancer. It has been estimated that of the many point mutations that cause inherited human diseases, 10% produce aberrant splicing of the gene containing the mutation.

The plasticity of RNA splicing also means that the cell can easily regulate the pattern of RNA splicing. Earlier in this section we saw that alternative splicing can give rise to different proteins from the same gene and that this is a common strategy to enhance the coding potential of genomes. Some examples of alternative splicing are constitutive; that is, the alternatively spliced mRNAs are produced continually by cells of an organism. However, in many cases, the cell regulates the splicing patterns so that different forms of the protein are produced at different times and in different tissues (see **Figure 6-27**). In Chapter 7, we return to this issue to discuss some specific examples of regulated RNA splicing.

Spliceosome-catalyzed RNA Splicing Evolved from RNA Self-splicing Mechanisms

When the spliceosome was first discovered, it puzzled molecular biologists. Why do RNA molecules instead of proteins perform important roles in splice-site recognition and in the chemistry of splicing? Why is a lariat intermediate used rather than the apparently simpler alternative of bringing the 5' and 3' splice sites together in a single step, followed by their direct cleavage and rejoining? The answers to these questions reflect the way in which the spliceosome has evolved.

As discussed in the final section of this chapter, it is likely that early cells used RNA molecules rather than proteins as their major catalysts and that they stored their genetic information in RNA rather than in DNA sequences. RNA-catalyzed splicing reactions presumably had critical roles in these early cells. As evidence, some *self-splicing RNA* introns (that is, intron sequences in RNA whose splicing out can occur in the absence of proteins or any other RNA molecules) remain today, for example, in the nuclear rRNA genes of the ciliate *Tetrahymena*, in a few bacteriophage T4 genes, and in some mitochondrial and chloroplast genes. In these cases, the RNA molecule folds into a specific three-dimensional structure that brings the intron-exon junctions together and directly catalyzes the two transesterification reactions. A self-splicing intron sequence can be identified in a test tube by incubating a pure RNA molecule that contains the intron sequence and observing the splicing reaction. Because the basic chemistry as well as the structure of the active site of some self-splicing RNAs is so similar to those of the pre-mRNA spliceosome, the much more involved

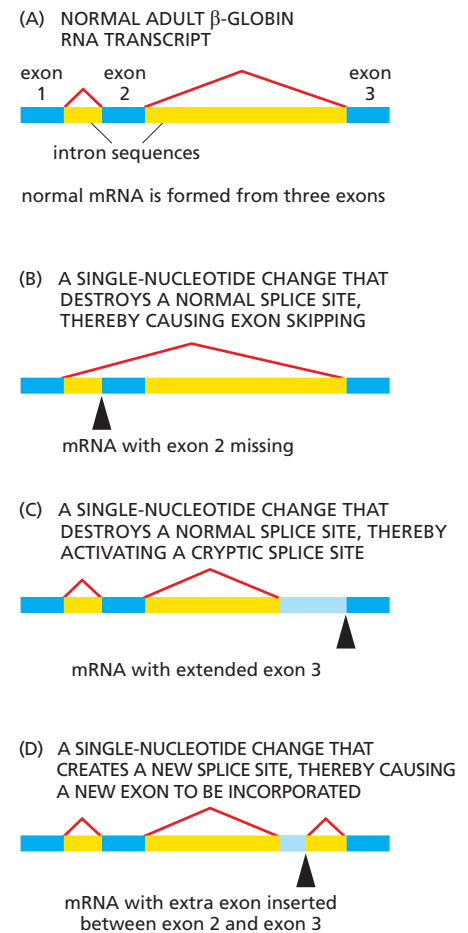
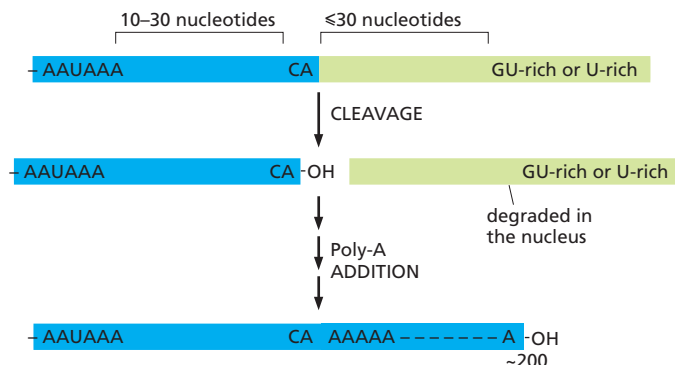


Figure 6-35 Abnormal processing of the β -globin primary RNA transcript in humans with the disease β -thalassemia.

In the examples shown, the disease (a severe anemia due to aberrant hemoglobin synthesis) is caused by splice-site mutations found in the genomes of affected individuals. The dark blue boxes represent the three normal exon sequences; the red lines connect the 5' and 3' splice sites that are used. The light blue boxes depict new nucleotide sequences included in the final mRNA molecule as a result of the mutation denoted by the black arrowhead. Note that when a mutation leaves a normal splice site without a partner, an exon is skipped as in panel B or one or more abnormal cryptic splice sites nearby is used as the partner site as in panel C.



process of pre-mRNA splicing described earlier very likely evolved from a simpler, ancestral form of RNA self-splicing.

RNA-processing Enzymes Generate the 3' End of Eukaryotic mRNAs

We have seen that the 5' end of the pre-mRNA produced by RNA polymerase II is capped almost as soon as it emerges from the RNA polymerase. Then, as the polymerase continues its movement along a gene, spliceosomes assemble on the RNA and delineate the intron and exon boundaries. The long C-terminal tail of the RNA polymerase coordinates these processes by transferring capping and splicing components directly to the RNA as it emerges from the enzyme. In this section, we describe how a similar mechanism ensures that the 3' end of the pre-mRNA is properly processed as RNA polymerase II reaches the end of a gene.

The position of the 3' end of each mRNA molecule is specified by signals encoded in the DNA nucleotide sequence (Figure 6-36). These signals are transcribed into RNA as the RNA polymerase II moves through them, and they are then recognized (as RNA) by a series of RNA-binding proteins and RNA-processing enzymes (Figure 6-37). Two multisubunit proteins (called CstF, cleavage stimulation factor; and CPSF, cleavage and polyadenylation specificity factor) are of special importance. Both of these proteins travel with the RNA polymerase tail and are transferred to the 3'-end processing sequence on an RNA molecule as it emerges from the RNA polymerase.

Once the two proteins bind to their recognition sequences on the emerging RNA molecule, additional proteins assemble with them to cleave the RNA (releasing it from the RNA polymerase) and complete the 3' end of the mRNA. Once the RNA is cleaved, an enzyme called *poly-A polymerase* (PAP) adds approximately 200 A nucleotides, one at a time, to the 3' end produced by the cleavage (see Figure 6-37). The precursor for these additions is ATP, and the same type of 5'-to-3' bonds are formed as in conventional RNA synthesis. But unlike other RNA polymerases, poly-A polymerase does not require a template; hence, the poly-A tail of eukaryotic mRNAs is not directly encoded in the genome. As the poly-A tail is synthesized, proteins called poly-A-binding proteins assemble onto it and, by a poorly understood mechanism, they help determine the final length of the tail.

The RNA polymerase II continues to transcribe after the 3' end of a eukaryotic pre-mRNA molecule has been cleaved, in some cases for hundreds of nucleotides. However, two factors increase the likelihood that an RNA polymerase will terminate transcription shortly after it has synthesized the RNA signal for 3'-end

Figure 6-36 The consensus nucleotide sequences in RNA that direct cleavage and polyadenylation to form the 3' end of a eukaryotic mRNA. These sequences are encoded in the genome, and specific proteins recognize them—as RNA—after they are transcribed. As shown in Figure 6-37, the hexamer AAUAAA is bound by CPSF, and the GU-rich element beyond the cleavage site is bound by CstF; the CA sequence is bound by a third protein factor required for the cleavage step. Like other consensus nucleotide sequences discussed in this chapter (see Figure 6-12), the sequences shown in the figure represent optimal sequences; in reality, a variety of related cleavage and polyadenylation signals occur in nature.

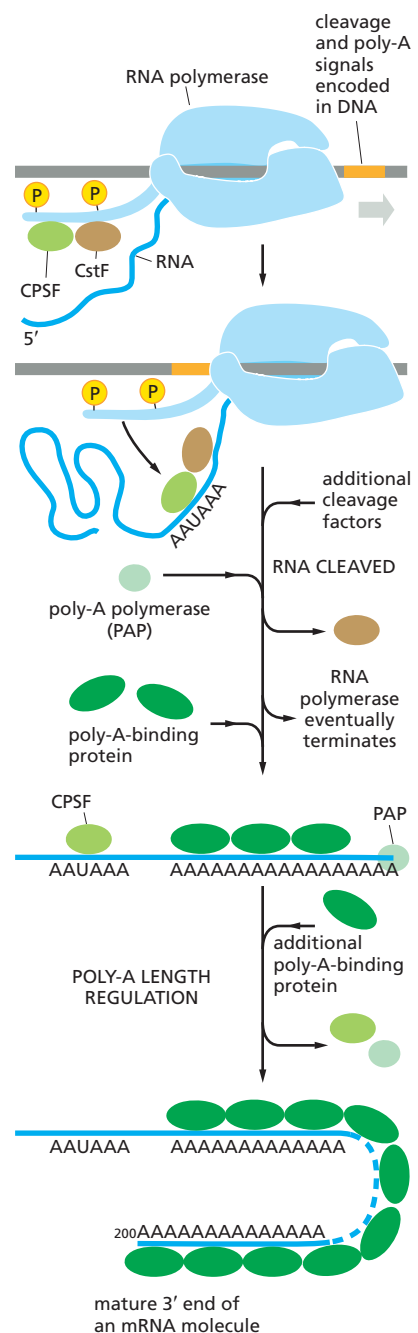


Figure 6-37 Some of the major steps in generating the 3' end of a eukaryotic mRNA. This process is much more complicated than the analogous process in bacteria, where the RNA polymerase simply stops at a termination signal and releases both the 3' end of its transcript and the DNA template (see Figure 6-11).

cleavage and polyadenylation. First, the recruitment of the many proteins needed for 3'-end processing (which occurs while some of them are still bound to the RNA polymerase tail) causes a conformational change in the polymerase, slowing it down and decreasing its processivity. Second, once 3'-end cleavage has occurred, the newly synthesized RNA emerging from the polymerase lacks a 5' cap; this unprotected RNA is rapidly degraded by a 5'→3' exonuclease, and, when it catches up to the polymerase, it causes the RNA polymerase to release its grip on the template and terminate transcription.

In the simplest case, a gene carries a single site for 3' RNA cleavage and polyadenylation. However, many genes have several such sites and can therefore produce a variety of mRNAs that differ in their 3' ends. As will be discussed in the next chapter, cells can regulate 3'-end processing to produce different proteins from the same gene in a manner analogous to alternative splicing.

Mature Eukaryotic mRNAs Are Selectively Exported from the Nucleus

Eukaryotic pre-mRNA synthesis and processing take place in an orderly fashion within the cell nucleus. But of the pre-mRNA that is synthesized, only a small fraction—the mature mRNA—is of further use to the cell. Most of the rest—excised introns, broken RNAs, aberrantly processed pre-mRNAs, and accidentally transcribed portions of the genome—is not only useless but potentially dangerous. How does the cell distinguish between the relatively rare mature mRNA molecules it wishes to keep and the overwhelming amount of useless debris?

The answer is that the RNAs are distinguished by the proteins bound to them. For example, we have seen that acquisition of cap-binding complexes, exon junction complexes, and poly-A-binding proteins marks the completion of capping, splicing, and poly-A addition, respectively. A properly completed mRNA molecule is also distinguished by the proteins it lacks. For example, the long-term presence of an snRNP protein would signify incomplete or aberrant splicing. Only when the proteins present on an mRNA molecule collectively signify that processing was successfully completed is the mRNA exported from the nucleus into the cytosol, where it can be translated into protein. Improperly processed mRNAs and other RNA debris are retained in the nucleus, where they are eventually degraded by the nuclear **RNA exosome**, a large protein complex whose interior is rich in 3'-to-5' RNA exonucleases (**Figure 6-38**). Indeed, the default fate of RNAs in the nucleus is degradation; only those bearing the proper constellation of proteins are spared.

Of all the proteins that assemble on pre-mRNA molecules as they emerge from transcribing RNA polymerases, the most abundant are the hnRNPs (heterogeneous nuclear ribonucleoproteins). Some of these proteins (there are approximately 30 different ones in humans) unwind the hairpin helices in the RNA so that splicing and other signals on the RNA can be read more easily. Others

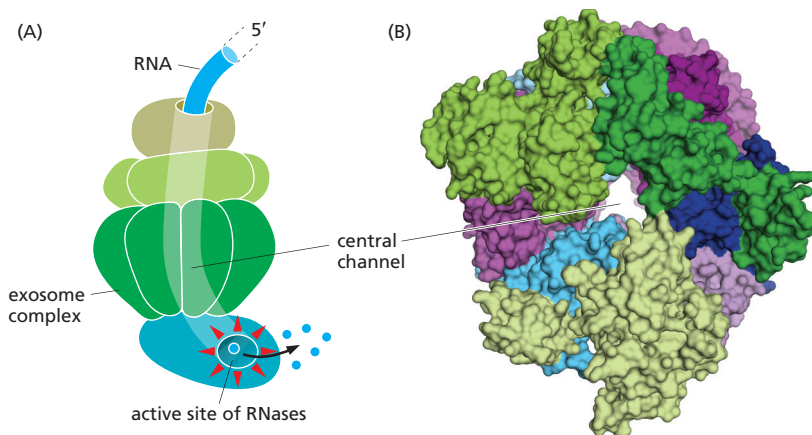


Figure 6-38 Structure of the nuclear RNA exosome. (A) RNA is fed into one end, passes through the central channel, and is degraded by RNases at the other end. (B) Structure of the central channel of the human RNA exosome viewed end-on. Nine different protein subunits (each represented by a different color) make up this large ring structure. Eukaryotic cells have both a nuclear exosome and a cytoplasmic exosome; both forms include the central channel but differ in their additional subunits. The nuclear RNA exosome degrades aberrant RNAs (including excised intron sequences and incorrectly spliced RNAs) before they are exported to the cytosol. It also processes certain types of RNA (for example, the ribosomal RNAs) to produce their final form. The cytoplasmic form of the RNA exosome is responsible for degrading mRNAs in the cytosol and is thus crucial in determining the lifetime of each mRNA molecule. (A, adapted from C. Kilchert et al., *Nat. Rev. Mol. Cell Biol.* 17:227, 2016. With permission from Springer Nature; B, PDB code: 2NN6.)

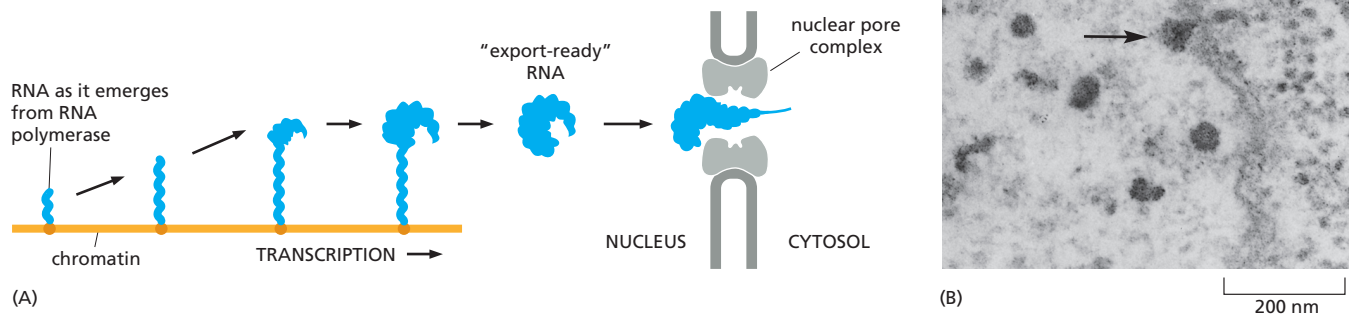


Figure 6-39 Transport of a large mRNA molecule through the nuclear pore complex. (A) The maturation of an mRNA molecule as it is synthesized by RNA polymerase and packaged by a variety of nuclear proteins. This drawing of an unusually large and abundant insect RNA, called the Balbiani Ring mRNA, is based on electron microscope micrographs such as that shown in (B). (A, adapted from B. Daneholt, *Cell* 88:585–588, 1997; B, © 1966 B.J. Stevens and H. Swift. Originally published in *J. Cell Biol.* <https://doi.org/10.1083/jcb.31.1.55>. With permission from Rockefeller University Press.)

preferentially package the RNA contained in the very long intron sequences found in complex organisms (see Figure 6-31); these may also help to distinguish the debris left over from RNA processing from fully mature mRNAs.

Successfully processed mRNAs are guided through the **nuclear pore complexes (NPCs)**—aqueous channels in the nuclear membrane that directly connect the nucleoplasm and cytosol (Figure 6-39). Small molecules (less than 40,000 daltons or about 5 nm in diameter) can diffuse freely through these channels. However, most of the macromolecules in cells, including mRNAs complexed with proteins, are far too large to pass through the channels without a special process. The cell uses energy to actively transport such macromolecules in both directions through the nuclear pore complexes.

As explained in detail in Chapter 12, macromolecules are moved through nuclear pore complexes by *nuclear transport receptors*, which, depending on the identity of the macromolecule, escort it from the nucleus to the cytoplasm or vice versa. For mRNA export to occur, a specific nuclear transport receptor must be loaded onto the mRNA, a step that, in many organisms, takes place in concert with 3' cleavage and polyadenylation.

The export of mRNA-protein complexes from the nucleus can be readily observed with the electron microscope for the unusually abundant mRNA of the insect *Balbiani Ring genes*. As these genes are transcribed, the newly formed pre-mRNA is seen to be packaged by proteins, including hnRNPs, SR proteins, and components of the spliceosome. This protein-pre-mRNA complex undergoes a series of structural transitions, probably reflecting RNA-processing events, culminating in a curved fiber. This curved fiber diffuses through the nucleoplasm, enters the nuclear pore complex (with its 5' cap proceeding first), and then undergoes additional structural transitions as it moves through the pore (see Figure 6-39). Such observations reveal that the pre-mRNA-protein and mRNA-protein complexes are dynamic structures that gain and lose specific proteins during RNA synthesis, processing, and export (Figure 6-40).

The journey of an individual mRNA molecule from the nucleus to the cytosol can also be tracked by fluorescently labeling it and observing it over time. A typical mRNA molecule that is released from its site of transcription spends several minutes randomly diffusing in the nucleus until it encounters a nuclear pore complex. During this time, RNA-processing events presumably continue, with the mRNA shedding previously bound proteins and acquiring new ones. Once it arrives at the entrance to the pore, the “export-ready” mRNA molecule hovers for several seconds, during which time the completion of RNA processing likely occurs, and it then is transported through the pore very rapidly. Some mRNA-protein complexes are

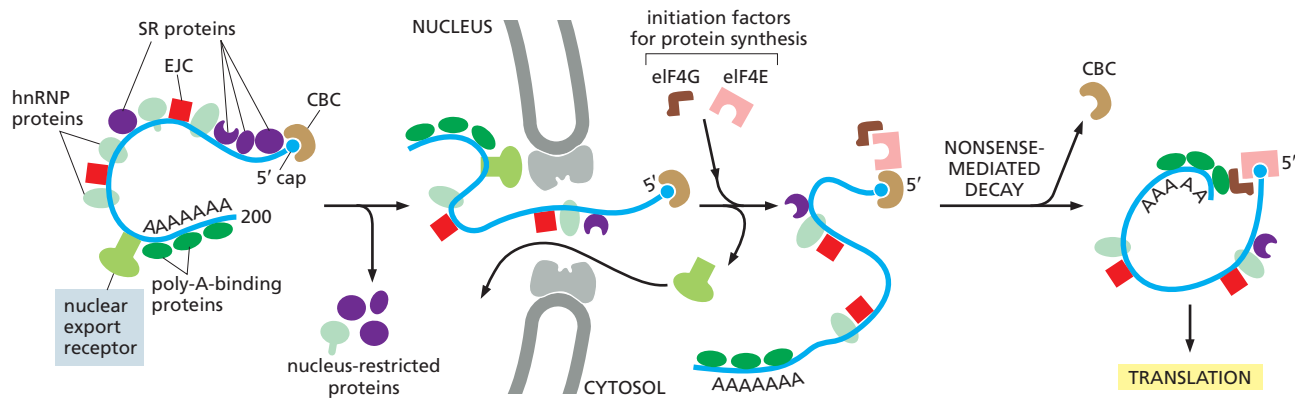


Figure 6-40 Schematic illustration of an export-ready mRNA molecule and its transport through the nuclear pore. As indicated, some proteins travel with the mRNA as it moves through the pore, whereas others remain in the nucleus. Some of the nuclear proteins that are lost are eventually replaced by cytosolic versions, such as those that bind the 5' cap. In some species (humans, for example), there are different poly-A-binding proteins in the nucleus and the cytosol; other species have only a single poly-A-binding protein. The nuclear export receptor for mRNAs is a complex of proteins that binds to an mRNA molecule once it has been correctly spliced and polyadenylated. After the mRNA has been exported to the cytosol, this export receptor dissociates from the mRNA and is re-imported into the nucleus, where it can be used again.

very large, and how they are moved through the nuclear pore complexes so rapidly (in about 10 milliseconds) remains a mystery.

Some of the proteins deposited on the mRNA while it is still in the nucleus can affect the fate of the mRNA after it is transported to the cytosol. Thus, the stability of an mRNA in the cytosol, the efficiency with which it is translated into protein, and its ultimate destination in the cell can all be determined by proteins acquired in the nucleus that remain bound to the mRNA in the cytosol.

Before discussing what next happens to the exported mRNAs, we briefly consider how the synthesis and processing of noncoding RNA molecules occur. There are many types of noncoding RNAs produced by cells (see Table 6-1, p. 327), but here we focus on the rRNAs, which are critically important for the translation of mRNAs into protein.

Noncoding RNAs Are Also Synthesized and Processed in the Nucleus

Of all the RNAs in a typical cell, only a few percent are mRNA. The bulk of RNA performs structural and catalytic functions (see Table 6-1). The most abundant RNAs in cells are the ribosomal RNAs (rRNAs), constituting approximately 80% of the RNA in rapidly dividing cells. As discussed later in this chapter, these RNAs form the core of the ribosome. Unlike bacteria—in which a single RNA polymerase synthesizes all RNAs in the cell—eukaryotes have a separate, specialized polymerase, RNA polymerase I, that is dedicated to producing rRNAs. RNA polymerase I is similar structurally to the RNA polymerase II discussed previously; however, the absence of a C-terminal tail in polymerase I helps to explain why its transcripts are neither capped nor polyadenylated.

Because multiple rounds of translation of each mRNA molecule can provide an enormous amplification in the production of protein molecules, many of the proteins that are very abundant in a cell can be synthesized from genes that are present in a single copy per haploid genome (see Figure 6-3). In contrast, the RNA components of the ribosome are final gene products, and a growing mammalian cell must synthesize approximately 10 million copies of each type of ribosomal RNA in each cell generation to construct its 10 million ribosomes. The cell can produce adequate quantities of ribosomal RNAs only because it contains multiple copies of the **rRNA genes** that code for **ribosomal RNAs (rRNAs)**. Even *E. coli* needs seven copies of its rRNA genes to meet the cell's need for ribosomes. Human cells contain about 200 rRNA gene copies per haploid genome, spread

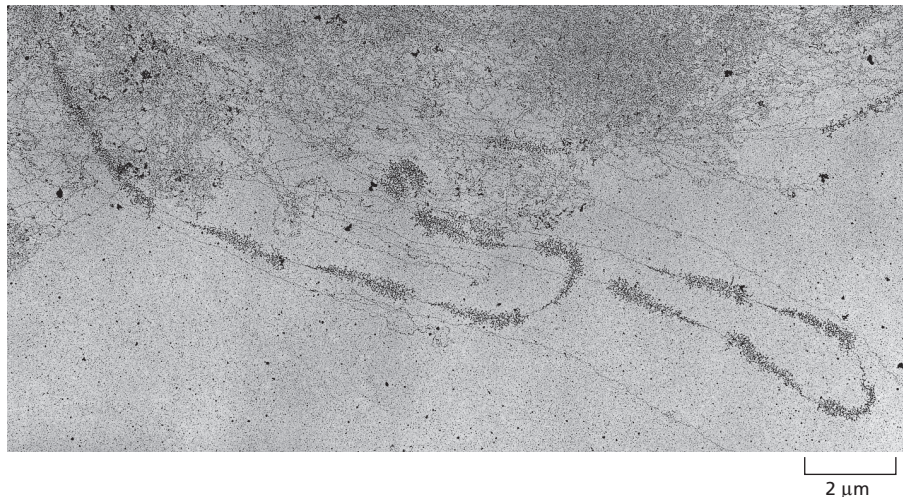


Figure 6–41 Transcription from tandemly arranged rRNA genes, as seen in the electron microscope. The pattern of alternating transcribed gene and nontranscribed spacer is readily seen. A higher-magnification view of rRNA genes is shown in Figure 6–10. (From V.E. Foe, *Cold Spring Harb. Symp. Quant. Biol.* 42:723–740, 1978. With permission from Cold Spring Harbor Laboratory Press.)

out in small clusters on five different chromosomes (see Figure 4–12), while cells of the frog *Xenopus* contain about 600 rRNA gene copies per haploid genome in a single cluster on one chromosome (Figure 6–41).

There are four types of eukaryotic rRNAs, each present in one copy per ribosome. Three of the four rRNAs (18S, 5.8S, and 28S) are made by chemically modifying and cleaving a single large precursor rRNA (Figure 6–42); the fourth (5S RNA) is synthesized from a separate cluster of genes by a different polymerase, RNA polymerase III, and it does not require chemical modification.

Extensive chemical modifications occur in the 13,000-nucleotide-long precursor rRNA before the mature 18S, 5.8S, and 28S rRNAs are cleaved out of it. These include about 100 methylations of the 2'-OH positions on nucleotide sugars and 100 isomerizations of uridine nucleotides to pseudouridine (Figure 6–43A). The functions of these modifications are not understood in detail, but they probably aid in ribosome assembly, and they may also subtly affect the operation of completed ribosomes. Each modification is made at a specific position in the precursor rRNA, specified by “guide RNAs,” which position themselves on the precursor rRNA through base-pairing and thereby bring an RNA-modifying enzyme to the appropriate position (Figure 6–43B). Other guide RNAs promote cleavage of the precursor rRNAs into the mature rRNAs, probably by causing conformational changes in the precursor rRNA that expose these sites to nucleases. All of these guide RNAs are members of a large class of RNAs called **small nucleolar RNAs (snoRNAs)**, so named because these RNAs perform their functions in a subcompartment of the nucleus called the nucleolus. Many snoRNAs are encoded in the introns of other genes, especially those encoding ribosomal proteins. They are synthesized by RNA polymerase II and processed from excised intron sequences.

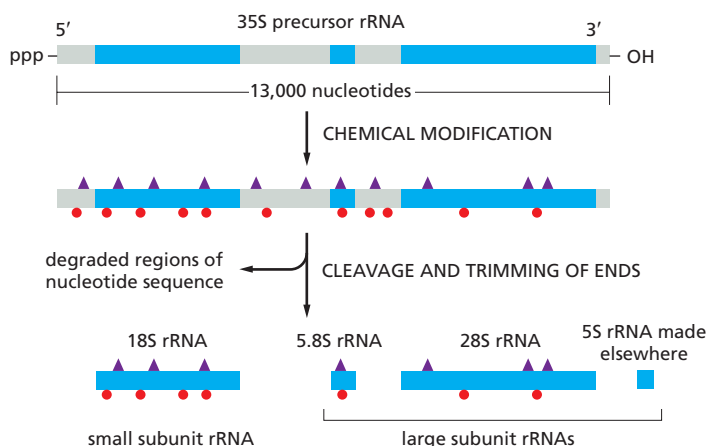


Figure 6–42 The chemical modification and nucleolytic processing of a eukaryotic precursor rRNA molecule into three separate ribosomal RNAs. Two types of chemical modifications (see Figure 6–43) are made to the precursor rRNA before it is cleaved. Nearly half of the nucleotide sequences in this precursor rRNA are discarded and degraded in the nucleus by the RNA exosome. The processing of the ribosomal RNAs begins while they are still being transcribed; the nascent transcripts also begin to be assembled with ribosomal proteins (see Figure 6–10). The rRNAs are named according to their “S” values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.

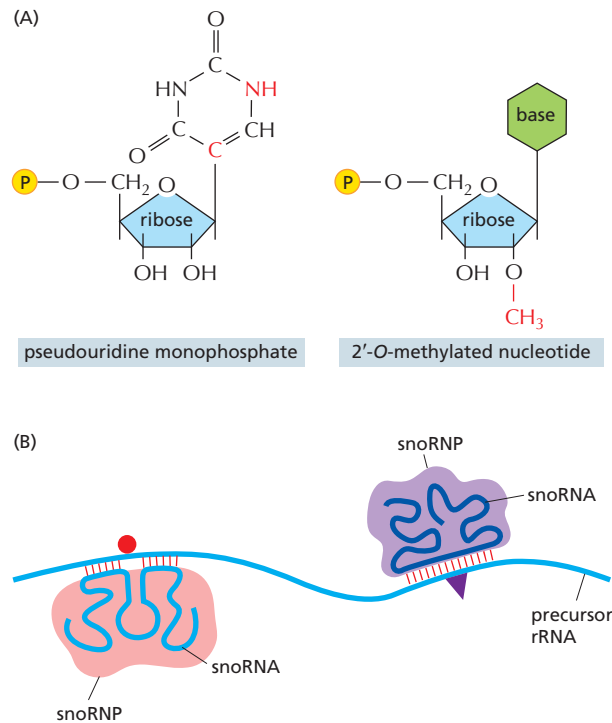


Figure 6-43 Modifications of the precursor rRNA by guide RNAs. (A) Two prominent covalent modifications made to rRNA; the differences from the initially incorporated nucleotide are indicated by *red* atoms. Pseudouridine is an isomer of uridine; the base has been “rotated” and is attached to the *red* C rather than to the *red* N of the sugar (compare to Figure 6-5B). (B) As indicated, snoRNAs determine the sites of modification by base-pairing to complementary sequences on the precursor rRNA. The snoRNAs are bound to proteins, and the complexes are called snoRNPs (small nucleolar ribonucleoproteins). The snoRNPs contain both the guide sequences and the enzymes that modify the rRNA.

The Nucleolus Is a Ribosome-producing Factory

The nucleolus is the most obvious structure seen in the nucleus of a eukaryotic cell when viewed in the light microscope. It was so closely scrutinized by early cytologists that an 1898 review could list some 700 references. We now know that the **nucleolus** is the site for the synthesis and processing of rRNAs and the assembly of ribosomes. Unlike many of the major organelles in the cell, the nucleolus is not bound by a membrane (**Figure 6-44**); instead, it is a huge *biomolecular condensate* of macromolecules, including the rRNA genes themselves, precursor

Figure 6-44 Electron micrograph of a thin section of a nucleolus in a human fibroblast, showing its three distinct zones. (A) View of entire nucleus. (B) Higher-power view of the nucleolus. It is believed that processing of the rRNAs and their assembly into the two subunits of the ribosome proceeds outward from the dense fibrillar component to the surrounding granular components (see Figure 6-46). (Courtesy of E.G. Jordan and J. McGovern.)

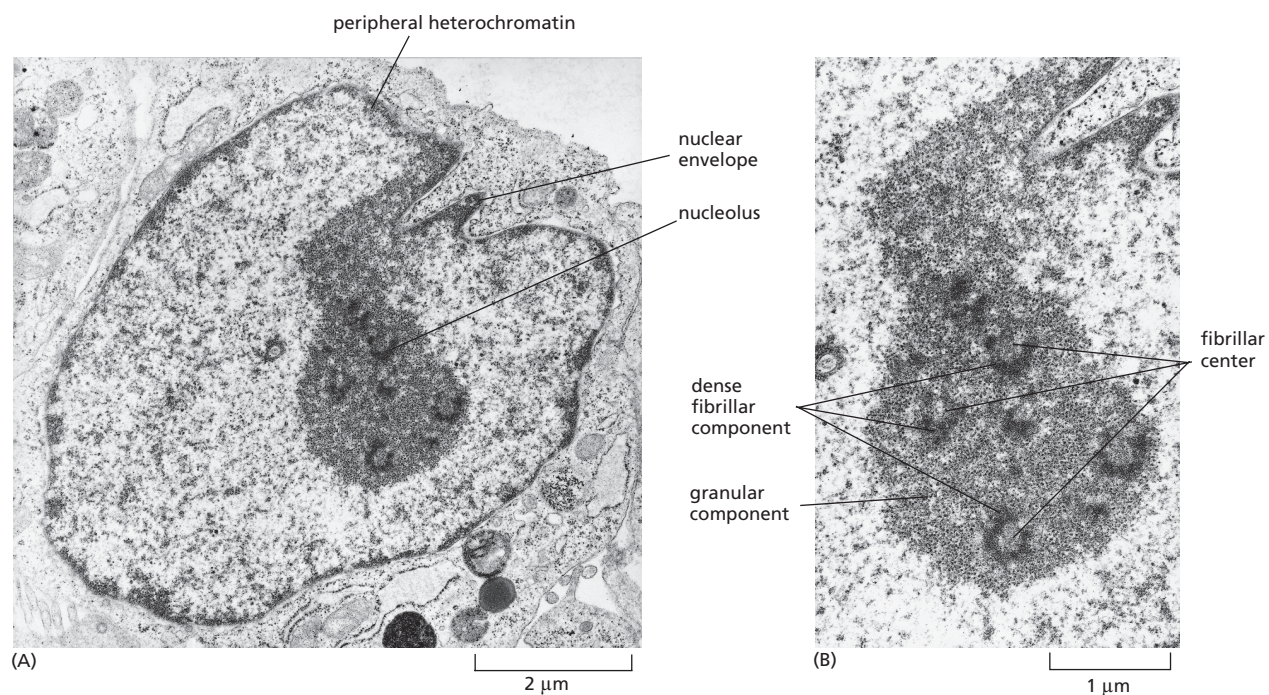
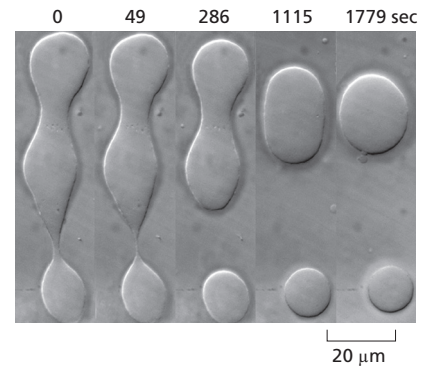


Figure 6–45 Nucleoli exhibit fluidlike behavior when observed *in vitro*. A time course showing the fate of three nucleoli from frog oocytes that have begun to fuse with one another. The lower fusion joint eventually breaks while the other enlarges, completing the fusion event. This experiment was carried out *in vitro* under mineral oil, and the nucleoli were observed using differential-interference-contrast microscopy (see Chapter 9). (Courtesy of C.P. Brangwynne et al., *Proc. Natl. Acad. Sci. USA* 108:4334–4339, 2011).



rRNAs, mature rRNAs, rRNA-processing enzymes, snoRNPs, a large set of assembly factors (including ATPases, GTPases, protein kinases, and RNA helicases), ribosomal proteins, and partly assembled ribosomes. We discuss the formation of membraneless organelles in Chapter 12; here, we note that their assembly is likely driven by the type of phase transitions discussed in Chapter 3 (see Figure 3–77). The close, but loose, association of all these components, which allows the assembly of ribosomes to occur rapidly and smoothly, endows the nucleolus with liquid-like properties (Figure 6–45).

The rRNA genes themselves have an important role in forming the nucleolus (Figure 6–46). In a diploid human cell, the rRNA genes are distributed into 10 clusters, located near the tips of five different chromosome pairs (see Figure 4–12). During interphase, these 10 chromosomes contribute DNA loops (containing the rRNA genes) to the nucleolus; in M phase, when the chromosomes condense, the nucleolus fragments and then disappears. Then, in the telophase part of mitosis, as chromosomes return to their semi-dispersed state, the tips of the 10 chromosomes re-form small nucleoli, which progressively coalesce into a single nucleolus (Figure 6–47 and Figure 6–48). As might be expected, the size of the nucleolus reflects the number of ribosomes that the cell is producing. Its size therefore varies greatly in different cells and can change in a single cell, occupying nearly 25% of the total nuclear volume in cells that are making unusually large amounts of protein.

Ribosome assembly is a complex process, requiring, in addition to the proteins and RNA molecules that compose the finished ribosome, more than 200 proteins that aid the assembly of the finished ribosome. These include *chaperones* (discussed later in this chapter), ATP-dependent RNA helicases, nucleases, and a wide variety of RNA-binding proteins. In addition, assembly requires a number of small RNA molecules, such as the two snoRNAs of Figure 6–43. In many respects, building a ribosome resembles the process by which a spliceosome is formed, as we discussed earlier in the chapter. In particular, many ATP-driven RNA structural rearrangements occur as assembly proceeds. However, a key difference is that a new spliceosome must be constructed and disassembled for each splicing event,

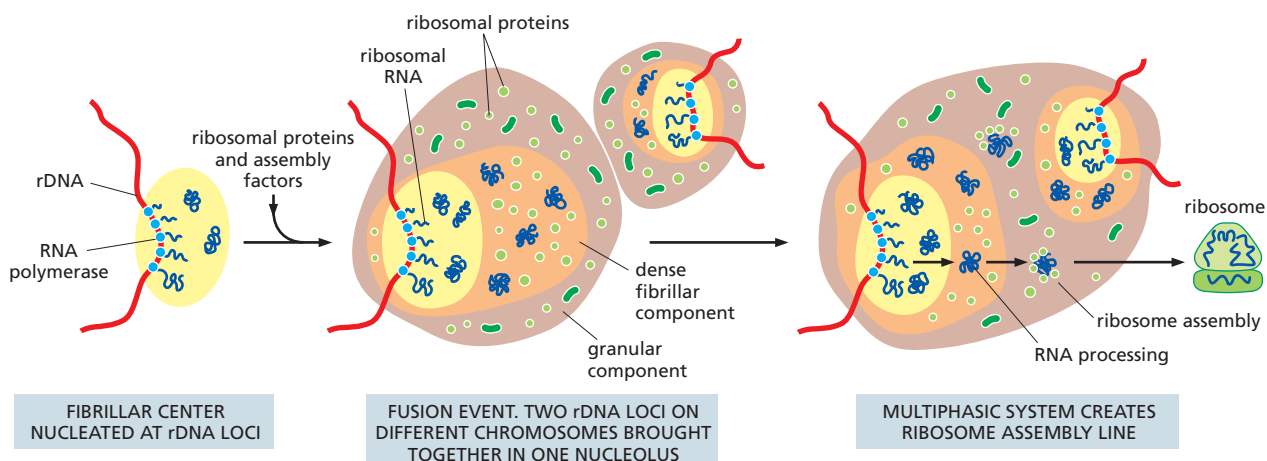


Figure 6–46 Schematic diagram of nucleolus formation after mitosis. According to this model, the nucleolus is formed from three distinct condensates, each with a different set of components. This arrangement is proposed to promote the orderly assembly of RNA–protein complexes, much like that observed on an assembly line. (Adapted from A.R. Strom and C.P. Brangwynne, *J. Cell Sci.* 132:jcs235093, 2019.)

while a ribosome, once formed, is stable and is used repeatedly to translate many mRNAs into protein. In human cells, it is estimated that each ribosome makes, on average, 3000 individual proteins in its lifetime. Ribosome assembly is understood in great detail, and only a few of the key features are summarized in [Figure 6-49](#).

In addition to its central role in ribosome biogenesis, the nucleolus is the site where other noncoding RNAs are produced and other RNA-protein complexes are assembled. For example, the U6 snRNP, a key component in pre-mRNA splicing (see [Figure 6-29](#)), is composed of one RNA molecule and seven proteins. The U6 snRNA is chemically modified by snoRNAs and assembled with its proteins in the nucleolus. Other important RNA-protein complexes, including telomerase (encountered in [Chapter 5](#)) and the signal-recognition particle (which we discuss in [Chapter 12](#)), are also assembled in the nucleolus. Finally, the tRNAs (transfer RNAs) that carry the amino acids for protein synthesis are processed there as well; like the rRNA genes, the genes encoding tRNAs are clustered in the nucleolus. Thus, the nucleolus can be thought of as a large factory at which different noncoding RNAs are transcribed, processed, and assembled with proteins to form a large variety of ribonucleoprotein complexes.

The Nucleus Contains a Variety of Subnuclear Biomolecular Condensates

Although the nucleolus is the most prominent structure in the nucleus, several other membraneless compartments have been observed and studied ([Figure 6-50](#)). These include Cajal bodies (named for the scientist who first described them in 1906) and interchromatin granule clusters (also called “speckles”). Like the nucleolus, these other compartments are highly dynamic depending on the needs of the cell, and their assembly is likely the result of the association of protein and RNA components involved in the synthesis, assembly, and storage of macromolecules involved in gene expression. Cajal bodies are sites where the snRNPs and snoRNPs undergo their final maturation steps, and where the snRNPs are recycled and their RNAs are “reset” after the rearrangements that occur during splicing (see pp. 344–345). In contrast, the interchromatin granule clusters are stockpiles of fully mature snRNPs and other RNA-processing components that are ready to be used in the production of mRNA.

Scientists have had difficulties in working out the exact function of these small compartments, in part because their appearances can change dramatically as cells traverse the cell cycle or respond to changes in their environment. Moreover, disrupting a particular type of nuclear body often has little effect on cell viability.

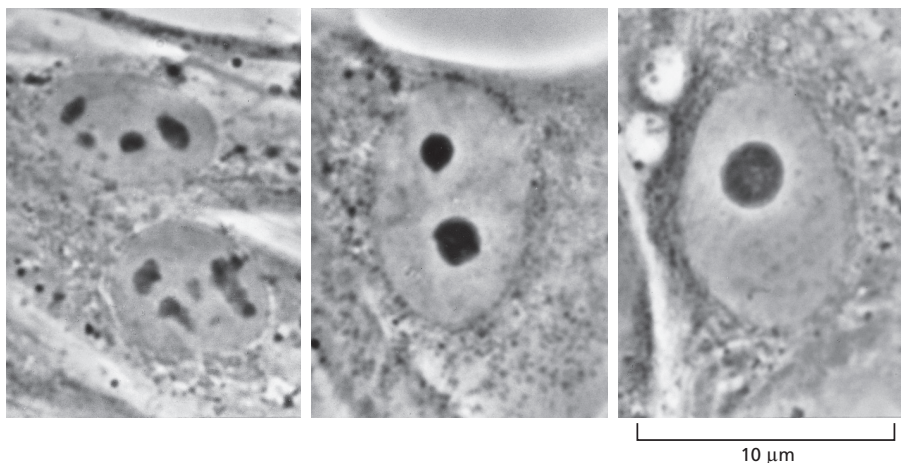


Figure 6-48 Nucleolar fusion *in vivo*. These light micrographs of human fibroblasts grown in culture show various stages of nucleolar fusion. After mitosis, each of the 10 human chromosomes that carry a cluster of rRNA genes begins to form a tiny nucleolus, but these rapidly coalesce as they grow to form the single large nucleolus typical of many interphase cells. (Courtesy of E.G. Jordan and J. McGovern.)

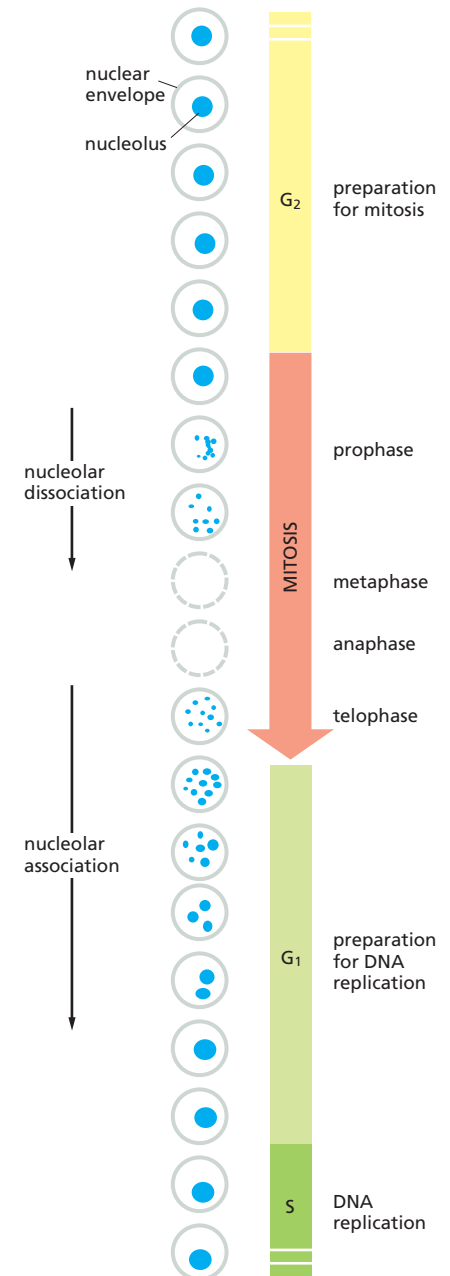


Figure 6-47 Changes in the appearance of the nucleolus in a human cell during the cell cycle. Only the cell nucleus is represented in this diagram. In most eukaryotic cells, the nuclear envelope breaks down during mitosis, as indicated by the dashed circles.

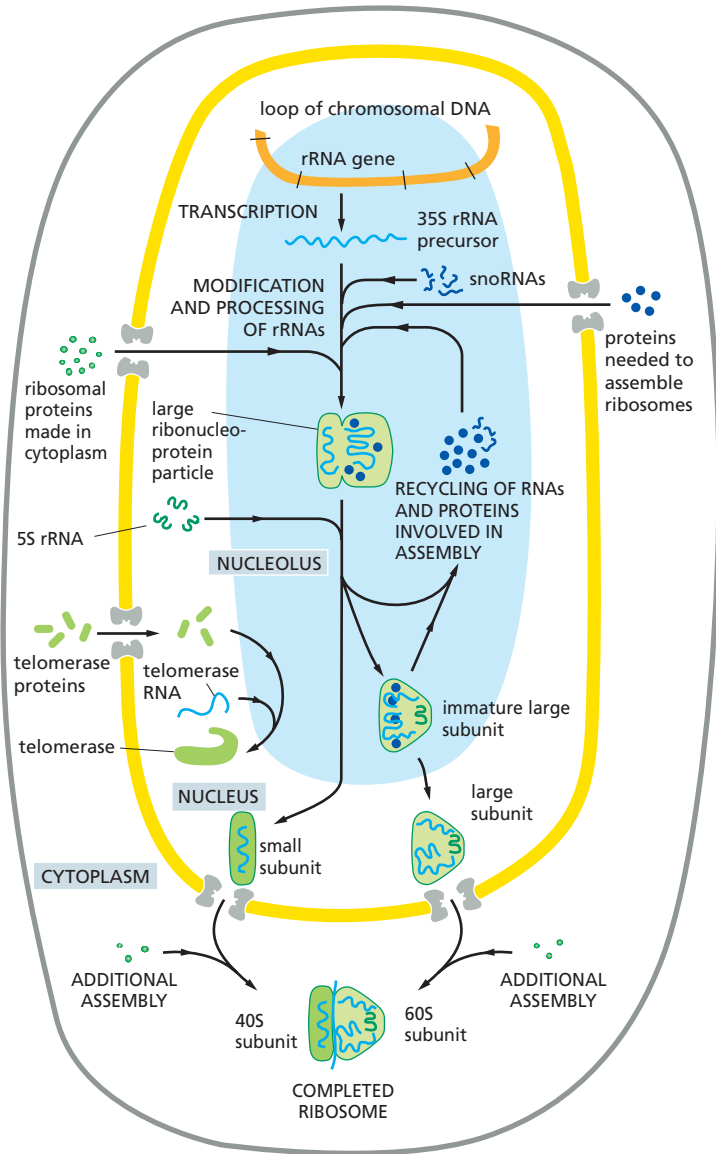


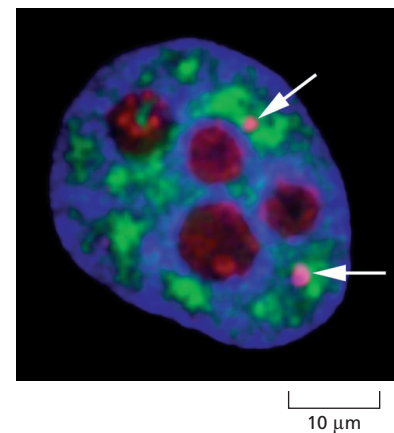
Figure 6–49 The function of the nucleolus in ribosome and other ribonucleoprotein synthesis. The 35S precursor rRNA is packaged in a large ribonucleoprotein particle containing many ribosomal proteins imported from the cytoplasm. While this particle remains at the nucleolus, selected components are added and others discarded as it is processed into immature large and small ribosomal subunits. The two ribosomal subunits attain their final functional form only after each is individually transported through the nuclear pores into the cytoplasm. Other ribonucleoprotein complexes, including telomerase shown here, are also assembled in the nucleolus. (Adapted from A.R. Strom and C.P. Brangwynne, *J. Cell Sci.* 132:jcs235093, 2019. With permission from the Company of Biologists.)

It seems that the main function of these aggregates is to bring components together at high concentration in order to speed up their assembly. For example, it is estimated that assembly of the U4/U6 snRNP (see Figure 6–29) occurs 10 times more rapidly in Cajal bodies than would be the case if the same number of components were dispersed throughout the nucleus. Consequently, Cajal bodies appear dispensable in many types of cells but are absolutely required in situations where cells must proliferate rapidly, such as in early vertebrate development. Here, protein synthesis (which depends on RNA splicing) must occur especially rapidly, and delays can be lethal.

Given the prominence of nuclear compartments in RNA processing, it might be expected that pre-mRNA splicing would occur in a particular location in the

Figure 6–50 Visualization of some prominent membraneless compartments in the nucleus.

The protein fibrillarin (red), a component of several snoRNPs, is present in both nucleoli and Cajal bodies; the latter are indicated by the arrows. The Cajal bodies (but not the nucleoli) are also highlighted by staining one of their main components, the protein coilin; the superposition of the snoRNP and coilin stains appears pink. Interchromatin granule clusters (green) have been revealed by using antibodies against a protein involved in pre-mRNA splicing. DNA is stained blue by the dye DAPI. (From J.R. Swedlow and A.I. Lamond, *Genome Biol.* 2:1–7, 2001. Micrograph courtesy of Judith Sleeman.)



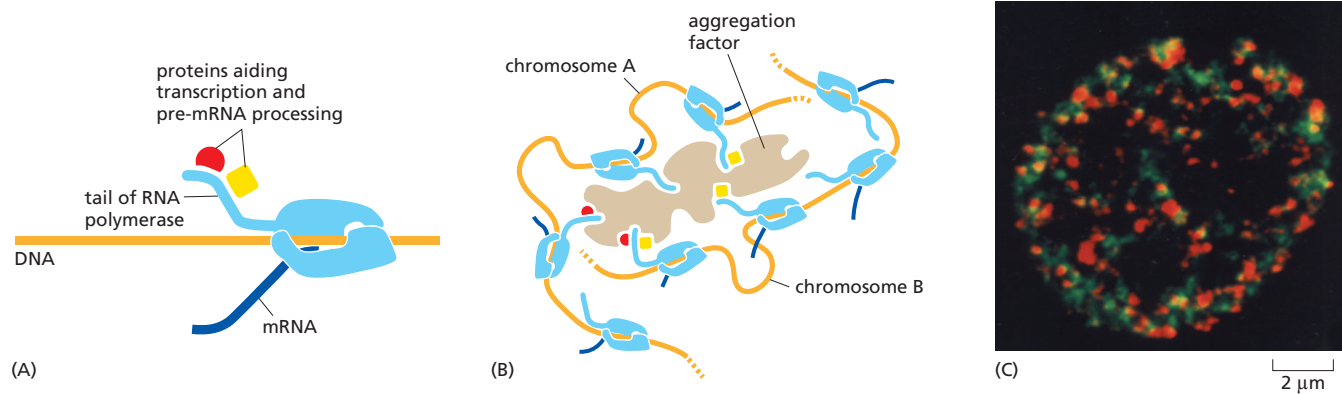


Figure 6–51 A model for an mRNA production factory. mRNA production is made more efficient in the nucleus by an aggregation of the many components needed for transcription and pre-mRNA processing, thereby producing a specialized biochemical factory. In (A), various components in the proximity of a transcribing RNA polymerase are carried on the tail (see Figure 6–23). In (B), a large number of RNA polymerase tails have been brought together to form a condensate that is highly enriched in the many components needed for the synthesis and processing of pre-mRNAs. Such a model can account for the several thousand sites of active RNA transcription and processing typically observed in the nucleus of a mammalian cell, each of which has a diameter of roughly 100 nm and is estimated to contain, on average, about 10 RNA polymerase II molecules in addition to many other proteins. (C) Here, mRNA production factories and DNA replication factories have been visualized in a mammalian cell by briefly incorporating differently modified nucleotides into each nucleic acid and detecting the RNA and DNA produced using antibodies, one (*green*) detecting the newly synthesized DNA and the other (*red*) detecting the newly synthesized RNA. (C, from D.G. Wansink et al., *J. Cell Sci.* 107:1449–1456, 1994. With permission from the Company of Biologists.)

nucleus, as it requires numerous RNA and protein components. However, as we have seen, the assembly of splicing components on pre-mRNA is co-transcriptional; thus, splicing must occur at many locations along chromosomes. Although a typical mammalian cell may be expressing on the order of 15,000 genes, transcription and RNA splicing takes place in only several thousand sites in the nucleus. These sites are highly dynamic and probably result from the association of transcription and splicing components to create small *factories*, the name given to specific condensates containing a high local concentration of selected components that create biochemical assembly lines (**Figure 6–51**). Indeed, it is thought that initial rounds of transcription and RNA processing are very slow and perhaps error-prone due to limiting concentrations of key components; only when a factory becomes fully assembled does mRNA production become rapid and accurate. Interchromatin granule clusters—which contain stockpiles of RNA-processing components—are often observed next to these sites of transcription, as though poised to replenish supplies. We can thus view the nucleus as organized into dynamic condensates of different sizes, with snRNPs, snoRNPs, and other nuclear components diffusing rapidly among them, so as to maintain high concentrations of the many components needed for each step of RNA production.

Summary

Before the synthesis of a particular protein can begin, the corresponding mRNA molecule must be produced by transcription. Bacteria contain a single type of RNA polymerase (the enzyme that carries out the transcription of DNA into RNA). An mRNA molecule is produced after this enzyme initiates transcription at a promoter, synthesizes the RNA by chain elongation, stops transcription at a terminator, and releases both the DNA template and the completed mRNA molecule. In eukaryotic cells, the process of transcription is much more complex, and there are three RNA polymerases—polymerase I, II, and III—that are related evolutionarily to one another and to the bacterial polymerase.

RNA polymerase II synthesizes eukaryotic mRNA. This enzyme requires a set of additional proteins, the general transcription factors, to initiate transcription on a DNA template. It requires still more proteins (including transcription activator

proteins, chromatin remodeling complexes, and histone-modifying enzymes) to initiate transcription on its chromatin templates inside the cell.

During the elongation phase of transcription, the nascent RNA undergoes three types of processing events: a special nucleotide is added to its 5' end (capping), intron sequences are removed from the middle of the RNA molecule (splicing), and the 3' end of the RNA is generated (cleavage and polyadenylation). Each of these processes is initiated by proteins that travel along with RNA polymerase II by binding to sites on its long, extended C-terminal tail. Splicing is unusual in that many assembly steps are required for each splicing event, and the catalytic site for the reaction is formed by RNA molecules rather than proteins. Only properly processed mRNAs are passed through nuclear pore complexes into the cytosol, where they are translated into protein.

For many genes, RNA, rather than protein, is the final product. In eukaryotes, the most abundant of these non-coding RNAs are transcribed by either RNA polymerase I or RNA polymerase III. RNA polymerase I makes the ribosomal RNAs, which are by far the most abundant RNAs in a cell. The rRNAs are chemically modified, cleaved, and assembled into the two ribosomal subunits in the nucleolus—a distinct membraneless organelle that also helps to process some smaller RNA-protein complexes in the cell. Additional biomolecular condensates in the nucleus (including Cajal bodies and interchromatin granule clusters) are sites where components involved in RNA processing are assembled, stored, and recycled. The high concentration of components in these and other biomolecular condensates ensures that the processes being catalyzed are rapid and efficient.

FROM RNA TO PROTEIN

In the preceding section, we saw that the final product of some genes is an RNA molecule itself, such as the RNAs present in the snRNPs and in ribosomes. However, most genes in a cell produce mRNA molecules that serve as intermediaries on the pathway to proteins. In this section, we examine how the cell converts the information carried in an mRNA molecule into a protein molecule. This feat of translation was a strong focus of attention for biologists in the late 1950s, when it was posed as the “coding problem”: How is the information in a linear sequence of nucleotides in RNA translated into the linear sequence of a chemically quite different set of units—the amino acids in proteins? This fascinating question stimulated great excitement. Here was a cryptogram set up by nature that, after more than 3 billion years of evolution, could finally be solved by one of the products of evolution—human beings. And indeed, not only was the code cracked step by step, but in the year 2000 the structure of the elaborate machinery by which cells read this code—the ribosome—was finally revealed in atomic detail.

An mRNA Sequence Is Decoded in Sets of Three Nucleotides

Once an mRNA has been produced by transcription and processing, the information present in its nucleotide sequence is used to synthesize a protein. Transcription is simple to understand as a means of information transfer: because DNA and RNA are chemically and structurally similar, the DNA can act as a direct template for the synthesis of RNA by complementary base-pairing. As the term “transcription” signifies, it is as if a message written out by hand is being converted, say, into a typewritten text. The language itself and the form of the message do not change, and the symbols used are closely related.

In contrast, the conversion of the information in RNA into protein represents a **translation** of the information into another language that uses quite different symbols. Moreover, because there are only 4 different nucleotides in mRNA and 20 different types of amino acids in a protein, this translation cannot be accounted for by a direct one-to-one correspondence between a nucleotide in RNA and an amino acid in protein. The nucleotide sequence of a gene, through the intermediary of mRNA, is instead translated into the amino acid sequence of a protein by rules that are known as the **genetic code**. This code was deciphered in the early 1960s.

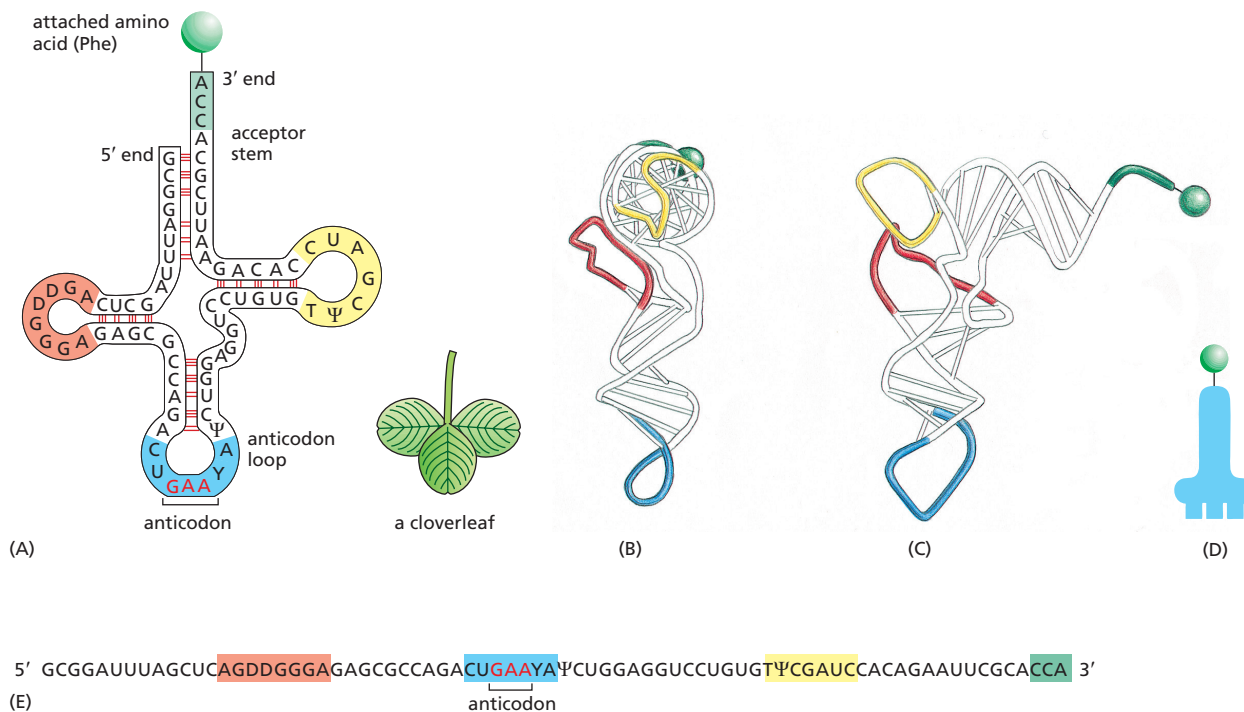


Figure 6-54 A tRNA molecule. A tRNA specific for the amino acid phenylalanine (Phe) is depicted in various ways. (A) The cloverleaf structure showing the complementary base-pairing (red lines) that creates the double-helical regions of the molecule. The anticodon is the sequence of three nucleotides that base-pairs with a codon in mRNA. The amino acid matching the codon–anticodon pair is attached at the 3' end of the tRNA. tRNAs contain some unusual bases, which are produced by chemical modification after the tRNA has been synthesized. For example, the bases denoted Ψ (pseudouridine—see Figure 6-43) and D (dihydrouridine—see Figure 6-57) are derived from uracil. (B and C) Views of the L-shaped molecule that are based on x-ray diffraction analysis. Although this diagram shows the tRNA for the amino acid phenylalanine, all other tRNAs have similar structures. (D) The tRNA icon we use in this book. (E) The linear nucleotide sequence of the molecule, color-coded to match the illustrations in A, B, and C.

amino acids have more than one tRNA, and some tRNAs are constructed so that they require accurate base-pairing only at the first two positions of the codon and can tolerate a mismatch (or *wobble*) at the third position (Figure 6-55). This wobble base-pairing explains why so many of the alternative codons for an amino acid differ only in their third nucleotide (see Figure 6-52). In bacteria, wobble base-pairings make it possible to fit the 20 amino acids to their 61 codons with as

tRNA	bacteria		eukaryotes	
	wobble codon base	possible anticodon bases	wobble codon base	possible anticodon bases
	U	A, G, or I	U	A, G, or I
	C	G or I	C	G or I
	A	U or I	A	U
	G	C or U	G	C

Figure 6-55 Wobble base-pairing between codons and anticodons. If the nucleotide listed in the first column is present at the third, or wobble, position of the codon, it can base-pair with any of the nucleotides listed in the second column. Thus, for example, when inosine (I) is present in the wobble position of the tRNA anticodon, the tRNA can recognize any one of three different codons in bacteria and either of two codons in eukaryotes. The inosine in tRNAs is formed from the deamination of adenosine (see Figure 6-57), a chemical modification that takes place after the tRNA has been synthesized. The nonstandard base pairs, including those made with inosine, are generally weaker than conventional base pairs. Codon–anticodon base-pairing is more stringent at positions 1 and 2 of the codon, where only conventional base pairs are permitted. The differences in wobble base-pairing interactions between bacteria and eukaryotes presumably result from subtle structural differences between bacterial and eukaryotic ribosomes, the molecular machines that perform protein synthesis. (Adapted from C. Guthrie and J. Abelson, in *The Molecular Biology of the Yeast *Saccharomyces*: Metabolism and Gene Expression*, pp. 487–528. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press, 1982.)

Figure 6-56 Structure of a tRNA-splicing endonuclease docked to a precursor tRNA. The endonuclease (a four-subunit enzyme) removes the tRNA intron (dark blue, bottom). A second enzyme, a multifunctional tRNA ligase (not shown), then joins the two tRNA halves together. (Courtesy of H. Li, C. Trotta, and J. Abelson. PDB code: 2A9L.)

few as 31 kinds of tRNA molecules. The exact number of different kinds of tRNAs, however, differs from one species to the next. For example, humans have nearly 500 tRNA genes that encode tRNAs with 48 different anticodons.

tRNAs Are Covalently Modified Before They Exit from the Nucleus

Like most other eukaryotic RNAs, tRNAs are covalently modified before they are allowed to exit from the nucleus. Eukaryotic tRNAs are synthesized by RNA polymerase III. Both bacterial and eukaryotic tRNAs are typically synthesized as larger precursor tRNAs, which are then trimmed to produce the mature tRNA. In addition, some tRNA precursors (from both bacteria and eukaryotes) contain introns that must be spliced out. This splicing reaction differs chemically from that of pre-mRNA splicing discussed earlier in the chapter; rather than generating a lariat intermediate, tRNA splicing uses a cut-and-paste mechanism that is catalyzed by proteins (**Figure 6-56**). Trimming and splicing both require the precursor tRNA to be correctly folded in its cloverleaf configuration. Because misfolded tRNA precursors will not be processed properly, the trimming and splicing reactions serve as quality-control steps in the generation of tRNAs. Those that do not pass the tests are degraded by the nuclear exosome (see **Figure 6-38**).

All tRNAs are modified chemically—nearly 1 in 10 nucleotides in each mature tRNA molecule is an altered version of a standard G, U, C, or A ribonucleotide. More than 50 different types of tRNA modifications are known; a few are shown in **Figure 6-57**. Some of the modified nucleotides lie within the anticodon—most notably inosine, produced by the deamination of adenosine—and affect the base-pairing of the anticodon, thereby facilitating the recognition of the appropriate mRNA codon by the tRNA molecule (see **Figure 6-55**). Other modifications affect the accuracy with which the tRNA is attached to the correct amino acid.

Specific Enzymes Couple Each Amino Acid to Its Appropriate tRNA Molecule

We have seen that, to read the genetic code in DNA, cells make a series of different tRNAs. We now consider how each tRNA molecule becomes linked to the one amino acid in 20 that is its appropriate partner. Recognition and attachment of

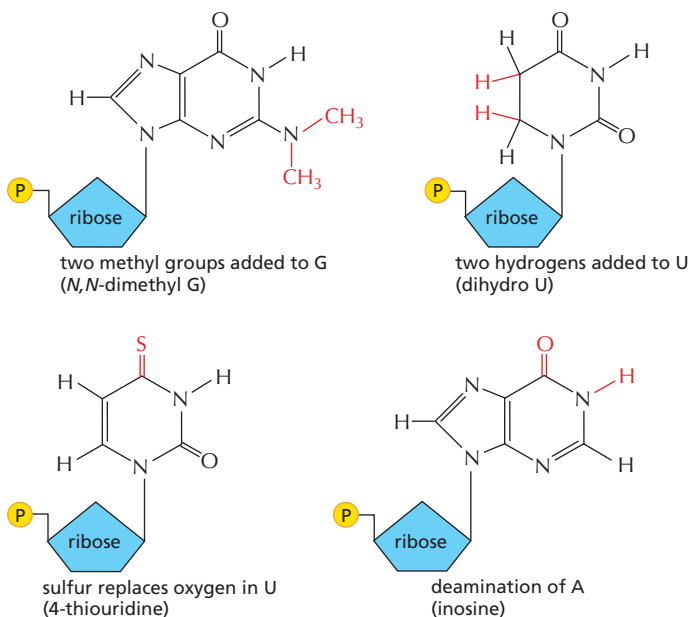
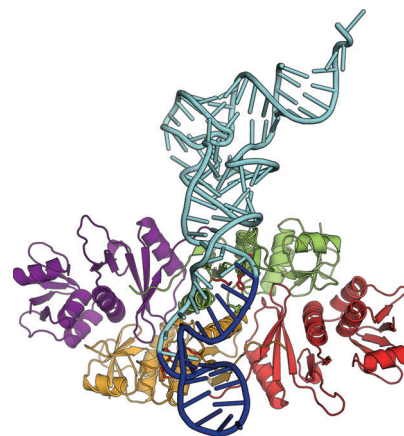


Figure 6-57 A few of the unusual nucleotides found in tRNA molecules. These nucleotides are produced by covalent modification of a normal nucleotide after it has been incorporated into an RNA chain. Two other types of modified nucleotides are shown in **Figure 6-43**. In most tRNA molecules, about 10% of the nucleotides are modified (see **Figure 6-54**). As shown in **Figure 6-55**, inosine is sometimes present at the wobble position in the tRNA anticodon.

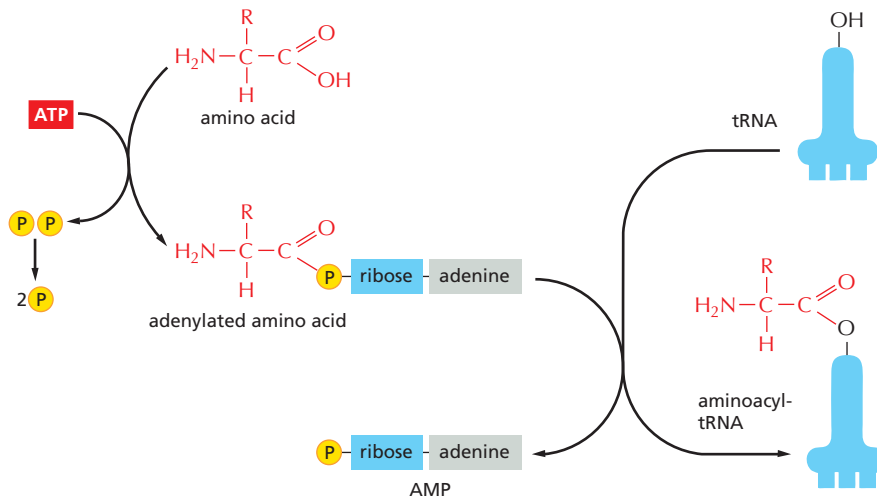


Figure 6–58 Amino acid activation by synthetase enzymes. An amino acid is attached to its corresponding tRNA in two steps by an aminoacyl-tRNA synthetase enzyme. As indicated, the energy of ATP hydrolysis is used in the reaction to produce a high-energy linkage. The amino acid is first activated through attachment of its carboxyl group directly to AMP, forming an *adenylated amino acid*; the linkage of the AMP, normally an unfavorable reaction, is driven by the hydrolysis of the ATP molecule that donates the AMP. Without leaving the synthetase enzyme, the AMP-linked carboxyl group on the amino acid is then transferred to a hydroxyl group on the sugar at the 3' end of the tRNA molecule. This transfer joins the amino acid by an activated ester linkage to the tRNA and forms the final aminoacyl-tRNA molecule. The synthetase enzyme is not shown in this diagram.

the correct amino acid depends on enzymes called **aminoacyl-tRNA synthetases**, which covalently couple each amino acid to its appropriate set of tRNA molecules (**Figure 6–58** and **Figure 6–59**). Most cells have a different synthetase enzyme for each amino acid (that is, 20 synthetases in all); one attaches glycine to all tRNAs that recognize codons for glycine, another attaches alanine to all tRNAs that recognize codons for alanine, and so on. Many bacteria, however, have fewer than 20 synthetases, and the same synthetase enzyme is responsible for coupling more

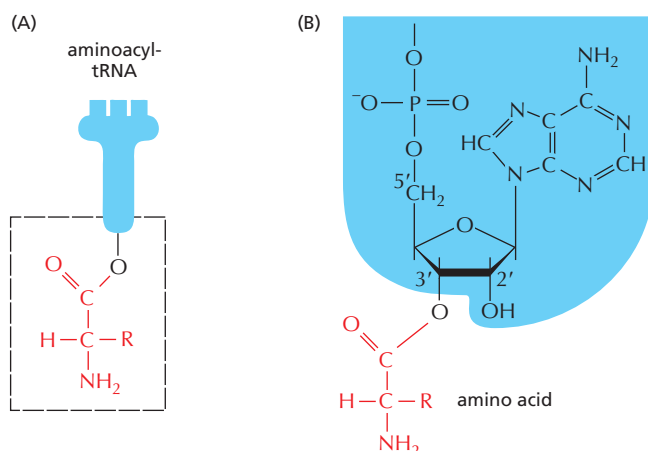


Figure 6–59 The structure of the aminoacyl-tRNA linkage. The carboxyl end of the amino acid forms an ester bond to ribose. Because the hydrolysis of this ester bond is associated with a large favorable change in free energy, an amino acid held in this way is said to be activated. (A) Schematic drawing of the structure. The amino acid is linked to the nucleotide at the 3' end of the tRNA (see Figure 6–54). (B) Actual structure corresponding to the boxed region in A. There are two major classes of synthetase enzymes: one links the amino acid directly to the 3'-OH group of the ribose, and the other links it initially to the 2'-OH group. In the latter case, a subsequent transesterification reaction shifts the amino acid to the 3' position. The "R" is a standard symbol used to represent the side chain of an amino acid.

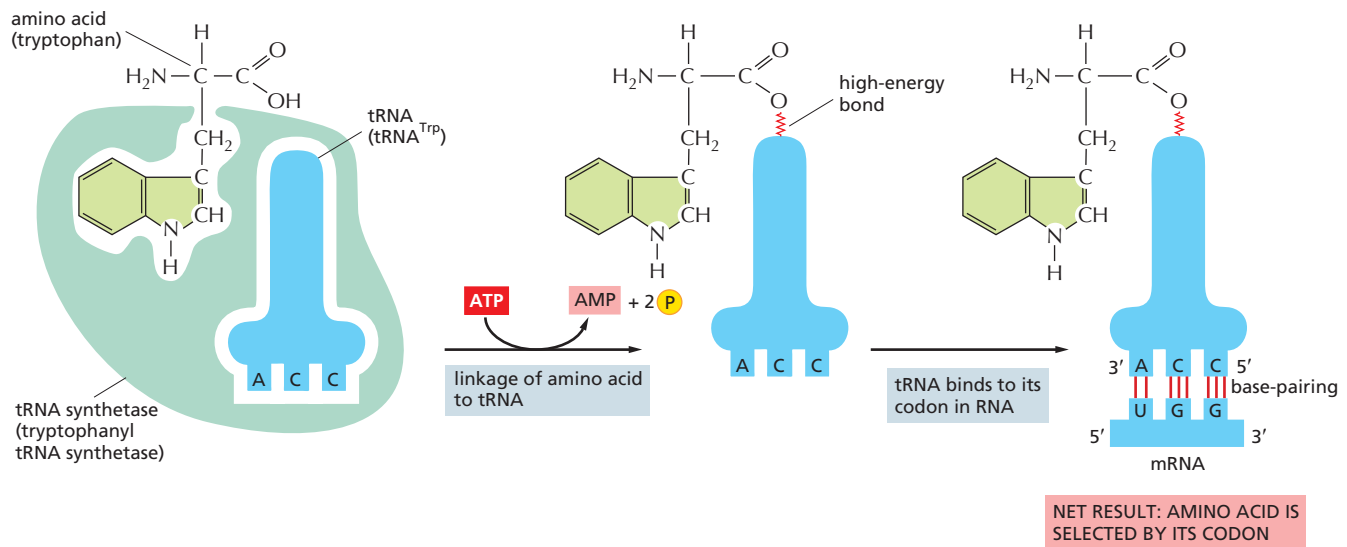


Figure 6–60 The genetic code is translated by means of two adaptors that act one after another. The first adaptor is the aminoacyl-tRNA synthetase, which couples a particular amino acid to its corresponding tRNA; the second adaptor is the tRNA molecule itself, whose *anticodon* forms base pairs with the appropriate *codon* on the mRNA. An error in either step would cause the wrong amino acid to be incorporated into a protein chain (**Movie 6.6**). In the sequence of events shown, the amino acid tryptophan (Trp) is selected by the codon UGG on the mRNA.

than one amino acid to the appropriate tRNAs. In these cases, a single synthetase places the identical amino acid on two different types of tRNAs, only one of which has an anticodon that matches the amino acid. A second enzyme then chemically modifies each “incorrectly” attached amino acid so that it now corresponds to the anticodon displayed by its covalently linked tRNA.

The synthetase-catalyzed reaction that attaches the amino acid to the 3' end of the tRNA is one of many reactions coupled to the energy-releasing hydrolysis of ATP (see pp. 70–72), and it produces a high-energy bond between the tRNA and the amino acid. The energy of this bond is used at a later stage in protein synthesis to link the amino acid covalently to the growing polypeptide chain.

The aminoacyl-tRNA synthetase enzymes and the tRNAs are equally important in the decoding process (**Figure 6–60**). This was established by an experiment in which one amino acid (cysteine) was chemically converted into a different amino acid (alanine) after it already had been attached to its specific tRNA. When such “hybrid” aminoacyl-tRNA molecules were used for protein synthesis in a cell-free system, the wrong amino acid was inserted at every point in the protein chain where that tRNA was used. Although, as we shall see, cells have several quality-control mechanisms to avoid this type of mishap, the experiment did establish that the genetic code is translated by two sets of adaptors that act sequentially. Each matches one molecular surface to another with great specificity, and it is their combined action that associates each sequence of three nucleotides in the mRNA molecule—that is, each codon—with its particular amino acid.

Editing by tRNA Synthetases Ensures Accuracy

Several mechanisms working together ensure that an aminoacyl-tRNA synthetase links the correct amino acid to each tRNA. Most synthetase enzymes select the correct amino acid by a two-step mechanism. The correct amino acid has the highest affinity for the active-site pocket of its synthetase and is therefore favored over the other 19; in particular, amino acids larger than the correct one are excluded from the active site. However, accurate discrimination between two similar amino acids, such as isoleucine and valine (which differ by only a methyl group), is very difficult to achieve in a single step. A second

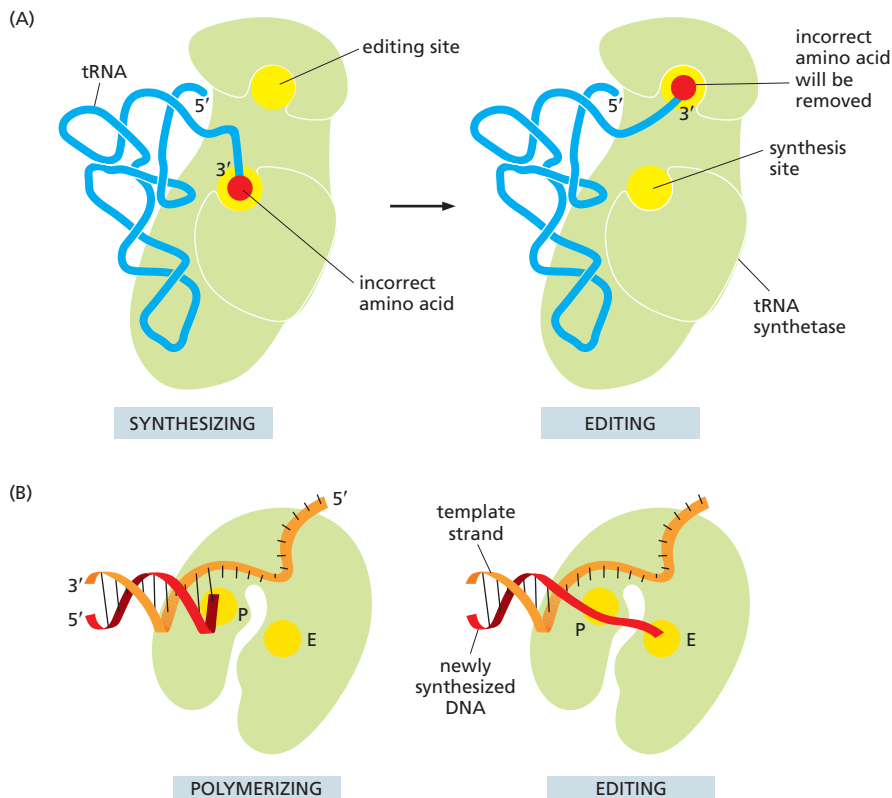


Figure 6-61 Hydrolytic editing in biology. (A) Aminoacyl-tRNA synthetases correct their own coupling errors through the hydrolytic editing of incorrectly attached amino acids. As described in the text, errors are selectively removed because the correct amino acid is rejected by the editing site on the synthetase. (B) The error-correction process performed by DNA polymerase, as described in the previous chapter. Here error removal depends on the wrong nucleotide mispairing with the DNA template (see Figure 5-9). (P, polymerization site; E, editing site.)

discrimination step occurs after the amino acid has been covalently linked to AMP (see Figure 6-58): when tRNA binds, the synthetase tries to force the adenylylated amino acid into a second editing pocket in the enzyme. The precise dimensions of this pocket exclude the correct amino acid, while allowing access by closely related amino acids. In the editing pocket, an amino acid is removed from the AMP (or from the tRNA itself if the aminoacyl-tRNA bond has already formed) by hydrolysis. This hydrolytic editing, which is analogous to the exonucleolytic proofreading by DNA polymerases, increases the overall accuracy of tRNA charging so that only about one mistake is made in 40,000 couplings (Figure 6-61).

The tRNA synthetase must also recognize the correct set of tRNAs, and extensive structural and chemical complementarity between the synthetase and the tRNA allows the synthetase to probe various features of the tRNA (Figure 6-62). Most tRNA synthetases directly recognize the matching tRNA anticodon; these synthetases contain three adjacent nucleotide-binding pockets, each of which is complementary in shape and charge to a nucleotide in the anticodon. For other synthetases, the nucleotide sequence of the amino acid-accepting arm (acceptor stem) is the key recognition determinant. In most cases, however, the synthetase “reads” the nucleotides at several different positions on the tRNA, thereby increasing the accurate linking of amino acids to their appropriate tRNAs.

Amino Acids Are Added to the C-terminal End of a Growing Polypeptide Chain

Having seen that amino acids are first coupled to specific tRNA molecules that serve as adaptors, we now turn to the mechanism that joins these amino acids together to form proteins. The fundamental reaction of protein synthesis is the formation of a peptide bond between the carboxyl group at the end of a growing polypeptide chain and a free amino group on an incoming amino acid.

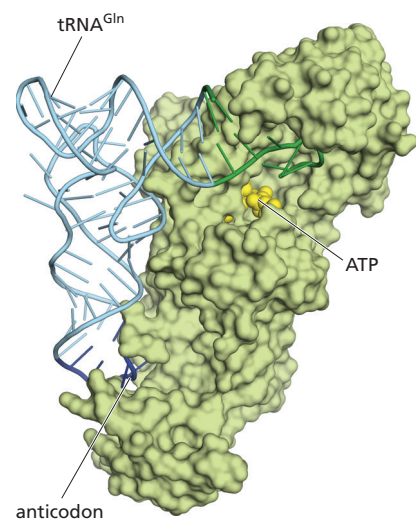


Figure 6-62 The recognition of a tRNA molecule by its aminoacyl-tRNA synthetase. For this tRNA (tRNA^{Gln}), specific nucleotides in both the anticodon (dark blue) and the amino acid-accepting arm (green) allow the correct tRNA to be recognized by the synthetase enzyme (yellow-green). The ATP molecule used in the coupling reaction is yellow. (Courtesy of Tom Steitz. PDB code: 1QRS.)

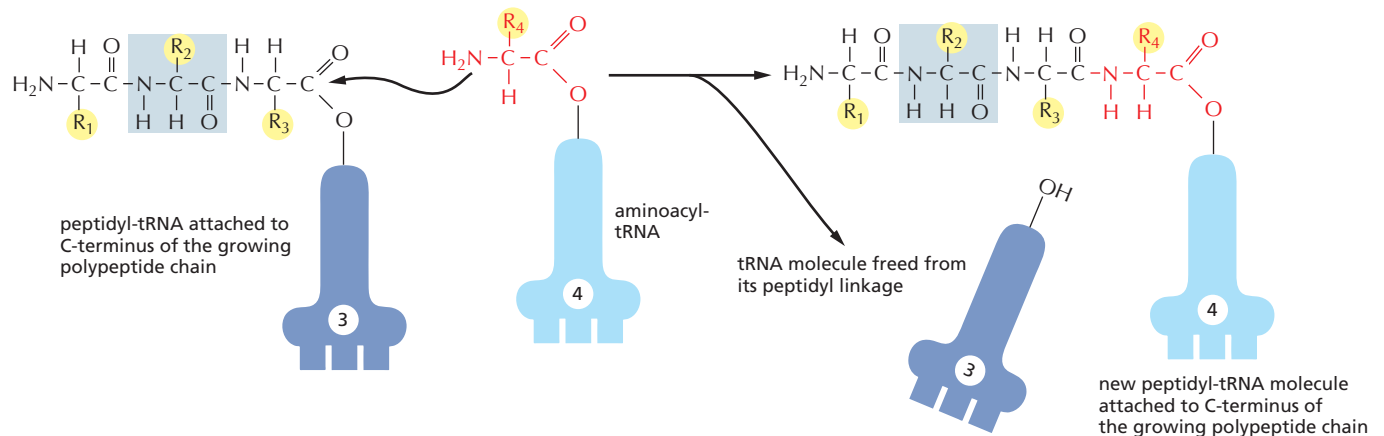


Figure 6-63 The incorporation of an amino acid into a protein. A polypeptide chain grows by the stepwise addition of amino acids to its C-terminal end. The formation of each peptide bond is energetically favorable because the growing C-terminus has been activated by the covalent attachment of a tRNA molecule. Note that the peptidyl-tRNA linkage that activates the growing end is regenerated during each addition. The amino acid side chains have been abbreviated as R_1 , R_2 , R_3 , and R_4 ; as a reference point, all of the atoms in the second amino acid in the polypeptide chain are shaded gray. The figure shows the addition of the fourth amino acid (red) to the growing chain.

Consequently, a protein is synthesized from its N-terminal end to its C-terminal end, one amino acid at a time. Throughout the entire process, the growing carboxyl end of the polypeptide chain remains activated by its covalent attachment to a tRNA molecule (forming a *peptidyl-tRNA*). Each addition disrupts this high-energy covalent linkage but immediately replaces it with an identical linkage on the most recently added amino acid (**Figure 6-63**). In this way, each amino acid added carries with it the activation energy for the addition of the next amino acid rather than the energy for its own addition—an example of the *polymer-end activation* mechanism for polymer synthesis described in Figure 2-44.

The RNA Message Is Decoded in Ribosomes

The synthesis of proteins is guided by information carried by mRNA molecules. To maintain the correct reading frame and to ensure accuracy (about 1 mistake every 10,000 amino acids), protein synthesis is performed by the **ribosome**, a complex catalytic machine made from more than 50 different proteins (the *ribosomal proteins*) and several RNA molecules, the ribosomal RNAs (rRNAs). A typical eukaryotic cell contains millions of ribosomes in its cytoplasm (**Figure 6-64**), and it takes approximately 1 minute to synthesize an average-sized

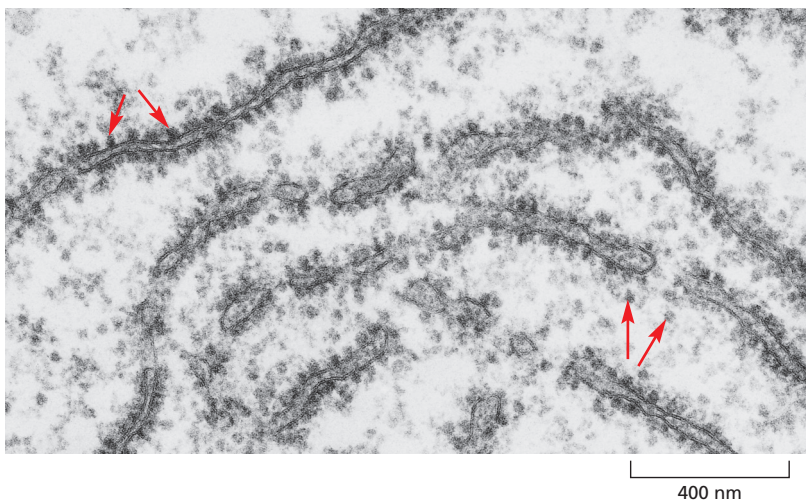
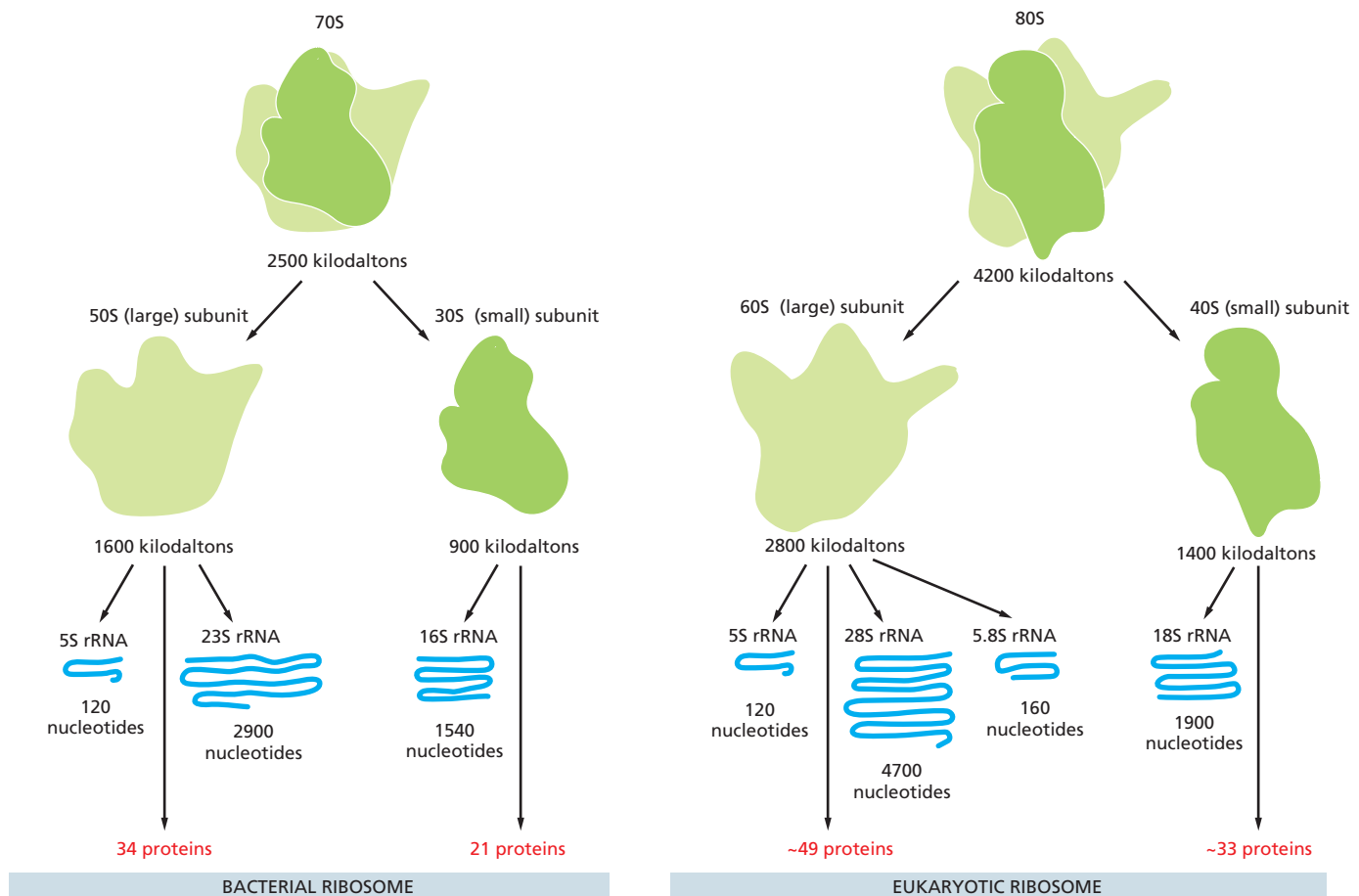


Figure 6-64 Ribosomes in the cytoplasm of a eukaryotic cell. This electron micrograph shows a thin section of a small region of cytoplasm. The ribosomes appear as black dots (red arrows). Some are free in the cytosol; others are attached to membranes of the endoplasmic reticulum. (Courtesy of Daniel S. Friend, by permission of E.L. Bear.)



protein. As we saw earlier in the chapter, the large and small ribosome subunits of eukaryotes are assembled in the nucleolus, where newly transcribed and modified rRNAs are brought into association with the ribosomal proteins that have been transported into the nucleus after their synthesis in the cytoplasm. These two ribosomal subunits are then exported to the cytoplasm, where they each undergo further assembly and join together as fully mature ribosomes to synthesize proteins (see Figure 6-49).

Eukaryotic and bacterial ribosomes have similar structures and functions, being composed of one large and one small subunit that fit together to form a complete ribosome with a mass of several million daltons (Figure 6-65). The small subunit provides the framework on which the tRNAs are accurately matched to the codons of the mRNA, while the large subunit catalyzes the formation of the peptide bonds that link the amino acids together into a polypeptide chain.

When not actively synthesizing proteins, the two subunits of the ribosome are separate. They join together on an mRNA molecule, usually near its 5' end, to initiate the synthesis of a protein. The mRNA is then pulled through the ribosome, three nucleotides at a time. As its codons enter the core of the ribosome, the mRNA nucleotide sequence is translated into an amino acid sequence using the tRNAs as adaptors to add each amino acid in the correct sequence to the growing end of the polypeptide chain. When a stop codon is encountered, the ribosome releases the finished protein, and its two subunits separate again. These subunits can then be used to start the synthesis of another protein on the same or another mRNA molecule. On average, a eukaryotic ribosome adds about four amino acids to a polypeptide chain every second; the ribosomes of bacterial cells operate even faster, at a rate of about 20 amino acids per second.

Figure 6-65 A comparison of bacterial and eukaryotic ribosomes. Despite differences in the number and size of their rRNA and protein components, both bacterial and eukaryotic ribosomes have nearly the same structure and they function similarly. Although the 18S and 28S rRNAs of the eukaryotic ribosome contain many nucleotides not present in their bacterial counterparts, these extra nucleotides are present as multiple insertions that form extra domains and leave the basic structure of the rRNA largely unchanged.

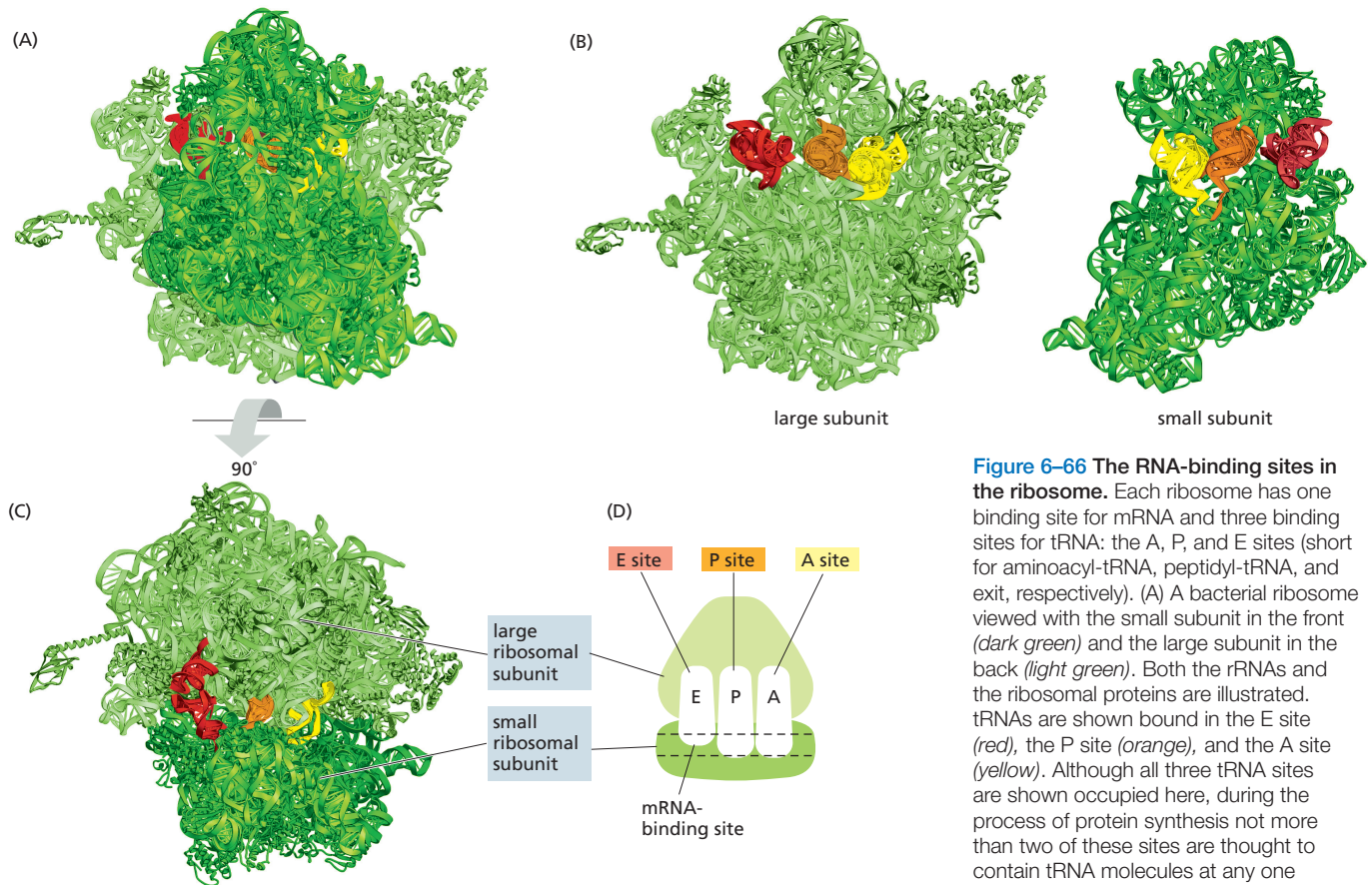


Figure 6-66 The RNA-binding sites in the ribosome. Each ribosome has one binding site for mRNA and three binding sites for tRNA: the A, P, and E sites (short for aminoacyl-tRNA, peptidyl-tRNA, and exit, respectively). (A) A bacterial ribosome viewed with the small subunit in the front (dark green) and the large subunit in the back (light green). Both the rRNAs and the ribosomal proteins are illustrated. tRNAs are shown bound in the E site (red), the P site (orange), and the A site (yellow). Although all three tRNA sites are shown occupied here, during the process of protein synthesis not more than two of these sites are thought to contain tRNA molecules at any one time (see Figure 6-68). (B) Large and small ribosomal subunits shown separately, and arranged as though the ribosome in panel A were opened like a book. (C) The entire ribosome in panel A rotated through 90° and viewed with the large subunit on top and small subunit on the bottom. (D) Schematic representation of the ribosome in the same orientation as that in panel C, which is how the ribosome will be depicted in subsequent figures. (A, B, and C, adapted from M.M. Yusupov et al., *Science* 292:883–896, 2001. With permission from AAAS.)

To choreograph the many coordinated movements required for efficient translation, a ribosome contains four binding sites for RNA molecules: one is for the mRNA and three (called the A site, the P site, and the E site) are for tRNAs (Figure 6-66). A tRNA molecule is held tightly at the A and P sites only if its anticodon forms base pairs with a complementary codon (allowing for wobble) on the mRNA molecule that is threaded through the ribosome (Figure 6-67). The A

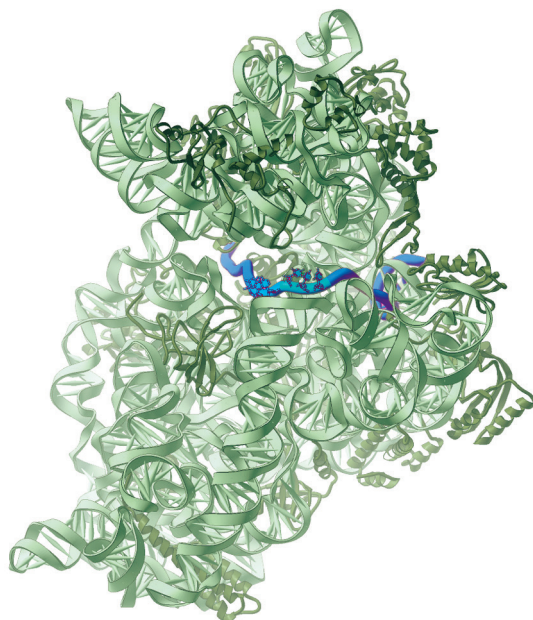


Figure 6-67 The path of mRNA (blue) through the small ribosomal subunit. The orientation is the same as that for the small subunit in the right-hand panel of Figure 6-66B, allowing the position of the three tRNA-binding sites to be compared to that of the mRNA shown here. (Courtesy of Harry F. Noller, based on data in G.Z. Yusupova et al., *Cell* 106:233–241, 2001.)

Figure 6–68 Translating an mRNA molecule. Each amino acid added to the growing end of a polypeptide chain is selected by complementary base-pairing between the anticodon on its attached tRNA molecule and the next codon on the mRNA chain. Because only one of the many types of tRNA molecules in a cell can base-pair with each codon, the codon determines the specific amino acid to be added to the growing polypeptide chain. The four-step cycle shown is repeated over and over during the synthesis of a protein. In step 1, an aminoacyl-tRNA molecule binds to a vacant A site on the ribosome. In step 2, a new peptide bond is formed. In step 3, the large subunit translocates relative to the small subunit, leaving the two tRNAs in hybrid sites: P on the large subunit and A on the small, for one tRNA; E on the large subunit and P on the small, for the other tRNA. In step 4, the small subunit translocates carrying its mRNA a distance of three nucleotides through the ribosome. This “resets” the ribosome with a fully empty A site, ready for the next aminoacyl-tRNA molecule to bind. As indicated, the mRNA is translated in the 5′-to-3′ direction, and the N-terminal end of a protein is made first, with each cycle adding one amino acid to the C-terminus of the polypeptide chain (Movie 6.7 and Movie 6.8).

and P sites are close enough together for their two tRNA molecules to be forced to form base pairs with adjacent codons on the mRNA molecule. This feature of the ribosome maintains the correct reading frame on the mRNA.

Once protein synthesis has been initiated, each new amino acid is added to the elongating chain in a cycle of reactions containing four major steps: tRNA binding (step 1), peptide bond formation (step 2), large subunit translocation (step 3), and small subunit translocation (step 4). As a result of the two translocation steps, the entire ribosome moves three nucleotides along the mRNA and is positioned to start the next cycle. **Figure 6–68** illustrates this four-step process, beginning at a point at which three amino acids have already been linked together, and there is a tRNA molecule in the P site on the ribosome, covalently joined to the C-terminal end of the short polypeptide. In step 1, a tRNA carrying the next amino acid in the chain binds to the ribosomal A site by forming base pairs with the mRNA codon positioned there, so that the P site and the A site contain adjacent bound tRNAs. In step 2, the carboxyl end of the polypeptide chain is released from the tRNA at the P site (by breakage of the high-energy bond between the tRNA and its amino acid) and joined to the free amino group of the amino acid linked to the tRNA at the A site, forming a new peptide bond. This central reaction of protein synthesis is catalyzed by a *peptidyl transferase* contained in the large ribosomal subunit. In step 3, the large subunit moves relative to the mRNA held by the small subunit, thereby shifting the acceptor stems of the two tRNAs to the E and P sites of the large subunit. In step 4, another series of conformational changes moves the small subunit and its bound mRNA exactly three nucleotides, ejecting the spent tRNA from the E site and resetting the ribosome so it is ready to receive the next aminoacyl-tRNA. Step 1 is then repeated with a new incoming aminoacyl-tRNA, and so on.

This four-step cycle is repeated each time an amino acid is added to the polypeptide chain, as the chain grows from its amino to its carboxyl end.

Elongation Factors Drive Translation Forward and Improve Its Accuracy

The basic cycle of polypeptide elongation shown in outline in **Figure 6–68** has an additional feature that makes translation especially efficient and accurate. Two *elongation factors* enter and leave the ribosome during each cycle, each hydrolyzing GTP to GDP and undergoing conformational changes in the process. These factors are called EF-Tu and EF-G in bacteria, and EF1 and EF2 in eukaryotes. **Figure 6–69** illustrates how their cycles of ribosome association, GTP hydrolysis, and ribosome dissociation contribute to the process of protein synthesis that was just outlined in **Figure 6–68**.

Under some conditions *in vitro*, ribosomes can be forced to synthesize proteins without the aid of these elongation factors and their GTP hydrolysis; but this synthesis is very slow, inefficient, and inaccurate. The coupling of GTP hydrolysis-driven changes in these elongation factors to transitions between

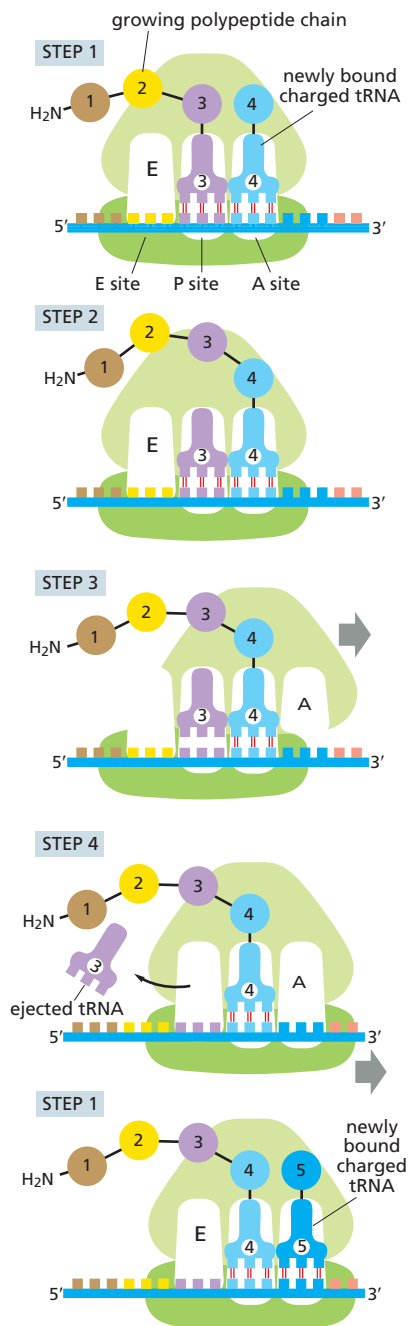


Figure 6–69 Detailed view of the translation cycle. The outline of translation presented in Figure 6–68 has been expanded to show the roles of the two elongation factors EF-Tu and EF-G, which drive translation in the forward direction. As explained in the text, EF-Tu provides two opportunities for proofreading of the codon–anticodon match. In this way, incorrectly paired tRNAs are preferentially rejected, and the accuracy of translation is improved. The binding of a molecule of EF-G to the ribosome and that elongation factor's subsequent hydrolysis of GTP lead to a rearrangement of the ribosome structure, moving the mRNA being decoded exactly three nucleotides through it (**Movie 6.9**).

different states of the ribosome speeds up protein synthesis enormously, ensuring that the many changes required occur only in the “forward” direction.

In addition to driving translation forward, EF-Tu increases its accuracy. As we discussed in Chapter 3, EF-Tu can simultaneously bind GTP and aminoacyl-tRNAs (see Figures 3–68 and 3–69), and it is in this form that the initial codon–anticodon interaction occurs in the A site of the ribosome. Because of the free-energy change associated with base-pair formation, a correct codon–anticodon match will bind more tightly than an incorrect interaction. However, this difference in affinity is relatively modest, and it cannot by itself account for the high accuracy of translation.

To increase the accuracy of aminoacyl-tRNA binding in the first step of protein synthesis, the ribosome and EF-Tu work together in the following ways. First, the 16S rRNA in the small subunit of the ribosome assesses the “correctness” of the codon–anticodon match by folding around it and probing its molecular details (**Figure 6–70**). When a correct match is found, the rRNA closes tightly around the codon–anticodon pair, causing a conformational change in the ribosome that triggers GTP hydrolysis by EF-Tu. Only when GTP is hydrolyzed does EF-Tu release its grip on the aminoacyl-tRNA and allow it to be used in protein synthesis. Because incorrect codon–anticodon matches do not readily trigger this conformational change, most of these errant tRNAs fall off the ribosome before they can be used in protein synthesis (see the first proofreading step in Figure 6–69).

After GTP is hydrolyzed and EF-Tu dissociates from the ribosome, there is a second opportunity for the ribosome to prevent an incorrect amino acid from being added to the growing chain. This arises due to a time delay before the amino acid carried by the tRNA moves into its correct position on the ribosome. Not only is this time delay shorter for correct than incorrect codon–anticodon pairs, but incorrectly matched tRNAs dissociate more rapidly than those that are correctly bound. Thus, most of the incorrectly bound tRNA molecules that remain (as well as a significant number of correctly bound molecules) will leave the ribosome without being used for protein synthesis (see the second proofreading step in Figure 6–69). The two proofreading steps, acting in series, are largely responsible for the 99.99% accuracy of the ribosome in translating RNA into protein.

Induced Fit and Kinetic Proofreading Help Biological Processes Overcome the Inherent Limitations of Complementary Base-Pairing

We have seen in this and the previous chapter that DNA replication, repair, transcription, RNA splicing, and translation all rely on complementary base-pairing—G with C, and A with T (or U). However, if only the difference in hydrogen bonding is considered, a correct versus incorrect match should differ in affinity only by a factor of 10- to 100-fold. These processes have an accuracy much higher than can be accounted for by this difference. Although the mechanisms used to “squeeze out” additional specificity from complementary base-pairing differ from one process to the next, two principles exemplified by the ribosome appear to be general.

The first is **induced fit**. We have seen that, before an amino acid is added to a growing polypeptide chain, the ribosome folds around the codon–anticodon

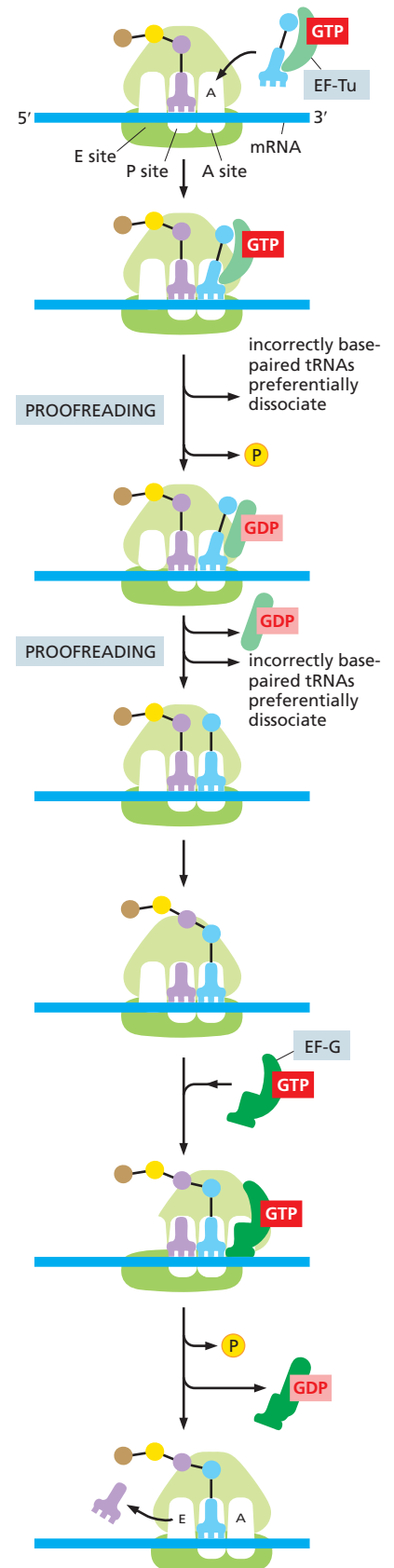
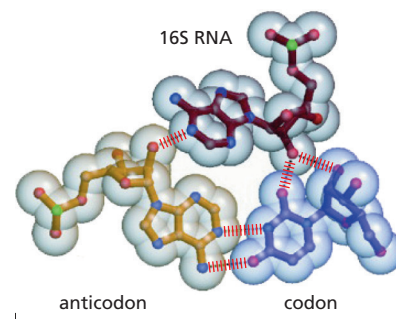


Figure 6–70 Recognition of correct codon–anticodon matches by the small-subunit rRNA of the ribosome. Shown here is the interaction between a nucleotide of the small-subunit rRNA and the first nucleotide pair of a correctly paired codon–anticodon. Similar interactions form between other nucleotides of the rRNA and the second and third positions of the codon–anticodon pair. The small-subunit rRNA can form this network of hydrogen bonds only when an anticodon is correctly matched to a codon. As explained in the text, this codon–anticodon monitoring by the small-subunit rRNA increases the accuracy of protein synthesis. (From J.M. Ogle et al., *Science* 292:897–902, 2001. With permission from AAAS.)



interaction, and only when the match is correct is this folding completed and the reaction allowed to proceed. Thus, the codon–anticodon interaction is thereby checked twice—once by the initial complementary base-pairing and a second time by the folding of the ribosome, which depends on the correctness of the match. This same principle of induced fit is seen in transcription by RNA polymerase; here, an incoming nucleoside triphosphate initially forms a base pair with the template; at this point the enzyme folds around the base pair (thereby assessing its correctness) and, in doing so, creates the active site of the enzyme. The enzyme then covalently adds the nucleotide to the growing chain. Because their geometry is “wrong,” incorrect base pairs impair this induced fit, and they are therefore likely to dissociate before being incorporated into the growing chain.

A second principle used to increase the specificity of complementary base-pairing is called **kinetic proofreading**. We have seen that after the initial codon–anticodon pairing and conformational change of the ribosome, GTP is hydrolyzed. This creates an irreversible step and starts the clock on a time delay during which the aminoacyl-tRNA moves into the proper position for catalysis. During this delay, those incorrect codon–anticodon pairs that have somehow slipped through the induced-fit scrutiny have a higher likelihood of dissociating than correct pairs. There are two reasons for this: (1) the interaction of the wrong tRNA with the codon is weaker, and (2) the delay is longer for incorrect matches than for correct matches.

In its most general form, kinetic proofreading refers to a time delay that begins with an irreversible step such as ATP or GTP hydrolysis, during which an incorrect substrate is more likely to dissociate than a correct one. In this case, kinetic proofreading thus increases the specificity of complementary base-pairing far above what is possible from simple thermodynamic associations alone. In fact, kinetic proofreading can increase the fidelity of a reaction from one error in 10^x to as little as one error in 10^{2x} , assuming that dissociation rate differences underlie the specificity of the molecular interactions involved.

The increase in specificity produced by kinetic proofreading comes at an energetic cost in the form of ATP or GTP hydrolysis. Kinetic proofreading operates in biological processes that range from DNA replication and DNA repair to RNA splicing and protein translation, helping to make life possible by greatly increasing the specificity of biochemical reactions.

Accuracy in Translation Requires a Large Expenditure of Free Energy

Translation by the ribosome is a compromise between the opposing constraints of accuracy and speed. We have seen, for example, that the accuracy of translation (one mistake per 10^4 amino acids joined) requires time delays each time a new amino acid is added to a growing polypeptide chain, producing an overall speed of translation of 20 amino acids incorporated per second in bacteria. Mutant bacteria with a specific alteration in the small ribosomal subunit have longer delays and translate mRNA into protein with an accuracy considerably higher than this; however, protein synthesis is so slow in these mutants that the bacteria are barely able to survive.

We have also seen that attaining the observed accuracy of protein synthesis requires the expenditure of a great deal of free energy; this is expected, because,

as discussed in Chapter 2, there is a price to be paid for any increase in order in the cell. In most cells, protein synthesis consumes more energy than any other biosynthetic process. At least four high-energy phosphate bonds are split to make each new peptide bond: two are consumed in charging a tRNA molecule with an amino acid (see Figure 6-58), and two more drive steps in the cycle of reactions occurring on the ribosome during protein synthesis itself (see Figure 6-69). In addition, extra energy is consumed each time that an incorrect amino acid linkage is hydrolyzed by a tRNA synthetase (see Figure 6-61) and whenever an incorrect tRNA enters the ribosome, triggers GTP hydrolysis, and is rejected (see Figure 6-69). To be effective, any proofreading mechanism must also allow an appreciable fraction of correct interactions to be removed, and this adds an even greater energy cost to proofreading.

The Ribosome Is a Ribozyme

The ribosome is a large complex composed, by mass, of two-thirds RNA and one-third protein. The determination, in 2000, of the entire three-dimensional conformation of its large and small subunits was a major triumph of modern structural biology. The ribosomal RNAs are folded into highly compact, precise three-dimensional structures that form the compact core of the ribosome and determine its overall shape (Figure 6-71). These observations confirmed earlier evidence that rRNAs—and not proteins—are responsible for the ribosome's overall structure, its ability to position tRNAs on the mRNA, and its catalytic activity in forming covalent peptide bonds.

In marked contrast to the central positions of the rRNAs, the ribosomal proteins are generally located on the surface and fill in the gaps and crevices of the

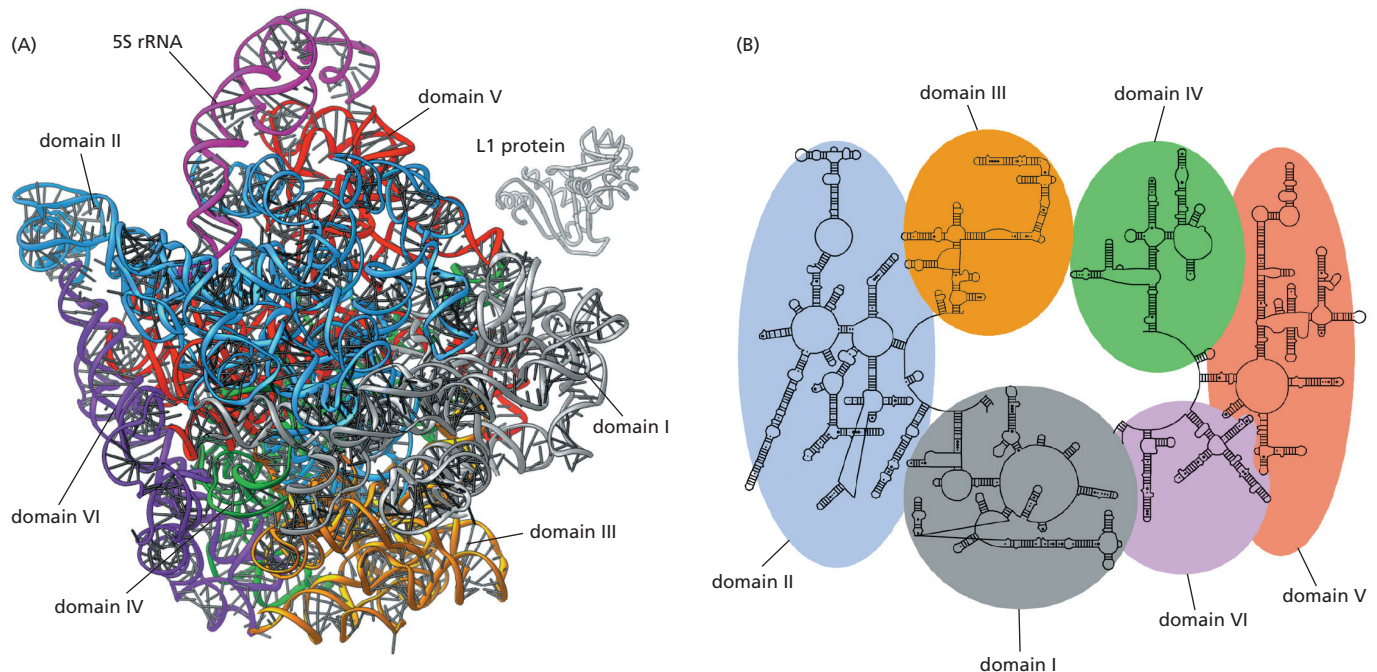


Figure 6-71 Structure of the rRNAs in the large subunit of a bacterial ribosome, as determined by x-ray crystallography.

(A) Three-dimensional conformations of the large-subunit rRNAs (5S and 23S) as they appear in the ribosome. One of the protein subunits of the ribosome (L1) is also shown as a reference point, because it forms a characteristic protrusion on the ribosome. (B) Schematic diagram of the secondary structure of the 23S rRNA, showing the extensive network of base-pairing. The structure has been divided into six “domains” whose colors correspond to those in A. The secondary-structure diagram is highly schematized to represent as much of the structure as possible in two dimensions. To do this, several discontinuities in the RNA chain have been introduced, although in reality the 23S rRNA is a single RNA molecule. For example, the base of domain III is continuous with the base of domain IV even though a gap appears in the diagram. (Adapted from N. Ban et al., *Science* 289:905–920, 2000. With permission from AAAS.)

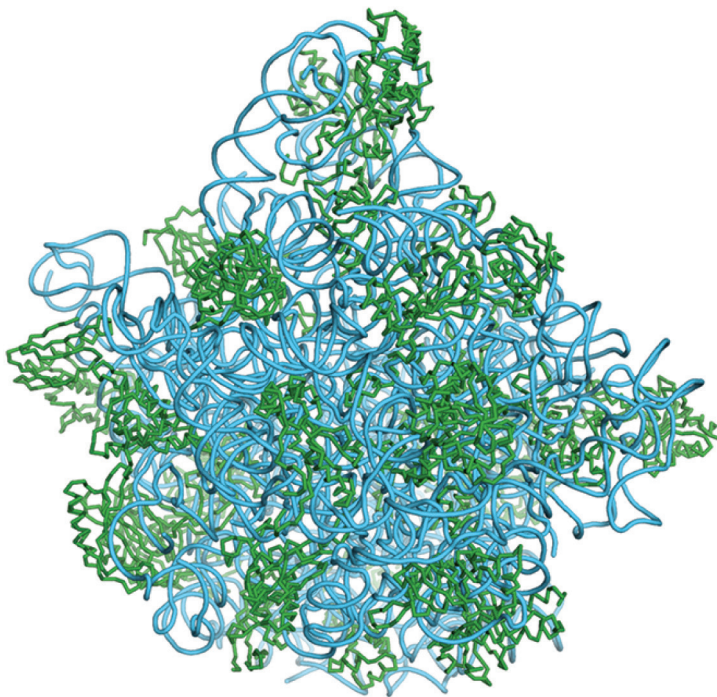


Figure 6-72 Location of the protein components of the bacterial large ribosomal subunit. The rRNAs (5S and 23S) are shown in *blue* and the proteins of the large subunit in *green*. This view is toward the backside of the ribosome relative to Figure 6-66A; the interface with the small subunit is facing into the page. (PDB code: 1FFK.)

folded RNA (**Figure 6-72**). Some of these proteins send out extended regions of polypeptide chain that penetrate short distances into holes in the RNA core (**Figure 6-73**). The main role of the ribosomal proteins seems to be to stabilize the RNA core, while permitting the changes in rRNA conformation that are necessary for this RNA to catalyze efficient protein synthesis. As discussed earlier, eukaryotic ribosome assembly takes place primarily in the nucleolus; ribosomal proteins are synthesized in the cytoplasm and brought to the nucleolus largely in unfolded form, escorted by specific *chaperones*. Only as ribosomes are assembled do their proteins assume their final, folded state, which is highly dependent on the rRNA structural framework.

Not only are the A, P, and E binding sites for tRNAs formed primarily by ribosomal RNAs, but the catalytic site for peptide bond formation is also formed by RNA, as the nearest amino acid is located more than 1.8 nm away. This discovery came as a surprise to biologists because, unlike proteins, RNA does not contain easily ionizable functional groups that can be used to catalyze sophisticated reactions such as peptide bond formation. Moreover, metal ions, which are often used by RNA molecules to catalyze chemical reactions (as is the case for RNA splicing, discussed earlier), were not observed at the active site of the ribosome. Instead, it is believed that the 23S rRNA forms a highly structured pocket that, through a network of hydrogen bonds, precisely orients the two reactants (the growing peptide chain and an aminoacyl-tRNA) and thereby greatly accelerates their covalent joining. An additional surprise came from the discovery that the tRNA in the P site contributes an important -OH group to the active site and participates directly in the catalysis. This mechanism may ensure that catalysis occurs only when the P-site tRNA is properly positioned in the ribosome.

RNA molecules that possess catalytic activity are known as **ribozymes**. We saw earlier in this chapter that the spliceosome is also a ribozyme, although its catalytic site is formed from several different RNA molecules rather than a single RNA, as is the case in the ribosome. In the final section of this chapter, we consider what the ability of RNA molecules to function as catalysts might mean for the early evolution of living cells. For now, we merely note that there is good reason to suspect that RNA rather than protein molecules served as the first catalysts for living cells. Thus the ribosome, with its RNA core, is suspected to be a relic of an earlier time in life's history—when protein synthesis evolved in cells that were run almost entirely by ribozymes.

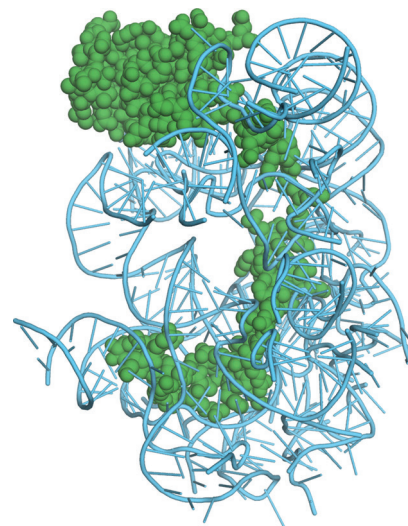


Figure 6-73 How proteins help shape ribosomal RNA. Shown here is the L15 protein in the large subunit of the bacterial ribosome. The globular domain of the protein lies on the surface of the ribosome, and an extended region penetrates deeply into the RNA core of the ribosome. The protein is shown in *green* and a portion of the ribosomal RNA core is shown in *blue*. (From D. Klein et al., *J. Mol. Biol.* 340:141–177, 2004. PDB code: 1S72.)

Nucleotide Sequences in mRNA Signal Where to Start Protein Synthesis

The initiation and termination of translation share properties of the translation elongation cycle described earlier but have additional features. The site at which protein synthesis begins on the mRNA is especially crucial, because it sets the reading frame for the whole length of the message. An error of one nucleotide either way at this stage would cause every subsequent codon in the message to be misread, resulting in a nonfunctional protein with a garbled sequence of amino acids. The initiation step is also important because, for most genes, it is the last point at which the cell can decide whether the mRNA is to be translated to produce a protein. The efficiency of this step is thus one determinant of the rate at which any particular protein will be synthesized. We shall see in Chapter 7 how regulation of this step occurs.

The translation of an mRNA usually begins with the codon AUG, and a special tRNA is required to start translation. This **initiator tRNA** always carries the amino acid methionine (in bacteria, a modified form of methionine—formylmethionine—is used), with the result that all newly made proteins have methionine as the first amino acid at their N-terminus, the end of a protein that is synthesized first. (This methionine is usually removed later by a specific protease.) The initiator tRNA is specially recognized by initiation factors because it has a nucleotide sequence distinct from that of the tRNA that normally carries methionine.

In eukaryotes, the initiator tRNA-methionine complex (Met-tRNA_i) is first loaded into the small ribosomal subunit along with additional proteins called **eukaryotic initiation factors**, or **eIFs**. Of all the aminoacyl-tRNAs in the cell, only the methionine-charged initiator tRNA is capable of tightly binding the small ribosomal subunit without the complete ribosome being present, and unlike other tRNAs, it binds directly to the P site (Figure 6-74). Next, the small ribosomal subunit binds to the 5' end of an mRNA molecule, which is recognized by virtue of its 5' cap that has previously bound two initiation factors, eIF4E and eIF4G (see Figure 6-40). The small ribosomal subunit then moves forward (5' to 3') along the mRNA, searching for the first AUG; additional initiation factors that act as ATP-powered helicases facilitate this movement. In about 90% of mRNAs, translation begins at the first AUG encountered by the small subunit. At this point, the initiation factors dissociate, allowing the large ribosomal subunit to assemble with the complex and complete the ribosome. The initiator tRNA remains at the P site, leaving the A site vacant. Protein synthesis is therefore ready to begin (see Figure 6-69).

The nucleotides immediately surrounding the start site in eukaryotic mRNAs influence the efficiency of AUG recognition during the above scanning process. If this recognition site differs substantially from the consensus recognition sequence (5'-ACCAUGG-3', known as the Kozak sequence after its discoverer), scanning ribosomal subunits will sometimes ignore the first AUG codon in the mRNA and move to the second or third AUG codon instead. Cells frequently use this phenomenon, known as “leaky scanning,” to produce two or more proteins, differing in their N-termini, from the same mRNA molecule. This mechanism allows some genes to produce the same protein with and without a signal sequence attached at its N-terminus, for example, so that the protein is directed to two different compartments in the cell.

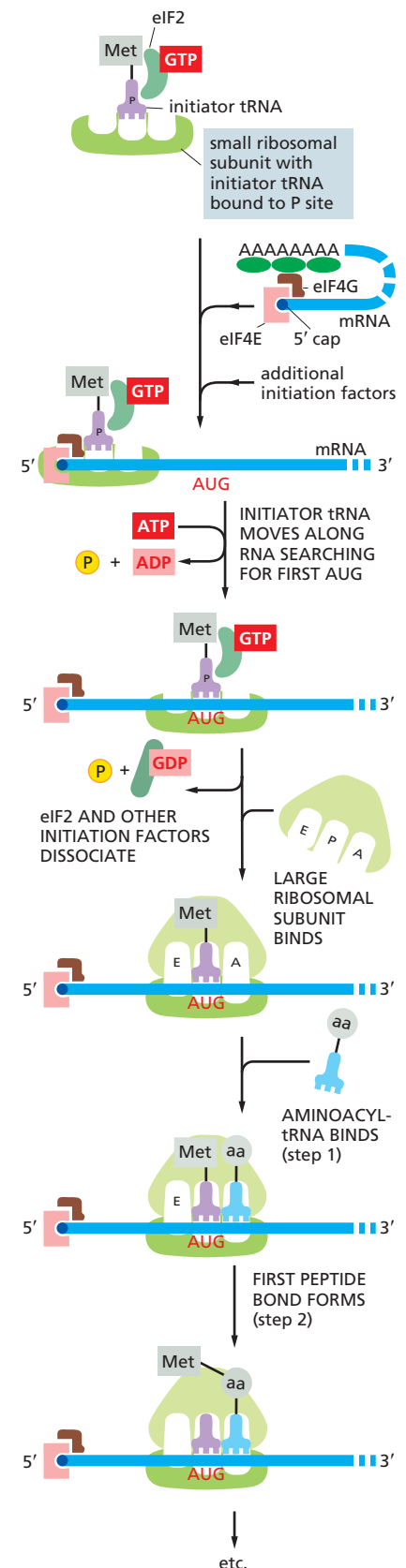


Figure 6-74 The initiation of protein synthesis in eukaryotes. Only three of the many translation initiation factors required for this process are shown. In addition to an initiating AUG codon, efficient translation initiation requires the poly-A tail of the mRNA bound by poly-A-binding proteins. In this way, the translation apparatus ascertains that both ends of the mRNA are intact before initiating protein synthesis. The communication between the 5' and 3' ends of the mRNA is mediated, at least in part, by interactions between the poly-A-binding proteins and eIF4G, as shown. This interaction appears to be transient; once translation begins, the 5' and 3' ends of the mRNA dissociate. Although only one GTP-hydrolysis event is shown in the figure, a second is known to occur just before the large and small ribosomal subunits join. In the last two steps shown in the figure, the ribosome has begun the standard elongation cycle, depicted in Figure 6-68.

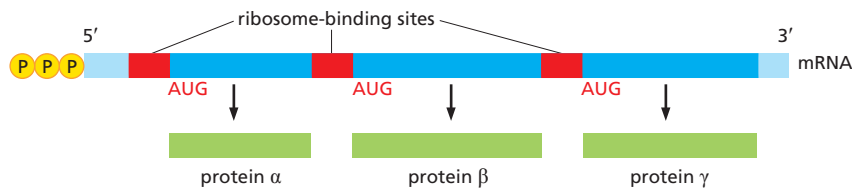


Figure 6–75 Structure of a typical bacterial mRNA molecule. Unlike eukaryotic ribosomes, which typically require a capped 5' end on the mRNA, prokaryotic ribosomes initiate translation at ribosome-binding sites (Shine–Dalgarno sequences), which can be located anywhere along an mRNA molecule. This property of their ribosomes permits bacteria to synthesize more than one type of protein from a single mRNA molecule.

Although protein synthesis in eukaryotes usually begins at an AUG codon, there are exceptions. For some proteins, translation begins at a codon that differs from AUG by a single base, particularly in the third position. In these cases, the normal initiator tRNA (carrying methionine) is used, but because the codon–anticodon match is not perfect, translation of these proteins is typically less efficient than translation of those that begin with an AUG. This deviation from the norm allows the cell to make very low amounts of some proteins compared with others. In a few rare cases, however, translation can begin with an entirely different tRNA. For example, a few proteins begin with a CUG codon, and a leucine tRNA (the perfect codon–anticodon match) begins translation with leucine as the first amino acid. In the next chapter, we will discuss how these and other deviations from the standard translation initiation process can be used to regulate protein synthesis in response to signals from the environment.

The mechanism for selecting a start codon in bacteria is fundamentally different from that in eukaryotes. Bacterial mRNAs have no 5' caps to signal the ribosome where to begin searching for the start of translation. Instead, each bacterial mRNA contains a specific ribosome-binding site (called the Shine–Dalgarno sequence, named after its discoverers) that is located a few nucleotides upstream of the AUG at which translation is to begin. This nucleotide sequence, with the consensus 5'-AGGAGGU-3', forms base pairs with the 16S rRNA of the small ribosomal subunit to position the initiating AUG codon in the ribosome. A set of translation initiation factors orchestrates this interaction, as well as the subsequent assembly of the large ribosomal subunit to complete the ribosome.

A bacterial ribosome can readily assemble directly on a start codon that lies in the interior of an mRNA molecule, so long as a ribosome-binding site precedes it by several nucleotides. As a result, bacterial mRNAs are often *polycistronic*; that is, they encode several entirely different proteins, each of which is translated from the same mRNA molecule (Figure 6–75). In contrast, a eukaryotic mRNA generally encodes only a single protein, or more accurately, a single set of closely related proteins. We will see in the next chapter that there are some exceptions to this generalization, where a eukaryotic mRNA can carry information for two or more distinct proteins.

Stop Codons Mark the End of Translation

The end of the protein-coding message is signaled by the presence of one of three *stop codons* (UAA, UAG, or UGA) (see Figure 6–52). These are not recognized by a tRNA and do not specify an amino acid, but instead signal to the ribosome to stop translation. Proteins known as *release factors* bind to any ribosome with a stop codon positioned in the A site, forcing the peptidyl transferase in the ribosome to catalyze the addition of a water molecule instead of an amino acid to the peptidyl-tRNA (Figure 6–76). This reaction frees the carboxyl end of the growing polypeptide chain from its attachment to a tRNA molecule, and as only this attachment normally holds the growing polypeptide to the ribosome, the completed protein chain is released into the cytoplasm. The ribosome then releases its bound mRNA molecule and separates into the large

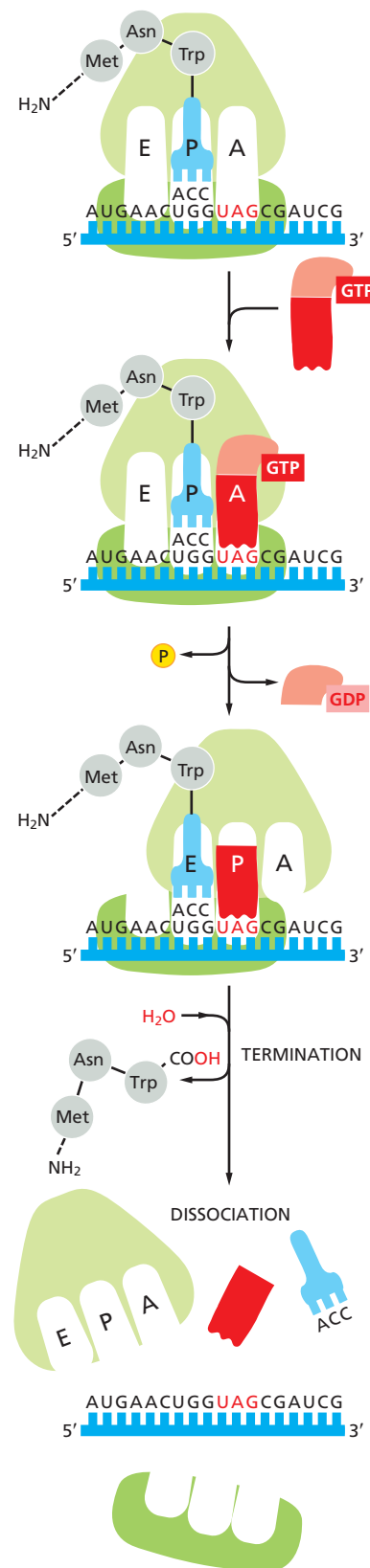


Figure 6–76 The final phase of eukaryotic protein synthesis. The binding of a two-subunit release factor to an A site bearing a stop codon terminates translation. Release factors are proteins that resemble tRNAs in their overall shape and charge distribution.

and small subunits. These subunits can then assemble on this or another mRNA molecule to begin a new round of protein synthesis.

During translation, the nascent polypeptide moves through a large, water-filled exit tunnel (approximately $10 \text{ nm} \times 1.5 \text{ nm}$) in the large subunit of the ribosome. The walls of this tunnel, made primarily of 23S rRNA, are a patchwork of tiny hydrophobic surfaces embedded in a more extensive hydrophilic surface. This structure is not complementary to any peptide and thus provides a “Teflon” coating through which a polypeptide chain can easily slide. We will see later in the chapter that, although some protein folding can occur in the exit tunnel, most of it takes place as the newly synthesized protein emerges from the ribosome.

Proteins Are Made on Polyribosomes

The synthesis of most protein molecules takes between 20 seconds and several minutes. During this short period, multiple initiations generally take place on each mRNA molecule being translated. As soon as the preceding ribosome has translated enough of the nucleotide sequence to move out of the way, the 5′ end of the mRNA is threaded into a new ribosome. For this reason, most mRNA molecules, and particularly those being translated at high rates, are found in the form of *polyribosomes* (or *polysomes*): large cytoplasmic assemblies made up of several ribosomes spaced as close as 80 nucleotides apart along a single mRNA molecule (**Figure 6-77**). The multiple initiations allow the cell to make many more protein molecules in a given time than would be possible if each protein had to be completed before the next could start. It is estimated that, in a typical human cell, about one-third of the mRNAs lack ribosomes altogether, and the remainder have 10–20 ribosomes per mRNA.

There Are Minor Variations in the Standard Genetic Code

As discussed in Chapter 1, the genetic code (shown in Figure 6-52) applies to all three major branches of life, providing important evidence for the common ancestry of all life on Earth. Although rare, there are exceptions to this code. For example, *Candida albicans*, the most prevalent fungal pathogen of humans, translates the codon CUG as serine, whereas nearly all other organisms translate it as leucine. And in some ciliates (unicellular eukaryotes that propel themselves using cilia), the three conventional stop codons specify particular amino

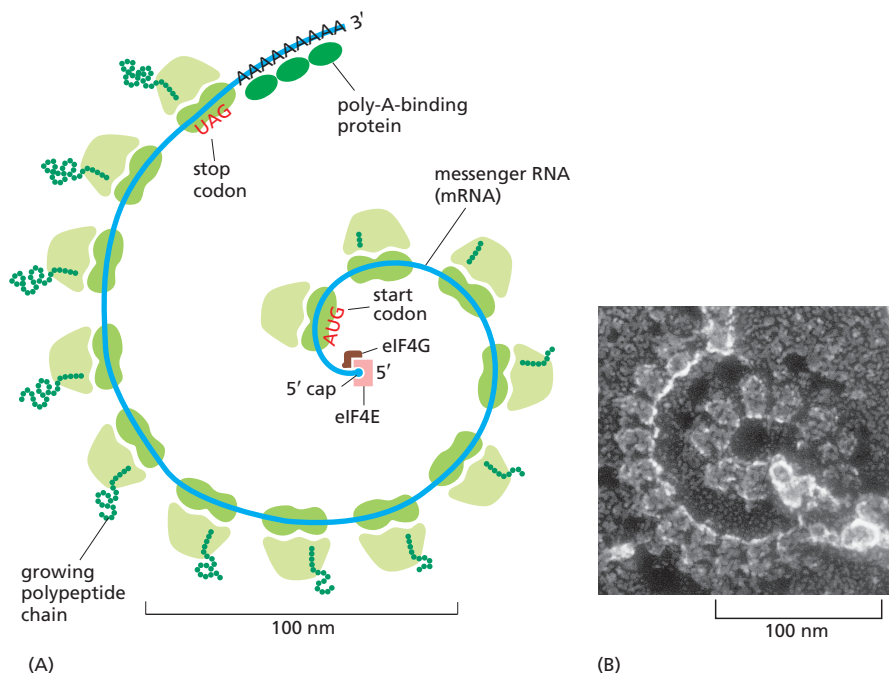


Figure 6-77 A polyribosome.

(A) Schematic drawing showing how a series of ribosomes can simultaneously translate the same eukaryotic mRNA molecule. (B) Electron micrograph of a polyribosome from a eukaryotic cell (**Movie 6.10**). (B, courtesy of John Heuser.)

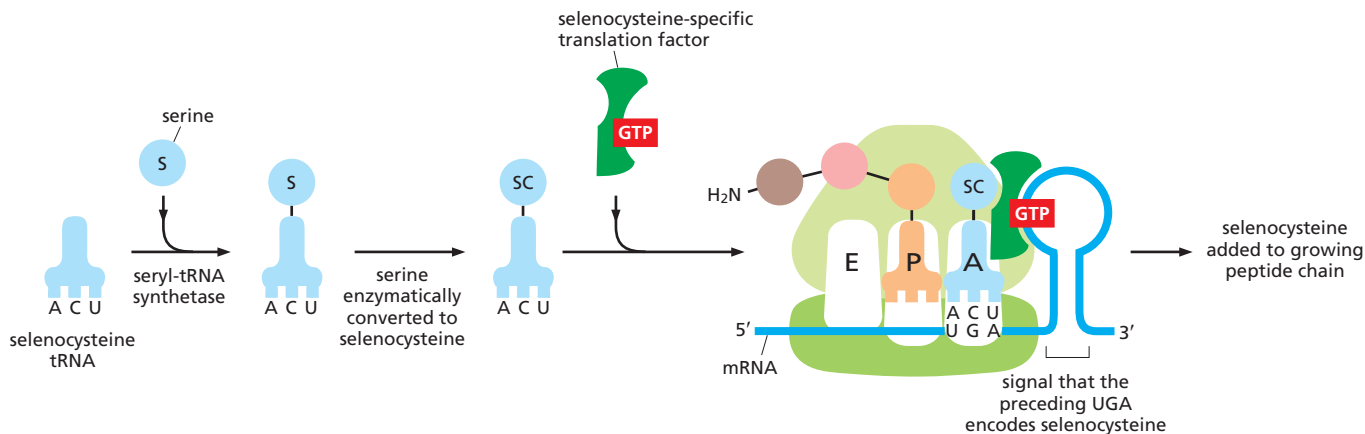


Figure 6–78 Incorporation of selenocysteine into a growing polypeptide chain. A specialized tRNA is charged with serine by the normal seryl-tRNA synthetase, and the serine is subsequently converted enzymatically to selenocysteine. A specific RNA structure in the mRNA (a stem-and-loop structure with a particular nucleotide sequence) signals that selenocysteine is to be inserted at the neighboring UGA codon. As indicated, this event requires the participation of a selenocysteine-specific translation factor. After the addition of selenocysteine, translation continues until a conventional stop codon is encountered.

acids, and the end of translation is instead signaled by the 3' end of the mRNA. Mitochondria of many species (which have their own genomes and encode much of their translational apparatus) routinely deviate from the standard code. For example, in mammalian mitochondria AUA is translated as methionine, whereas in the cytosol of the cell it is translated as isoleucine (see Table 14–4, p. 864).

A common type of variation, sometimes called *translation recoding*, is used to incorporate selenocysteine into proteins. In this case, neighboring nucleotide sequence information present in an mRNA changes the meaning of the genetic code at a particular site in the mRNA molecule. The standard code allows cells to manufacture proteins using only 20 amino acids. However, bacteria, archaea, and eukaryotes have selenocysteine available as a twenty-first amino acid that can be incorporated directly into a growing polypeptide chain. Selenocysteine, which contains a selenium atom in place of the sulfur atom of cysteine, is essential for the efficient function of a variety of enzymes. It is enzymatically produced from a serine attached to a special tRNA molecule that base-pairs with the UGA codon, a codon normally used to signal a translation stop. The mRNAs for those proteins in which selenocysteine is to be inserted at a UGA codon carry an additional nearby nucleotide sequence in the mRNA that triggers this recoding event (Figure 6–78).

Inhibitors of Prokaryotic Protein Synthesis Are Useful as Antibiotics

Many of the most effective antibiotics used in modern medicine are compounds that inhibit bacterial protein synthesis. Although chemists have improved these compounds, most were originally isolated from bacteria and fungi, where they are thought to have arisen over evolutionary time because of the warfare between competing microbes. Because many of these drugs are selective for bacterial ribosomes, humans can take high dosages without undue toxicity. Many of these antibiotics lodge in pockets in the ribosomal RNAs and simply “gum up” the smooth operation of the ribosome; others block specific parts of the ribosome such as the exit tunnel (Figure 6–79). Table 6–4 lists some common antibiotics of this kind, along with several other inhibitors of protein synthesis, some of which act on eukaryotic cells and therefore cannot be used as antibiotics.

Because they block specific steps in the processes that lead from DNA to protein, many of the compounds listed in Table 6–4 are useful for cell biological studies. Among the most commonly used drugs in such investigations are *chloramphenicol*, *cycloheximide*, and *puromycin*, all of which specifically inhibit

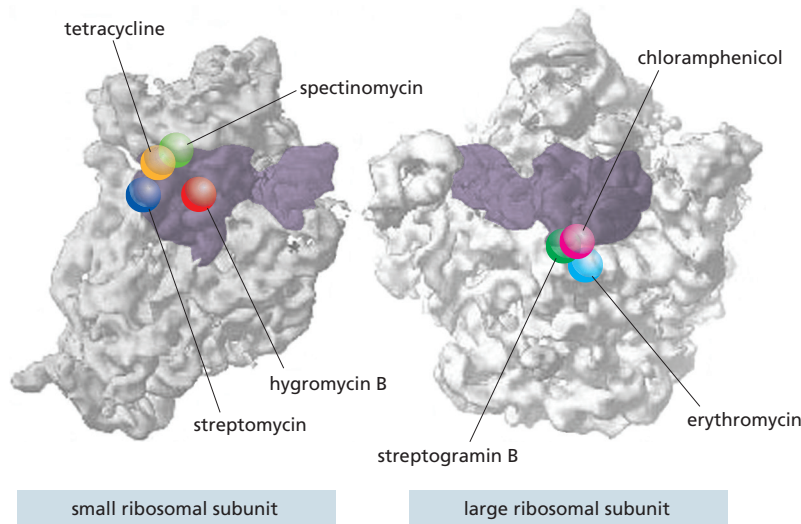


Figure 6-79 Binding sites for antibiotics on the bacterial ribosome. The small (*left*) and large (*right*) subunits of the ribosome are arranged as though the ribosome has been opened like a book. Antibiotic-binding sites are marked with colored spheres, and the bound tRNA molecules are shown in *purple* (see Figure 6-66). Most of the antibiotics shown bind directly to pockets formed by the ribosomal RNA molecules. Hygromycin B induces errors in translation, spectinomycin blocks the translocation of the peptidyl-tRNA from the A site to the P site, and streptogramin B prevents elongation of nascent peptides. Table 6-4 lists the inhibitory mechanisms of additional commonly used antibiotics. (Adapted from J. Poehlsgaard and S. Douthwaite, *Nat. Rev. Microbiol.* 3:870-881, 2005.)

protein synthesis. In a eukaryotic cell, for example, chloramphenicol inhibits protein synthesis on ribosomes only in mitochondria (and in chloroplasts in plants), presumably reflecting the bacterial origins of these organelles (discussed in Chapter 14). Cycloheximide, in contrast, affects only ribosomes in the cytoplasm. Puromycin is especially interesting because it is a structural analog of a tRNA molecule linked to an amino acid and is therefore a fascinating example of molecular mimicry. The ribosome mistakes it for an authentic charged tRNA and covalently incorporates it at the C-terminus of the growing peptide chain, causing premature termination and release of the polypeptide. As might be expected, puromycin inhibits protein synthesis in both bacteria and eukaryotes.

TABLE 6-4 Inhibitors of Protein or RNA Synthesis	
Inhibitor	Specific effect
Acting only on bacteria	
Tetracycline	Blocks binding of aminoacyl-tRNA to the A site of the ribosome
Streptomycin	Prevents the transition from translation initiation to chain elongation and also causes miscoding
Chloramphenicol	Blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6-68)
Erythromycin	Binds in the exit tunnel of the ribosome and thereby inhibits elongation of the peptide chain
Rifamycin	Blocks initiation of RNA chains by binding to RNA polymerase (prevents RNA synthesis)
Acting on bacteria and eukaryotes	
Puromycin	Causes the premature release of nascent polypeptide chains by its addition to the growing chain end
Actinomycin D	Binds to DNA and blocks the movement of RNA polymerase (prevents RNA synthesis)
Acting on eukaryotes but not bacteria	
Harringtonine	Blocks the A site of the 60S ribosome subunit after translation initiation, but before elongation (see Figure 6-74)
Anisomycin	Blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6-68)
α -Amanitin	Blocks mRNA synthesis by binding preferentially to RNA polymerase II
The ribosomes of eukaryotic mitochondria (and chloroplasts) often resemble those of bacteria in their sensitivity to inhibitors. Therefore, some of these antibiotics can have a deleterious effect on human mitochondria.	

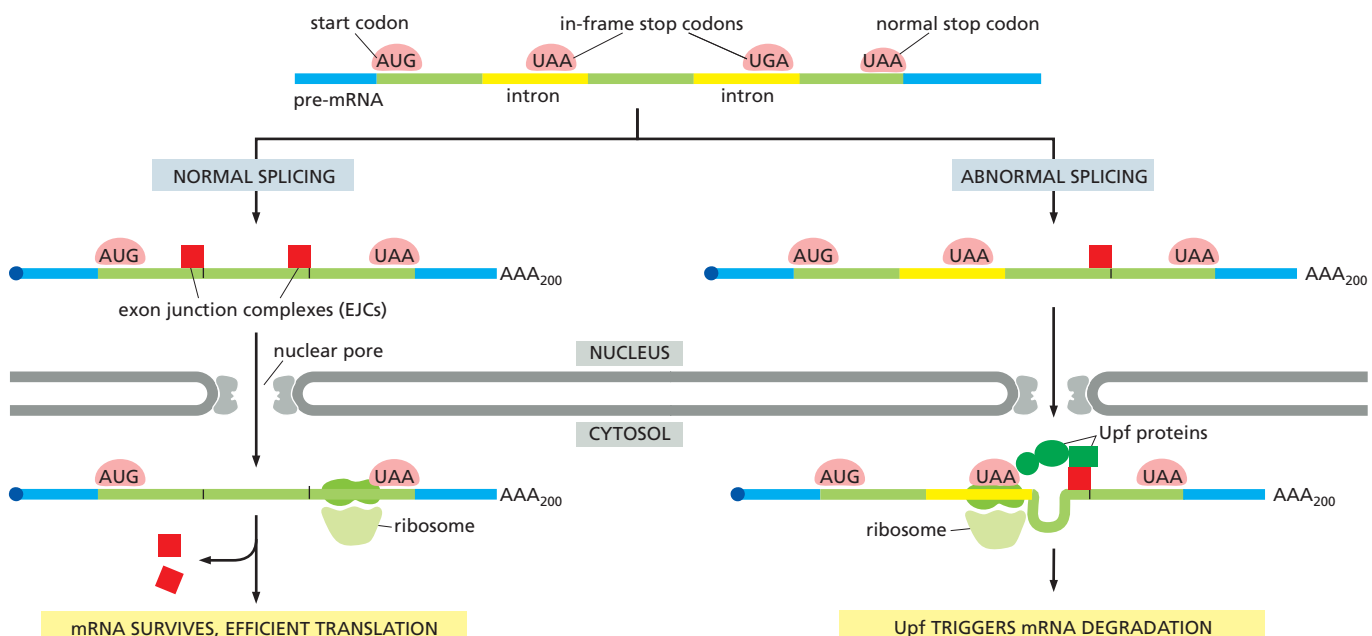
Quality-Control Mechanisms Act to Prevent Translation of Damaged mRNAs

In eukaryotes, mRNA production involves both transcription and a series of elaborate RNA-processing steps; as we have seen, these take place in the nucleus, segregated from ribosomes, and only when the processing is complete and the mRNAs deemed “export-ready” are they transported to the cytosol to be translated (see Figure 6-40). However, this scheme is not foolproof, and some incorrectly processed mRNAs are inadvertently sent to the cytosol. In addition, mRNAs that were flawless when they left the nucleus can become broken or otherwise damaged in the cytosol. The danger of translating damaged or incompletely processed mRNAs (which would produce truncated or otherwise aberrant proteins) is apparently so great that the cell has several backup measures to prevent this from happening. To avoid translating broken mRNAs, for example, the 5' cap and the poly-A tail are both recognized by the translation initiation machinery before translation begins (see Figure 6-74).

One of the most powerful mRNA surveillance systems, called **nonsense-mediated mRNA decay**, prevents defective mRNAs from escaping from the nuclear envelope. This mechanism is brought into play when the cell determines that an mRNA molecule has a nonsense (stop) codon (UAA, UAG, or UGA) in the “wrong” place. This situation is likely to arise in an mRNA molecule that has been improperly spliced, because aberrant splicing will usually result in the random introduction of a nonsense codon into the reading frame of the mRNA—especially in organisms, such as humans, that have large introns.

The nonsense-mediated mRNA decay mechanism begins as an mRNA molecule is being transported from the nucleus to the cytosol. As soon as its 5' end emerges from a nuclear pore, the mRNA is met by a ribosome, which begins to translate it. As translation proceeds, the exon junction complexes (EJCs) that are bound to the mRNA at each completed splice site (see Figure 6-29) are displaced by the moving ribosome. The normal stop codon should lie within the last exon, so by the time the ribosome reaches it, the mRNA should be free of EJCs. In this case, the mRNA “passes inspection” and is released to the cytosol where it can be translated in earnest (Figure 6-80). However, if the ribosome reaches a stop codon earlier, when EJCs remain bound, the mRNA molecule is rapidly degraded. In this way, the first round of translation allows the cell to test the fitness of each mRNA molecule as it exits the nucleus.

Figure 6-80 Nonsense-mediated mRNA decay. As shown on the *right*, the failure to correctly splice a pre-mRNA often introduces a premature stop codon into the reading frame for the protein. These abnormal mRNAs are destroyed by the nonsense-mediated decay mechanism. To activate this mechanism, an mRNA molecule, bearing the exon junction complexes (EJCs) that mark successfully completed splices, is quickly met by a ribosome that performs a “test” round of translation. As the mRNA passes through the ribosome, the EJCs are stripped off, and successful mRNAs are released to undergo multiple rounds of translation (*left side*). However, if an in-frame stop codon is encountered before the final EJC is reached (*right side*), the mRNA undergoes nonsense-mediated decay, which is triggered by the Upf proteins (green) that bind to each EJC. This mechanism ensures that nonsense-mediated decay is triggered only when the premature stop codon is in the same reading frame as that of the normal protein. (Adapted from J. Lykke-Andersen et al., *Cell* 103:1121–1131, 2000.)



Nonsense-mediated decay may have been especially important in evolution, allowing eukaryotic cells to more easily explore new genes formed by DNA rearrangements, mutations, or alternative patterns of splicing—by selecting only those mRNAs for translation that can produce a full-length protein. Nonsense-mediated decay is also important in cells of the developing immune system, where the extensive DNA rearrangements that occur (see Figure 24–28) often generate premature termination codons. The surveillance system degrades the mRNAs produced from these faulty rearranged genes, thereby avoiding the potential toxic effects of truncated proteins.

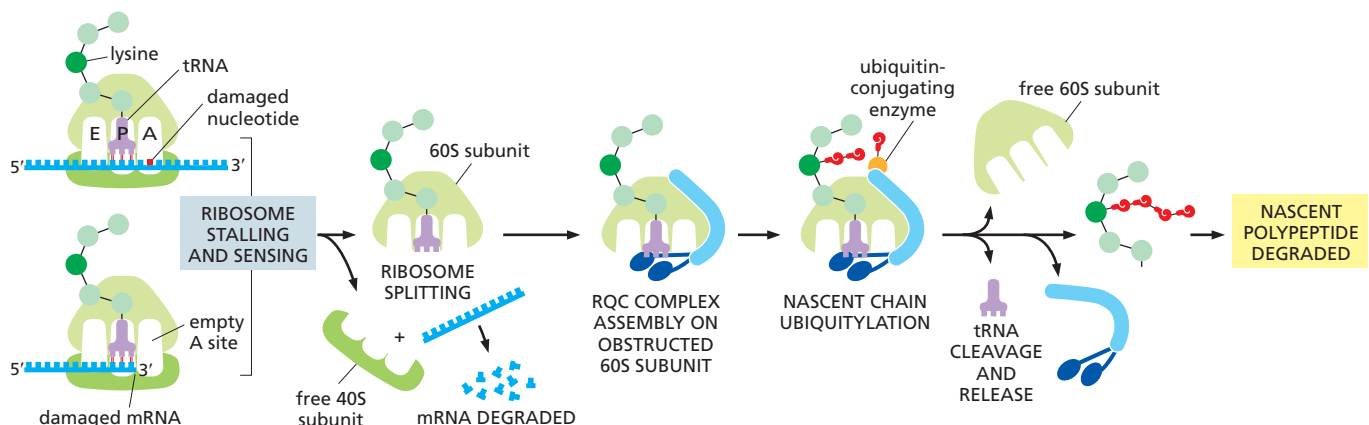
The nonsense-mediated surveillance pathway also plays an important role in mitigating the symptoms of many inherited human diseases. As we have seen, inherited diseases are usually caused by mutations that spoil the function of a key protein, such as hemoglobin or one of the blood-clotting factors. Approximately one-third of all genetic disorders in humans result from mutations that change a normal codon into a stop (nonsense) codon or mutations (such as frameshift mutations or splice-site mutations) that place nonsense mutations into the gene's reading frame. In individuals that carry one mutant and one functional gene, nonsense-mediated decay eliminates the aberrant mRNA, and thereby prevents a potentially toxic protein from being made. Without this safeguard, individuals with one functional and one mutant “disease gene” would likely suffer much more severe symptoms.

Stalled Ribosomes Can Be Rescued

Even when an mRNA molecule passes all the initial inspections, additional problems can arise as it is being translated in the cytosol, and the cell has several ways to deal with them. In a sense, translating ribosomes constantly act as sensors for mRNA “health,” and when one stalls, those upstream of it collide with it (and each other), generating a string of nonfunctioning ribosomes. For example, if an mRNA becomes broken and thereby lacks an in-frame stop codon, the ribosome will translate to the 3' end of the RNA but will not be released. Damaged RNA bases (which cannot form stable codon–anticodon interactions), stable mRNA secondary structures, and stretches of rare codons (that is, codons that have very low concentrations of their matching tRNAs) can also stall ribosomes. Although some of these problems can be overcome simply by allowing the ribosome sufficient time, many are dealt with more aggressively by a set of mechanisms known collectively as *ribosome-associated quality control* (Figure 6–81). Although the detailed pathways differ for the different types of barriers that stall ribosomes, three steps are common: the mRNA is degraded, the nascent protein is degraded, and the stalled ribosome is disengaged so it can be used again.

All of these steps make conceptual sense: the first removes a damaged mRNA so it will cause no further problems, the second prevents an aberrant protein from

Figure 6–81 Eukaryotic ribosome-associated quality control. A ribosome can stall at damaged bases, rare codons, or other barriers (*top left*); here, the signal to correct the problem is generated by a collision between the stalled ribosome and the one immediately upstream of it. Ribosomes also stall at the ends of broken mRNAs (*bottom left*); in this case the empty A site is the cue for rescue. Although the mechanisms differ somewhat depending on the nature of the stall, in general a stalled ribosome is split (immediately releasing the 40S subunit for reuse), the mRNA is degraded, and the 60S subunit, which contains a tRNA lodged in its P site attached to a partially synthesized protein, is rescued by the ribosome quality control (RQC) complex (indicated in blue). The nascent protein is ubiquitinated and the tRNA cleaved from it, freeing the protein from the ribosome. As will be described shortly, this ubiquitylation serves as a signal for the protein to be destroyed. (Adapted from C.A.P. Joazeiro, *Nat. Rev. Mol. Cell Biol.* 20:368–383, 2019. With permission from Springer Nature.)



being released into the cell, and the third rescues a valuable RNA–protein machine complex that requires many resources to assemble. It is estimated that each ribosome in a human cell synthesizes approximately 3000 protein molecules, and the ability to rescue stalled ribosomes (as opposed to degrading them) makes a major contribution to their longevity.

The Ribosome Coordinates the Folding, Enzymatic Modification, and Assembly of Newly Synthesized Proteins

The process of gene expression is not over when the genetic code has been used to create the sequence of amino acids that constitutes a protein. To be useful to the cell, this new polypeptide chain must fold up into its unique three-dimensional conformation, bind any small-molecule cofactors required for its activity, be appropriately modified by protein kinases and other protein-modifying enzymes, and assemble correctly with the other protein subunits with which it functions (Figure 6–82).

The information needed for all of the steps listed above is ultimately contained in the sequence of amino acids that the ribosome produces when it translates an mRNA molecule into a polypeptide chain. As discussed in Chapter 3, when a protein folds into a compact structure, it buries most of its hydrophobic residues in an interior core. In addition, large numbers of noncovalent interactions form between various parts of the molecule. It is the sum of all of these energetically favorable arrangements that determines the final folding pattern of the polypeptide chain—as the conformation of lowest free energy (see p. 121).

For some proteins, folding begins in the exit tunnel of the ribosome. Although too narrow to accommodate the folding of complete proteins, the exit tunnel widens toward its end, and simple structures such as short α helices can form as the growing peptide is pushed through by the translation process. But the major portion of protein folding begins as the protein emerges from the ribosome exit tunnel, where the new protein is also met by enzymes that modify its N-terminus. We saw earlier in the chapter that translation of a typical mRNA always begins with a methionine (see Figure 6–74); however, in most eukaryotic proteins (70% of cytosolic proteins), this amino acid is cleaved off and the resulting N-terminus modified by acetylation. The enzymes that carry out these and other modifications have a weak affinity for the ribosome and “hover” (that is, rapidly associate and dissociate with the ribosome) near the exit tunnel—in position to act on the growing protein as it first emerges.

Molecular Chaperones Help Guide the Folding of Most Proteins

Most proteins on their own do not fold correctly during their synthesis; instead, they require a special class of proteins called **molecular chaperones** to do so. Molecular chaperones are useful for cells because there are many different folding paths available to an unfolded or partially folded protein. Without these chaperones, some of these pathways would not lead to the correctly folded (and most stable) form; instead, the protein would become “kinetically trapped” in structures that are off-pathway. Some of these off-pathway conformations would aggregate and be left as irreversible dead ends, producing nonfunctional (and potentially dangerous) structures. As we will see shortly, chaperones help solve this problem by repeatedly binding and releasing a protein’s partially folded regions, giving them many chances to fold correctly.

Many molecular chaperones are called *heat-shock proteins* (designated *hsp*), because they are synthesized in dramatically increased amounts after a brief exposure of cells to an elevated temperature (for example, 42°C for cells that normally live at 37°C). This increase reflects the operation of a feedback system that responds to an increase in misfolded proteins (such as those produced by elevated temperatures) by boosting the synthesis of the chaperones that help these proteins refold. However, even under normal conditions, the heat-shock proteins

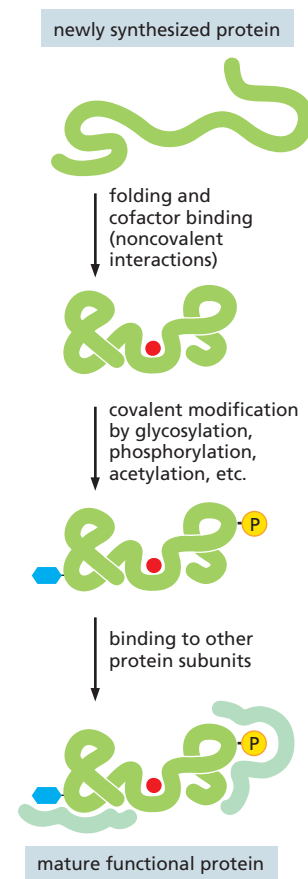


Figure 6–82 Steps in the creation of a functional protein. As indicated, translation of an mRNA sequence into an amino acid sequence on the ribosome is not the end of the process of forming a protein. To function, the completed polypeptide chain must fold correctly into its three-dimensional conformation, bind any cofactors required, and assemble with its partner protein chains, if any. Noncovalent bond formation drives these changes. As indicated, many proteins also require covalent modifications of selected amino acids. Although the most frequent modifications are protein glycosylation and protein phosphorylation, more than 200 different types of covalent modifications are known (see pp. 175–176). As described in the text, many of these steps begin while the protein is still being synthesized.

are so abundant that all proteins should be considered to be embedded in a rich soup of these molecules.

There are several major families of molecular chaperones, including the hsp60 and hsp70 proteins, with different family members functioning in different organelles. Thus, as discussed in Chapter 12, mitochondria contain their own hsp60 and hsp70 molecules that are distinct from those that function in the cytosol, and a special hsp70 (called *BiP*) helps to fold proteins in the endoplasmic reticulum.

In all, humans have thirteen hsp70 proteins, including a group that associates with the ribosome and helps nearly all emerging proteins fold correctly. They do this by rapidly binding and releasing short sequences (approximately five amino acids each) as a new protein is pushed through the ribosome exit tunnel. When one of these chaperones encounters a sequence rich in hydrophobic amino acids, which typically form the core of folded proteins, it clamps down on it, thereby delaying the folding of the emerging protein until enough of it has been made to begin to fold correctly. Hydrophobic stretches that emerge early from the ribosome are thereby prevented from aggregating with other hydrophobic surfaces until they can be properly folded into the core of the nascent protein. The hsp70 clamping and release reactions required are regulated by ATP hydrolysis catalyzed by the nucleotide-binding domain of hsp70 (**Figure 6–83**).

Most proteins are at least partially folded when they are released from the ribosome, but the final folding for many occurs as additional hsp70 molecules associate with them away from the ribosome. By undergoing multiple cycles of ATP hydrolysis, these hsp70 molecules allow proteins to complete their folding.

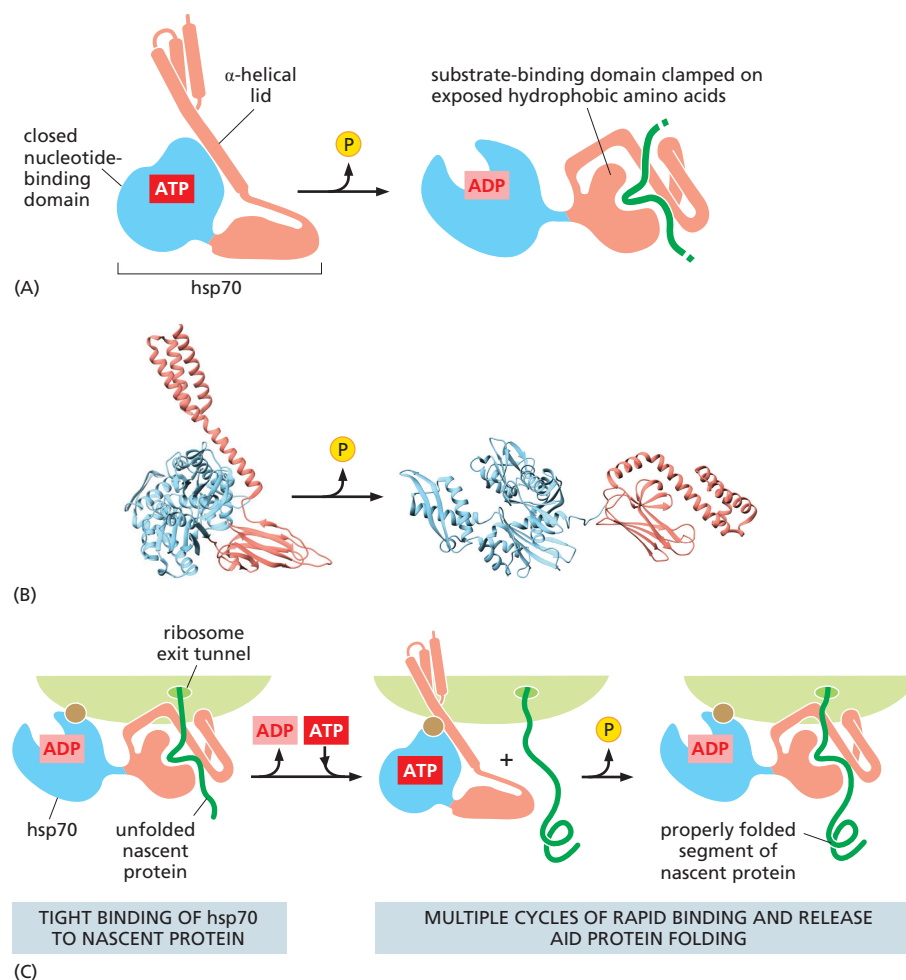


Figure 6–83 The hsp70 family of molecular chaperones. These are among the most abundant of all proteins, constituting several percent of total cell protein, and they thus play prominent roles inside the cell. (A) Short stretches of hydrophobic amino acids that are abnormally exposed in misfolded proteins (green) trigger the hydrolysis of ATP to ADP, causing hsp70 to close down on its substrate, trapping it in an extended conformation. The rebinding of another molecule of ATP will open the hsp70 “clamp,” releasing the substrate in its extended form. Repeated cycles of hsp70 clamping and release, often involving multiple hsp70 molecules on the same target, help fold the target protein by keeping its hydrophobic regions from aggregating. This can continue until these regions are assimilated into the core of the properly folded protein. (B) The three-dimensional structures of hsp70 that produce the mechanism schematically illustrated in A. (C) Some hsp70 family members bind to the ribosome and act early in the life of a newly synthesized protein. Aided by other proteins (not shown), ATP-bound hsp70 molecules rapidly bind and release the nascent protein as it emerges from the ribosome. (B, PDB codes: 2KH0 and 4JNE.)

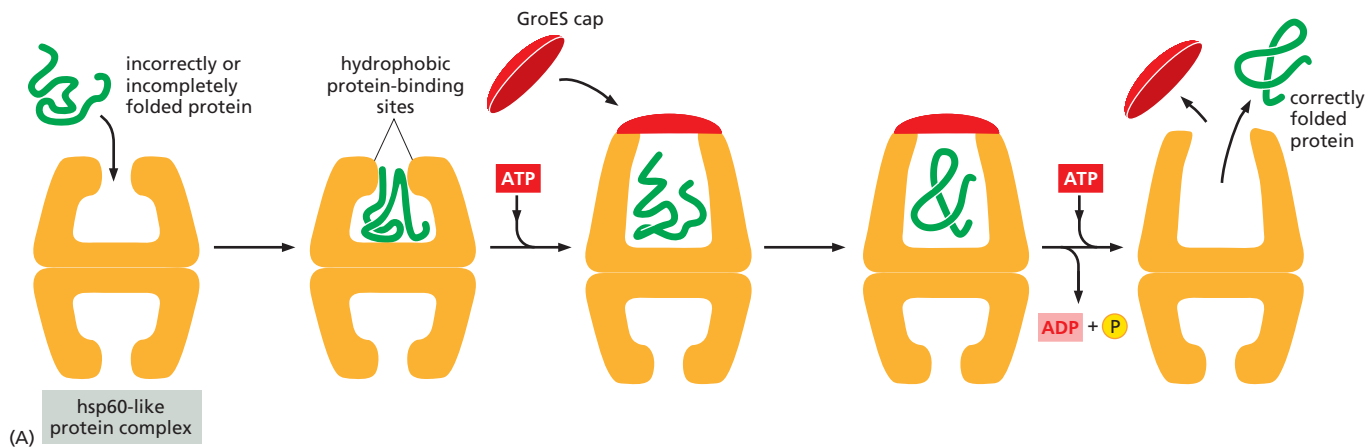
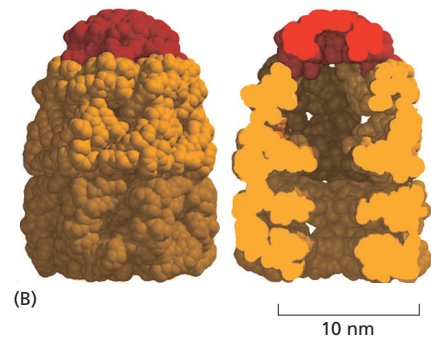


Figure 6–84 The structure and function of the hsp60 family of molecular chaperones. (A) A misfolded protein is initially captured by hydrophobic interactions with the exposed surface of the opening. This initial binding often helps to unfold a misfolded protein. The subsequent binding of ATP and a cap releases the substrate protein into an enclosed space, where it has a new opportunity to fold. After about 10 seconds, ATP hydrolysis occurs, weakening the binding of the cap. Subsequent binding of additional ATP molecules ejects the cap, and the protein is released. As indicated, only half of the symmetric barrel operates on a client protein at any one time. This hsp type of chaperone, known also as a *chaperonin*, is designated as hsp60 in mitochondria, TRiC in the cytosol of vertebrate cells, and GroEL in bacteria. (B) The structure of GroEL bound to its GroES cap, as determined by x-ray crystallography. On the left is shown the outside of the barrel-like structure, and on the right is a cross section through its center. (B, adapted from B. Bukau and A.L. Horwich, *Cell* 92:351–366, 1998. With permission from Elsevier.)



Some proteins produced by the cell cannot properly fold with help from the hsp70 proteins alone, and other types of chaperones are brought into play. An important class is exemplified by the hsp60 proteins. These form a large barrel-shaped structure that acts after a protein has been fully synthesized but before it has folded correctly. This type of chaperone, sometimes called a *chaperonin*, forms an “isolation chamber” for the folding process (Figure 6–84). As we have seen, incorrectly folded proteins are often characterized by exposed hydrophobic patches. The entrance to the hsp chamber is itself hydrophobic and attracts these patches. In conjunction with ATP binding, the lid to the chamber closes, forcing the incorrectly folded protein into the chamber interior and causing a twisting of the chamber subunits whereby some of the hydrophobic surfaces of the inner chamber are moved out of the way and replaced with hydrophilic surfaces. The substrate protein is thereby given a chance to fold into its final configuration (which generally favors hydrophilic amino acids on its outside) in the absence of any other proteins with which to aggregate. When ATP is hydrolyzed, the lid pops off, and the substrate protein, whether correctly folded or not, is released from the chamber.

Folding and maintaining the enormous diversity of proteins in cells require a wide range of chaperones with versatile surveillance and correction capabilities. Although our discussion has focused on only two types of chaperones, the cell has a variety of others. These include an hsp90 chaperone that can harness mechanical forces to help proteins fold as part of a collaborative chaperone network (Figure 6–85).

The hsp70, hsp60, and hsp90 chaperones often need many cycles of ATP binding and hydrolysis to fold a single polypeptide chain correctly. This energy is used to create movements in each of these chaperone “machines,” converting them back and forth between binding and releasing conformations. Thus, just as we saw for transcription, splicing, and translation, a great deal of free energy is used by cells to improve the accuracy of a biological process—in this case the correct folding of proteins (Movie 6.11).

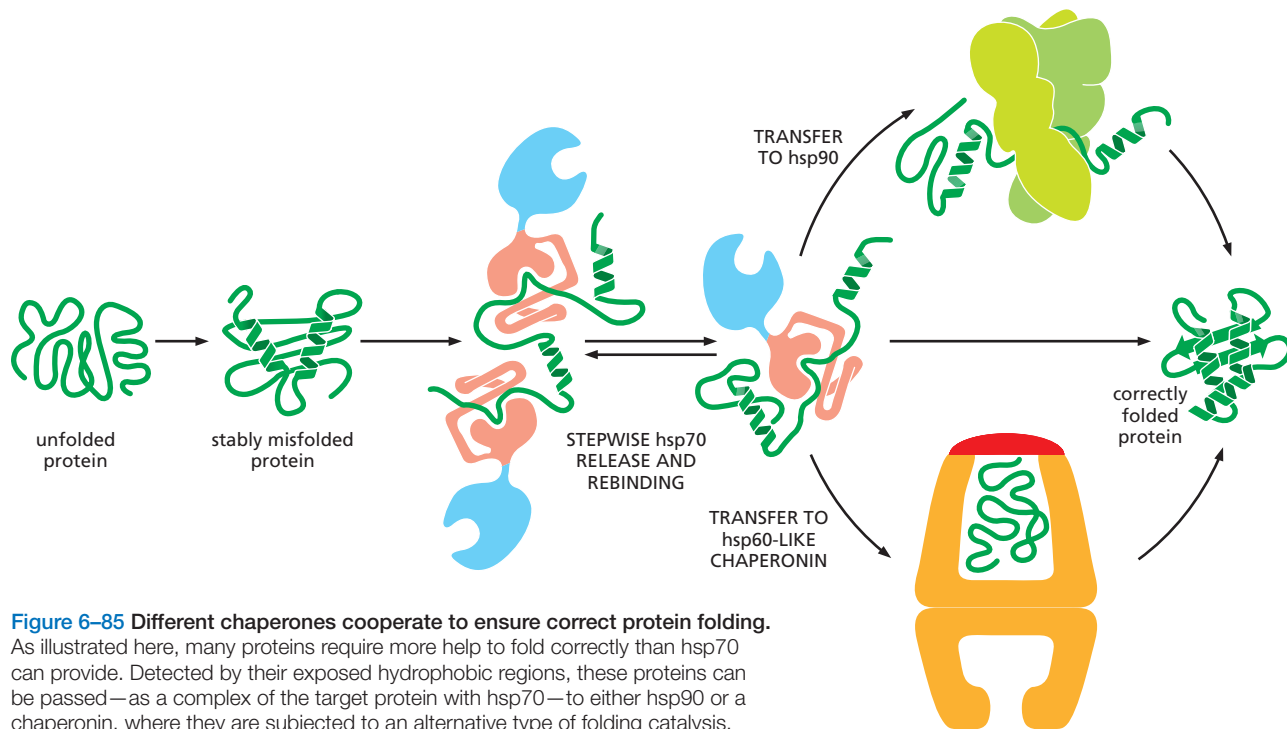


Figure 6–85 Different chaperones cooperate to ensure correct protein folding.

As illustrated here, many proteins require more help to fold correctly than hsp70 can provide. Detected by their exposed hydrophobic regions, these proteins can be passed—as a complex of the target protein with hsp70—to either hsp90 or a chaperonin, where they are subjected to an alternative type of folding catalysis.

Proper Folding of Newly Synthesized Proteins Is Also Aided by Translation Speed and Subunit Assembly

In addition to the highly abundant chaperones, two additional mechanisms help to correctly fold newly synthesized proteins. First, translation is not a smooth process; the ribosome moves in fits and starts, translating some RNA sequences rapidly and pausing at others. The pauses are typically caused by a series of rare codons (which require extra time for the corresponding low-abundance charged tRNAs to enter the ribosome by diffusion) or by mRNA secondary structures that form ahead of the ribosome, slowing down or temporarily blocking its smooth passage. It is thought that many of these pauses are not accidental; rather, they are spaced to allow extra time for “problem” portions of newly synthesized proteins to fold as they exit the ribosome. If true, it means that an mRNA molecule, in addition to coding for a protein, specifies changes in the speed of translation matched to its protein’s folding characteristics.

A second mechanism reflects the fact that a protein molecule does not typically work on its own; instead, it usually assembles with other subunits to form multisubunit structures. As a protein emerges from the ribosome, it often begins assembling with one or more of its fully folded partner subunits, which, by acting as complementary surfaces, help the newly synthesized protein adopt its correct three-dimensional structure. We have seen in both bacteria and eukaryotes that a given mRNA molecule can be simultaneously translated by many ribosomes. For protein complexes made up of identical subunits, fully synthesized protein molecules will therefore always be in proximity to aid in the folding of those being synthesized. In bacteria, different proteins can be translated from the same mRNA, providing an opportunity for different subunits of a protein complex to efficiently help each other fold. In eukaryotes, where only a single protein is typically produced from each mRNA, it has been proposed that mRNAs that code for different subunits of a protein complex are held in proximity, perhaps by information in the untranslated portion of the message, providing a high local concentration of mature subunits to aid in the assembly and folding of newly synthesized subunits.

Proteins That Ultimately Fail to Fold Correctly Are Marked for Destruction by Polyubiquitin

We have seen that the cell relies on many resources to ensure that a newly synthesized protein is folded properly. Yet occasionally, normal proteins simply fail to fold properly despite numerous attempts. In addition, errors in transcription, RNA splicing, and translation can produce aberrant proteins that will never fold properly. Finally, proteins that were once correctly folded can become damaged by chemical reactions (such as oxidation), which can cause their misfolding. In all these cases, the cell destroys the aberrant protein. Although the cell uses many cues to ascertain whether a given protein is misfolded, the most important is an exposed hydrophobic patch, the same signal that is recognized by chaperones. The continued presence of a chaperone is thus an indication that the protein has failed to fold. In this case, a series of specialized enzymes recognizes the hydrophobic patch (possibly with the aid of chaperones) and attaches the small protein ubiquitin to a nearby lysine (see Figure 3–65A). Additional ubiquitin molecules are subsequently added to build a polyubiquitin chain. As we saw in Chapter 3, ubiquitin modification of proteins is used for many purposes in the cell. The particular type of ubiquitin linkage that concerns us here is a chain of ubiquitin molecules linked together at lysine 48 (see Figure 3–65B), which is the distinguishing feature of the ubiquitin tag that marks a protein for destruction. This polyubiquitin chain delivers the protein to a protein-destruction machine, a complex protease known as the *proteasome*.

The Proteasome Is a Compartmentalized Protease with Sequestered Active Sites

Proteasomes are very abundant, constituting approximately 1% of the total proteins in cells. They are dispersed throughout the cytosol and the nucleus, but the proteasome also destroys aberrant proteins that have entered the endoplasmic reticulum (ER). In this case, an ER-based surveillance system detects proteins that have failed either to fold or to be assembled properly after they enter the ER and *retrotranslocates* them back to the cytosol for degradation by the proteasome (discussed in Chapter 12).

Each proteasome consists of a central hollow cylinder (the 20S core proteasome) formed from multiple protein subunits that assemble as a stack of four heptameric rings. Some of the subunits are proteases whose active sites face the cylinder's inner chamber, thus preventing them from running rampant through the cell. Each end of the cylinder is normally associated with a large protein complex (the 19S cap) that contains a six-subunit protein ring through which target proteins are threaded into the proteasome core, where they are degraded (Figure 6–86). The threading reaction, driven by ATP hydrolysis, unfolds the target

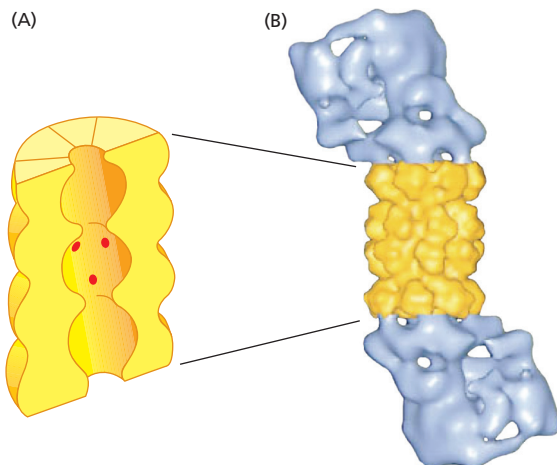


Figure 6–86 The proteasome. (A) A cutaway view of the structure of the central 20S cylinder, as determined by x-ray crystallography, with the active sites of the proteases indicated by red dots. (B) The entire proteasome, in which the central cylinder (yellow) is supplemented by a 19S cap (blue) at each end. The complex cap selectively binds proteins that are marked by ubiquitin for destruction; if the protein also contains a loosely structured region, the cap uses ATP hydrolysis to further unfold its polypeptide chain and feed it through the narrow channel (see Figure 6–87) into the inner chamber of the 20S cylinder for digestion to short peptides. (B, from W. Baumeister et al., *Cell* 92:367–380, 1998. With permission from Elsevier.)

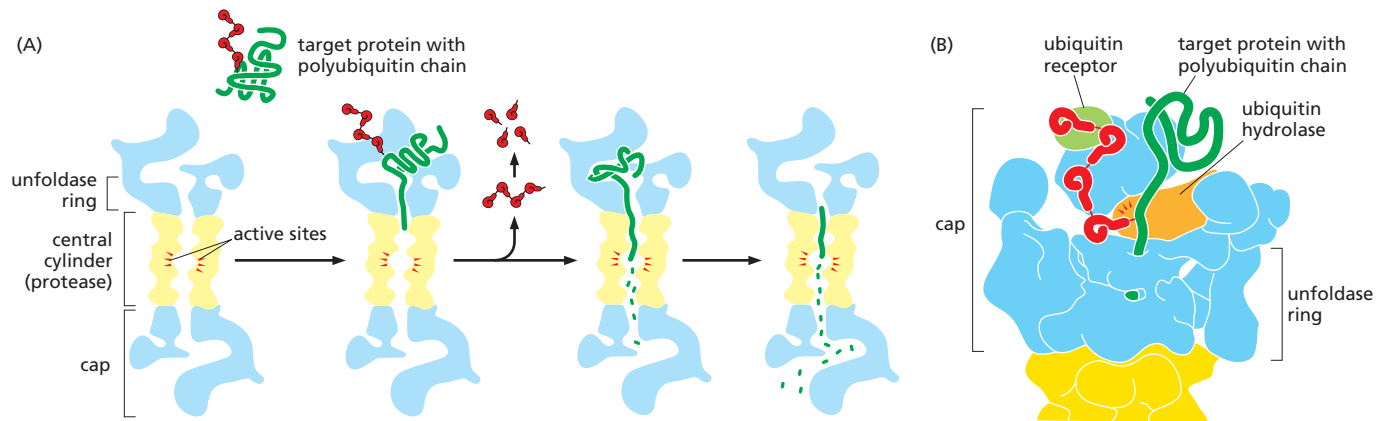


Figure 6-87 Processive protein digestion by the proteasome. (A) The proteasome cap recognizes proteins marked by a polyubiquitin chain (see Figure 3-65B). Most of these proteins also contain an unfolded region, and the cap translocates these proteins into the proteasome core, where they are digested. Before moving through the proteasome cap, the ubiquitin is cleaved from the substrate protein and is recycled. Translocation into the core of the proteasome is mediated by a ring of ATPases that unfold the substrate protein as it is threaded through the ring and into the proteasome core. (B) Detailed structure of the proteasome cap. The cap includes a ubiquitin receptor, which holds a ubiquitylated protein in place while attempts are made to pull it into the proteasome core, and a ubiquitin hydrolase, which cleaves ubiquitin from the protein. (A, from S. Prakash and A. Matouschek, *Trends Biochem. Sci.* 29:593–600, 2004. With permission from Elsevier; B, adapted from G.C. Lander et al., *Nature* 482:186–191, 2012.)

proteins as they move through the cap, exposing them to the proteases lining the proteasome core (Figure 6-87). The proteins that make up the ring structure in the proteasome cap belong to a large class of protein “unfoldases” known as AAA proteins. Many of them function as hexamers, and they share mechanistic features with the ATP-dependent DNA helicases that unwind DNA (Figure 6-88; see also Figure 5-14).

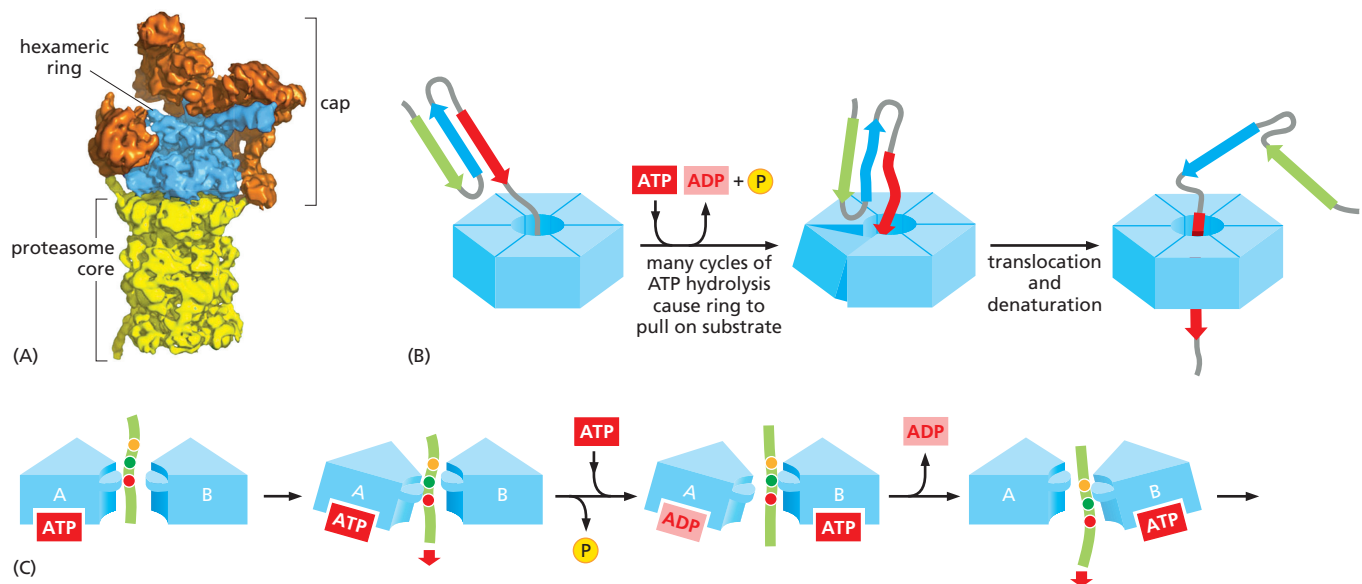


Figure 6-88 A hexameric protein unfoldase. (A) The proteasome cap includes a hexameric ring (blue) through which proteins to be destroyed are threaded. The hexameric ring (also called the unfoldase ring) is formed from six subunits, each belonging to the AAA family of proteins. (B) Model for the ATP-dependent unfoldase activity of AAA proteins. The ATP-bound form of a hexameric ring of AAA proteins grasps a substrate protein, and a conformational change, driven by ATP hydrolysis, further pulls the substrate and strains the ring structure. At this point, the substrate protein, which is being tugged upon, can partially unfold and enter further into the pore or it can maintain its structure and partially withdraw. Some protein substrates may require hundreds of cycles of ATP hydrolysis and dissociation before they are successfully pulled through the AAA protein ring; some proteins continue to resist these efforts and are ultimately de-ubiquitylated and released. (C) How the successive tilting of adjacent subunits (only two of which are shown), driven by ATP hydrolysis, is thought to pull the unfolded polypeptide chain through the hexameric ring into the proteasome. The motions of the ring subunits resemble a hand-over-hand pulling motion on the substrate protein, with the hands often slipping until the substrate is unfolded. Once unfolding occurs, the substrate protein moves relatively quickly through the ring by successive rounds of ATP hydrolysis. (A, adapted from G.C. Lander et al., *Nature* 482:186–191, published 2012 by Macmillan Publishers Ltd. Reproduced with permission of SNCSC; B, adapted from R.T. Sauer et al., *Cell* 119:9–18, 2004.)

A crucial property of the proteasome, and one reason for its complexity, is the *processivity* of its mechanism: in contrast to a “simple” protease that cleaves a substrate’s polypeptide chain just once before dissociating, the proteasome keeps the entire substrate bound until all of it is converted into short peptides. One would expect that a machine as efficient as the proteasome would be tightly regulated; in particular, it must be able to distinguish abnormal proteins from those that are properly folded. The 19S cap of the proteasome acts as a gate at the entrance to the inner proteolytic core, and only those proteins to be destroyed are threaded through the cap. We saw above that proteins marked for destruction are distinguished by carrying a particular type of polyubiquitin chain. This chain binds specifically to a receptor in the 19S cap of the proteasome, and here a second “check” is made by the cell to determine whether or not the protein will be destroyed. If the protein has, in addition to the ubiquitin mark, an unfolded region (which can span the mark), it is grasped tightly by the 19S cap, de-ubiquitylated, and pulled through the cap into the proteasome core. Ubiquitylated proteins that lack such a region are typically de-ubiquitylated and released back into solution.

As might be expected, there is competition for misfolded proteins between chaperones and the protein degradation machinery. Proteins that are folded quickly escape destruction (at least early in their life before they accumulate damage), whereas those that undergo many rounds of chaperone-assisted folding are more likely to be degraded. Some chaperones can directly hand off those proteins that remain improperly folded, after many attempts, to the protein destruction machinery. It is estimated that between 1 and 5% of all newly synthesized proteins fail to fold properly and are degraded by the proteasome.

Many Proteins Are Controlled by Regulated Destruction

One function of intracellular proteolytic mechanisms is to recognize and eliminate misfolded or otherwise abnormal proteins, as just described. Indeed, every protein in the cell eventually accumulates damage and is probably degraded by the proteasome. Yet another function of these proteolytic pathways is to confer short lifetimes on specific normal proteins whose concentrations must change promptly with alterations in the state of a cell. Some of these short-lived proteins are degraded rapidly at all times, while many others are *conditionally* short-lived; that is, they are metabolically stable under some conditions but become unstable upon a change in the cell’s state. For example, mitotic cyclins are long-lived throughout the cell cycle until their sudden degradation at the end of mitosis, as explained in Chapter 17.

How is such a regulated destruction of a protein controlled? Many such proteins contain short, unfolded regions, and a key step in regulating their destruction is the addition of ubiquitin. Several general mechanisms for controlling this step are illustrated in [Figure 6–89](#). Specific examples of each mechanism are discussed in later chapters. In one general class of mechanism ([Figure 6–89A](#)), the activity of a ubiquitin ligase is turned on either by E3 phosphorylation or by an allosteric transition in an E3 protein caused by its binding to a specific small or large molecule. For example, the anaphase-promoting complex/cyclosome (APC/C) is a multisubunit ubiquitin ligase that is activated by a cell-cycle-timed subunit addition at mitosis. The activated APC then recognizes specific amino acid sequences in mitotic cyclins and several other regulators of the metaphase–anaphase transition and ubiquitylates them, thereby sending them to the proteasome (see [Figure 17–18](#)). We saw another example of regulated destruction earlier in this chapter, where a nascent polypeptide extending from a stalled ribosome is ubiquitylated, targeting it for destruction. Here, the signal for bringing the ubiquitin ligase into proximity to its target is a “gummed up” large ribosome subunit (see [Figure 6–81](#)).

Alternatively, in response either to intracellular signals or to signals from the environment, a ubiquitylation site can be created in a protein ([Figure 6–89B](#)). One common way to create such a signal is by phosphorylation of a specific amino

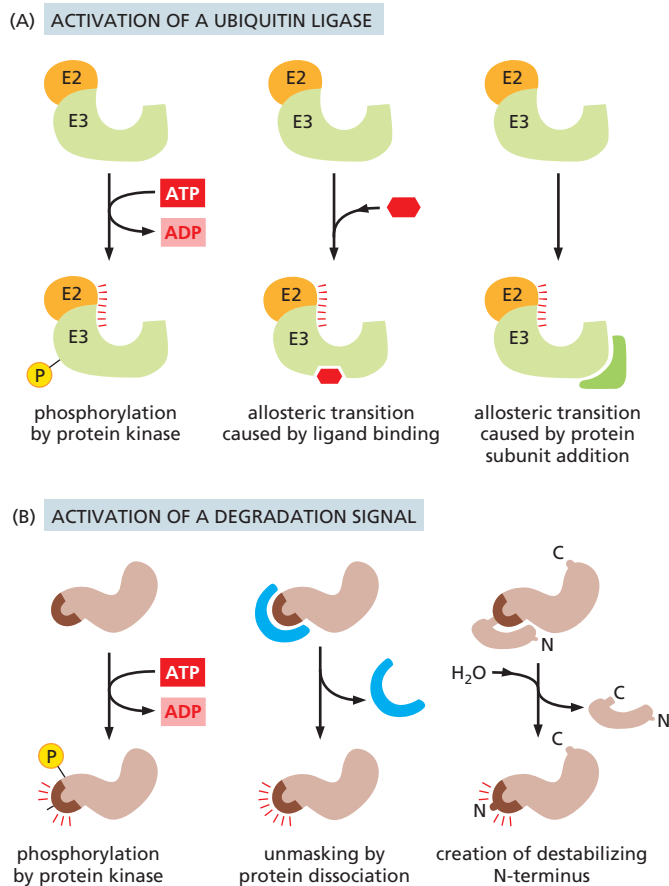


Figure 6–89 Two general ways of inducing the destruction of a specific protein. (A) Activation of a specific E3 molecule creates a new ubiquitin ligase. Eukaryotic cells have many different E3 molecules, each activated by a different signal. (B) Creation of an exposed destruction signal in the protein to be degraded. This signal binds a ubiquitin ligase, causing the addition of a polyubiquitin chain to a nearby lysine on the target protein. All six pathways shown are known to be used by cells to induce the movement of selected proteins into the proteasome for destruction.

acid sequence, which completes a ubiquitin ligase recognition site on the protein. Another is preexisting degradation signals that are unmasked by the regulated dissociation of a protein subunit. Finally, powerful degradation signals can be created by cleaving a single peptide bond, provided that this cleavage creates a new N-terminus that is recognized by a specific E3 protein as a “destabilizing” N-terminal residue. This E3 protein recognizes only certain amino acids at the N-terminus of a protein; thus, not all protein-cleavage events will lead to degradation of the C-terminal fragment produced.

In humans, nearly 70% of cytosolic proteins are acetylated on their N-terminal residue, and we now know that this modification is recognized by a specific E3 enzyme, which directs the ubiquitylation of the protein and sends it to the proteasome for degradation. Thus, the majority of human proteins carry their own signals for destruction. It has been proposed that when a protein is properly folded (and, before that, when it is in contact with a chaperone), this acetylated N-terminus is buried and therefore inaccessible to the E3. According to this idea, as a protein ages and becomes damaged (or if it fails to fold correctly from the start), this destruction signal becomes exposed, and the protein is destroyed.

There Are Many Steps from DNA to Protein

We have seen in this chapter that many different types of chemical reactions are required to produce a properly folded protein from the information contained in a gene (Figure 6–90). The final level of a useful protein in a cell therefore depends on the efficiency with which each of the many steps is performed. We also now know that the cell devotes enormous resources to selectively degrading proteins, particularly those that fail to fold properly or accumulate damage as they age. It is

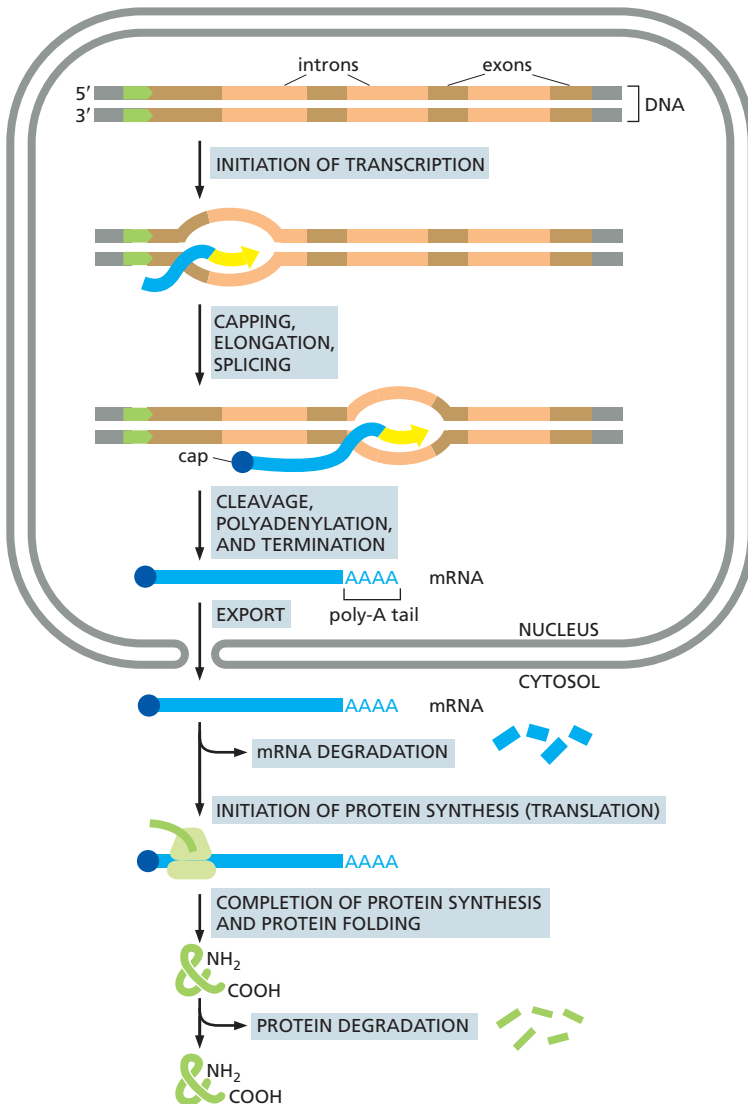


Figure 6–90 The production of a protein by a eukaryotic cell: a summary. The final level of each protein in a eukaryotic cell depends on the efficiency of each step depicted.

the balance between the rates of synthesis and degradation that determines the final amount of every protein in the cell.

In the following chapter, we shall see that cells have the ability to change the levels of their proteins according to their needs. In principle, any or all of the steps in Figure 6–90 could be regulated for each individual protein. As we shall see, there are examples of regulation at each step from gene to protein.

Summary

The translation of the nucleotide sequence of an mRNA molecule into protein takes place in the cytosol on a large ribonucleoprotein assembly called a ribosome. Each amino acid used for protein synthesis is first attached to a tRNA molecule that recognizes, by complementary base-pair interactions, a particular set of three nucleotides (codons) in the mRNA. As an mRNA is threaded through a ribosome, its sequence of nucleotides is then read from one end to the other in sets of three according to the genetic code.

To initiate translation, a small ribosomal subunit binds to the mRNA molecule at a start codon (AUG) that is recognized by a unique initiator tRNA molecule.

A large ribosomal subunit then binds to complete the ribosome and begin protein synthesis. During this phase, aminoacyl-tRNAs—each bearing a specific amino acid—bind sequentially to the appropriate codons in mRNA through complementary base-pairing between tRNA anticodons and mRNA codons. Each amino acid is added to the C-terminal end of the growing polypeptide in four sequential steps: aminoacyl-tRNA binding, followed by peptide bond formation, followed by two ribosome translocation steps. Elongation factors use GTP hydrolysis both to drive these reactions forward and to improve the accuracy of amino acid selection. The mRNA molecule progresses codon by codon through the ribosome in the 5'-to-3' direction until it reaches one of three stop codons. A release factor then binds to the ribosome, terminating translation and releasing the completed polypeptide.

Eukaryotic and bacterial ribosomes are closely related, despite differences in the number and size of their rRNA and protein components. The rRNA has the dominant role in translation, determining the overall structure of the ribosome, forming the binding sites for the tRNAs, matching the tRNAs to codons in the mRNA, and creating the active site of the peptidyl transferase enzyme that links amino acids together during translation.

Several distinct types of molecular chaperones, including hsp60 and hsp70, illustrate how the energy of ATP hydrolysis is used to help newly synthesized proteins assume their correct three-dimensional conformations. This protein-folding process competes with a control mechanism that destroys proteins that are abnormally folded by recognizing exposed hydrophobic patches and other unstructured regions. In this case, ubiquitin is covalently added to a misfolded protein by a ubiquitin ligase, and the resulting polyubiquitin chain is recognized by the cap on a proteasome that unfolds the protein as it is threaded into the interior of the proteasome for proteolytic degradation. A closely related proteolytic mechanism, based on special degradation signals recognized by ubiquitin ligases, is used to determine the lifetimes of many normally folded proteins, as well as to remove selected proteins from the cell in response to specific signals.

THE RNA WORLD AND THE ORIGINS OF LIFE

We have seen that the expression of hereditary information requires extraordinarily complex machinery and proceeds from DNA to protein through an RNA intermediate. This machinery presents a central paradox: if nucleic acids are required to synthesize proteins and proteins are required, in turn, to synthesize nucleic acids, how did such a system of interdependent components ever arise? A widely held view is that an **RNA world** existed on Earth before modern cells arose (Figure 6–91). According to this hypothesis, RNA not only stored genetic information but also directly catalyzed the chemical reactions in primitive cells—thus serving as enzymes (“ribozymes”). Only later in evolutionary time did DNA take over as the genetic material and proteins become the major catalysts and structural components of cells. But the transition out of the RNA world was never complete; as we have seen in this chapter, RNA still catalyzes several fundamental reactions in modern-day cells, which can be viewed as molecular holdovers from an earlier world.

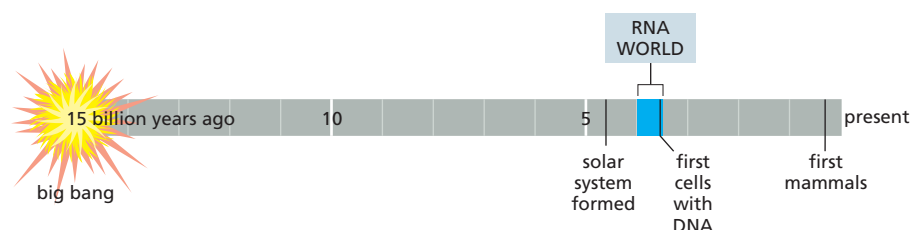


Figure 6–91 Timeline for the universe, highlighting the possible early existence of an “RNA world” in the evolution of Earth’s living systems.

The RNA world hypothesis relies on the fact that, among present-day biological molecules, RNA is unique in being able to act as both a carrier of genetic information and as a catalyst for chemical reactions. In this section, we discuss these properties of RNA and how they may have been especially important in early cells.

Single-Strand RNA Molecules Can Fold into Highly Elaborate Structures

We have seen in this chapter that RNA can carry genetic information in mRNAs, and we saw in Chapter 5 that the genomes of some viruses are composed solely of RNA. We have also seen that complementary base-pairing and other types of hydrogen bonds can occur between nucleotides in the same chain of RNA, causing an RNA molecule to fold up in a unique way that is determined by its nucleotide sequence (see, for example, Figures 6-54 and 6-71). Comparisons of many RNA structures have revealed conserved structural motifs, short elements that often appear as parts of larger structures (Figure 6-92).

Protein catalysts require a surface with unique contours and chemical properties on which a given set of substrates can react (discussed in Chapter 3). In exactly the same way, an RNA molecule with an appropriately folded shape can serve as a catalyst (Figure 6-93). Like some proteins, many of these ribozymes work by positioning metal ions at their active sites. This feature gives them a wider range of catalytic activities than can be provided by the limited chemical groups of a polynucleotide chain.

Ribozymes Can Be Produced in the Laboratory

Much of our inference about the RNA world has come from experiments in which large pools of RNA molecules of random nucleotide sequences are generated in the laboratory. Those rare RNA molecules with a property specified by the experimenter are then selected out and studied (Figure 6-94). Experiments of this type have created RNAs that can catalyze a wide variety of biochemical reactions (Table 6-5), with reaction rate enhancements only a few orders of magnitude lower than those of the “fastest” protein enzymes. Given these findings, it is not clear why protein catalysts greatly outnumber ribozymes in modern cells. Experiments have shown, however, that RNA molecules may have more difficulty than proteins in binding to flexible, hydrophobic substrates and in forming pockets specific for different small molecules. The availability of 20 types of amino acids

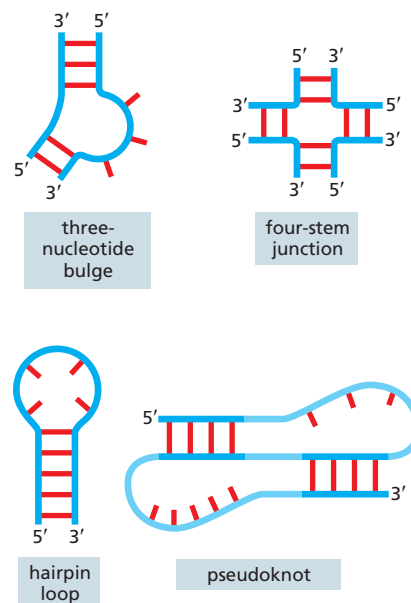


Figure 6-92 Some common elements of RNA structure. Conventional, complementary base-pairing interactions are indicated by red “rungs” in double-helical portions of the RNA.

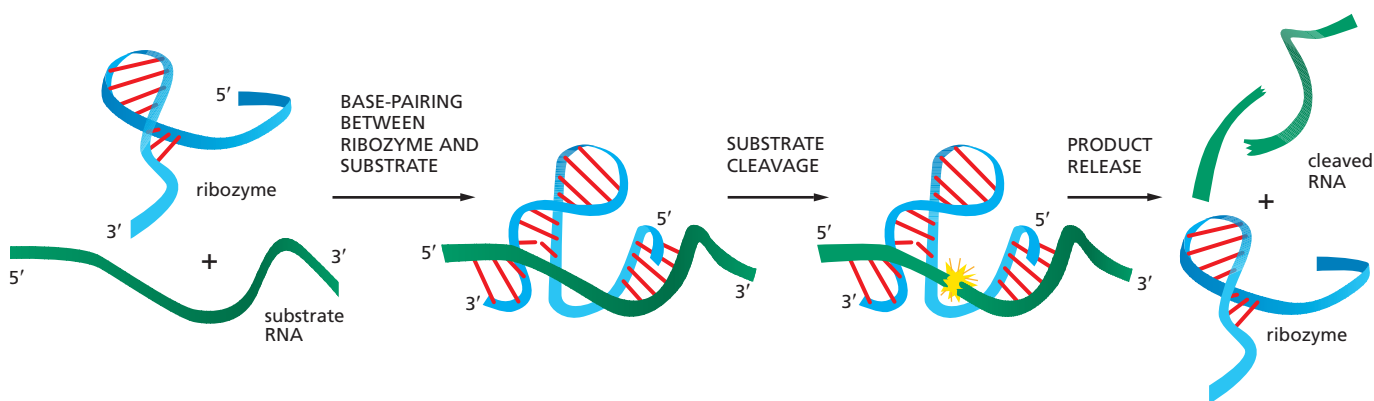


Figure 6-93 The activity of a well-studied ribozyme. This simple RNA molecule catalyzes the cleavage of a second RNA at a specific site. This ribozyme is found embedded in larger RNA genomes called viroids, which infect plants. The cleavage, which occurs in nature at a distant location on the same RNA molecule that contains the ribozyme, is a step in the replication of the viroid genome. Although not shown in the figure, the reaction requires a magnesium ion positioned at the active site. (Adapted from T.R. Cech and O.C. Uhlenbeck, *Nature* 372:39–40, 1994.)

Figure 6–94 *In vitro* selection of a synthetic ribozyme. Beginning with a large pool of different nucleic acid molecules synthesized in the laboratory, those rare RNA molecules that possess a specified catalytic activity can be isolated and studied. Although only one specific example (that of an autophosphorylating ribozyme) is shown, variations of this procedure have been used to generate many of the ribozymes listed in Table 6–5. For the strategy shown here, the RNA molecules are kept sufficiently dilute during the phosphorylation step to prevent the “cross-phosphorylation” of other RNA molecules. In reality, several repetitions of this procedure are necessary to select the very rare RNA molecules with this catalytic activity. Thus, the material initially eluted from the column is converted back into DNA, amplified many-fold (using reverse transcriptase and PCR, as explained in Chapter 8), transcribed back into RNA, and subjected to repeated rounds of selection. (Adapted from J.R. Lorsch and J.W. Szostak, *Nature* 371:31–36, 1994.)

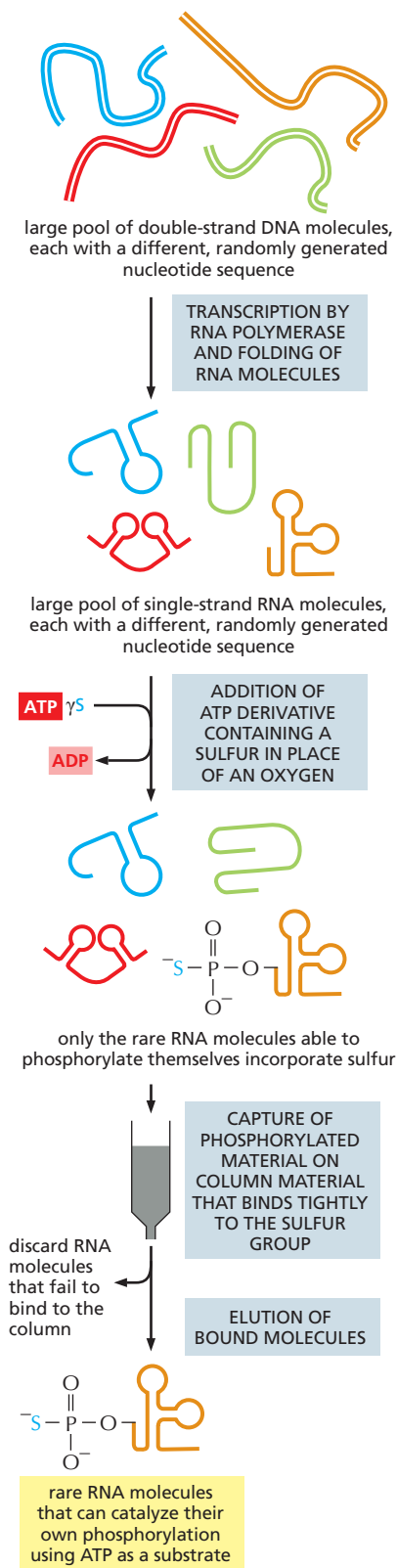
presumably provides proteins with a much greater number of binding and catalytic strategies.

RNA Can Both Store Information and Catalyze Chemical Reactions

RNA molecules have one property that contrasts with those of proteins: they can directly guide the formation of copies of their own sequence. This capacity depends on complementary base-pairing of their nucleotide subunits, which enables one RNA to act as a template for the formation of another. As we have seen in this and the preceding chapter, these complementary templating mechanisms lie at the heart of DNA replication and transcription in modern-day cells. But the efficient synthesis of RNA by such complementary templating mechanisms requires catalysts to promote the polymerization reaction: without catalysts, polymer formation would be slow, error-prone, and inefficient.

TABLE 6–5 Some Biochemical Reactions That Can Be Catalyzed by Ribozymes

Activity	Ribozymes
Peptide bond formation in protein synthesis	Ribosomal RNA
RNA cleavage, RNA ligation	Self-splicing RNAs; RNase P; also <i>in vitro</i> selected RNA
DNA cleavage	Self-splicing RNAs
RNA splicing	Self-splicing RNAs; RNAs of the spliceosome
RNA polymerization	<i>In vitro</i> selected RNA
RNA and DNA phosphorylation	<i>In vitro</i> selected RNA
RNA aminoacylation	<i>In vitro</i> selected RNA
RNA alkylation	<i>In vitro</i> selected RNA
Amide bond formation	<i>In vitro</i> selected RNA
Glycosidic bond formation	<i>In vitro</i> selected RNA
Oxidation–reduction reactions	<i>In vitro</i> selected RNA
Carbon–carbon bond formation	<i>In vitro</i> selected RNA
Phosphoamide bond formation	<i>In vitro</i> selected RNA
Disulfide exchange	<i>In vitro</i> selected RNA



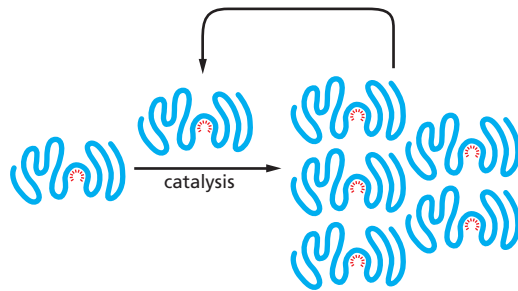


Figure 6–95 An RNA molecule that can catalyze its own synthesis. This hypothetical process would require synthesis of a second RNA strand complementary to the original strand (not shown) and the use of this second RNA molecule as a template to synthesize many molecules of RNA possessing the original sequence. The *red rays* represent the active site of this hypothetical RNA enzyme (ribozyme).

Because RNA has all the properties required of a molecule that could catalyze a variety of chemical reactions, including those that lead to its own synthesis (Figure 6–95), it has been proposed that RNAs served long ago as the catalysts for their own template-dependent RNA synthesis. Although self-replicating systems of RNA molecules have not been found in nature, scientists have made progress toward constructing such systems in the laboratory. Experiments of this type cannot prove that self-replicating RNA molecules were central to the origin of life on Earth, but they can help establish whether such a scenario is plausible.

Today, there is a widespread interest in investigating the exciting possibility that a primitive form of life once existed, or may even still exist, in some water-containing regions below the surface of Mars. Vehicles are being sent to promising sites on that planet to collect subterranean samples for eventual return to Earth, with the hope that their analysis will allow scientists to refine scenarios like that just described.

How Did Protein Synthesis Evolve?

The molecular processes underlying protein synthesis in present-day cells seem inextricably complex. Although we understand most of them, they do not make conceptual sense in the way that DNA transcription, DNA repair, and DNA replication do. It is especially difficult to imagine how protein synthesis evolved because it is now performed by a complex interlocking system of protein and RNA molecules; obviously, the proteins could not have existed until an early version of the translation apparatus was already in place. As attractive as the RNA world idea is for envisioning early life, it does not explain how the modern-day system of protein synthesis arose.

In modern cells, some short peptides (such as antibiotics) are synthesized without the ribosome; peptide synthetase enzymes assemble these peptides, with their proper sequence of amino acids, without mRNAs to guide their synthesis. It is plausible that this noncoded, primitive version of protein synthesis first developed in the RNA world, where it would have been catalyzed by RNA molecules. This idea presents no conceptual difficulties because, as we have seen, rRNA catalyzes peptide bond formation in present-day cells. Moreover, short simple peptides (for example, polylysine) have been shown to enhance the function of ribozymes created in the laboratory, raising the possibility that the first peptides were “selected” for their ability to help RNA molecules fold, assemble with each other, and catalyze reactions. These ideas, however, leave unexplained how the genetic code—which lies at the core of protein synthesis in today’s cells—might have arisen. We know that ribozymes created in the laboratory can perform specific aminoacylation reactions; that is, they can match specific amino acids to specific tRNAs. It is therefore possible that tRNA-like adaptors, each matched to a specific amino acid, could have arisen in the RNA world, marking the beginnings

Figure 6–96 The hypothesis that RNA preceded DNA and proteins in evolution. In the earliest cells, RNA molecules (or their close analogs) would have had combined genetic, structural, and catalytic functions. In present-day cells, DNA is the repository of genetic information, and proteins perform the vast majority of catalytic functions in cells. RNA primarily functions today as a go-between in protein synthesis, although it remains a catalyst for a small number of crucial reactions.

of a genetic code. And, although we can only speculate on the code's evolution, scientists have provided plausible scenarios, in which the first codons specified only 10 or so of today's 20 amino acids.

Once coded protein synthesis evolved, the transition to a protein-dominated world could proceed, with proteins eventually taking over the majority of catalytic and structural tasks because of their greater versatility, with 20 rather than 4 different subunits. Although these ideas are highly speculative, they are consistent with the known properties of RNA and protein molecules.

All Present-Day Cells Use DNA as Their Hereditary Material

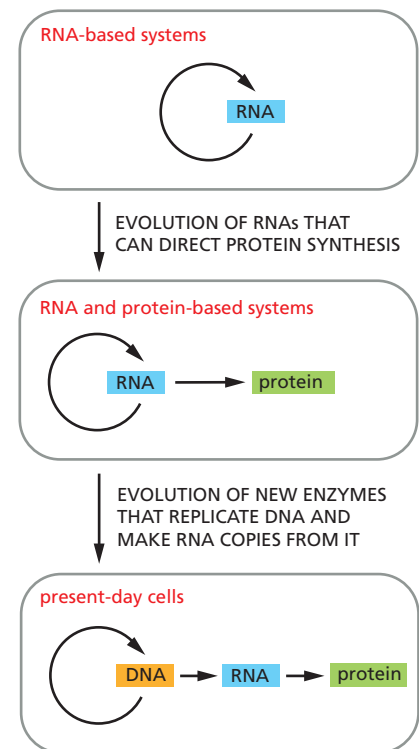
If the evolutionary speculations embodied in the RNA world hypothesis are correct, early cells would have differed fundamentally from the cells we know today in having their hereditary information stored in RNA rather than in DNA (Figure 6–96).

Evidence that RNA arose before DNA in evolution can be found in the chemical differences between them. Ribose, like glucose and other simple carbohydrates, can be formed from formaldehyde (HCHO), a simple chemical that is readily produced in laboratory experiments that attempt to simulate conditions on the primitive Earth. The sugar deoxyribose is harder to make, and in present-day cells it is produced from ribose in a reaction catalyzed by a protein enzyme, suggesting that ribose predates deoxyribose in cells. Presumably, DNA appeared on the scene later, but then proved more suitable than RNA as a permanent repository of genetic information. In particular, the deoxyribose in its sugar-phosphate backbone makes chains of DNA chemically more stable than chains of RNA, so that much greater lengths of DNA can be maintained without breakage. Because DNA and RNA use similar base-pairing rules for their template-dependent synthesis, the transition between the two is not difficult to envision. Consistent with this idea, a ribozyme has been created in the laboratory that can synthesize both RNA using a DNA template and DNA using an RNA template.

The other differences between RNA and DNA—the double-helical structure of DNA and the use of thymine rather than uracil—further enhance DNA stability by making the many unavoidable accidents that occur to the molecule much easier to repair, as discussed in detail in Chapter 5 (p. 288).

Summary

From our knowledge of present-day organisms and the molecules they contain, it seems likely that the development of the distinctive autocatalytic mechanisms fundamental to living systems began with the evolution of families of RNA molecules that could catalyze their own replication. DNA is thought to have been a later addition: as the accumulation of protein catalysts allowed more efficient and complex cells to evolve, the DNA double helix replaced RNA as a much more stable molecule for storing the increased amounts of genetic information required by such cells.



PROBLEMS

Which statements are true? Explain why or why not.

6-1 Errors in transcription are less dangerous to an organism than errors in DNA replication.

6-2 Because introns are largely genetic “junk,” they do not have to be removed precisely during RNA splicing.

6-3 Wobble pairing occurs between the first position in the codon and the third position in the anticodon.

6-4 During protein synthesis, the thermodynamics of base-pairing between tRNAs and mRNAs sets the upper limit for the accuracy with which protein molecules are made.

6-5 Protein enzymes are thought to greatly outnumber ribozymes in modern cells because they can catalyze a much greater variety of reactions, and all of them have faster rates than any ribozyme.

Discuss the following problems.

6-6 You have attached an RNA polymerase molecule to a glass slide and have allowed it to initiate transcription on a template DNA that has both strands tethered to a magnetic bead as shown in Figure Q6-1. If the DNA with its attached magnetic bead moves relative to the RNA polymerase as indicated by the arrows in the figure, in which direction will the bead rotate?

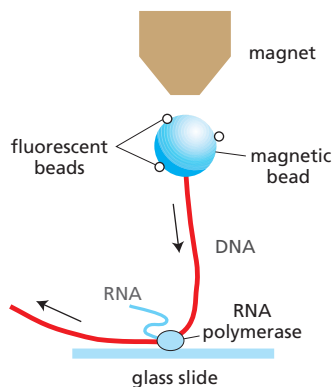


Figure Q6-1 System for measuring the rotation of DNA caused by RNA polymerase (Problem 6-6). The magnet holds the bead upright (but doesn't interfere with its rotation), and the attached tiny fluorescent beads allow the direction of motion to be visualized under the microscope. RNA polymerase is held in place by attachment to the glass slide. (Adapted from Y. Harada et al., *Nature* 409: 113–115, 2001. With permission from Springer Nature.)

6-7 In which direction along the template must the RNA polymerase in Figure Q6-2 be moving to have generated the supercoiled structures that are shown? Would you expect supercoils to be generated if the RNA polymerase were free to rotate about the axis of the DNA as it progressed along the template?



Figure Q6-2 Supercoils produced by a moving RNA polymerase (Problem 6-7).

6-8 The human α -tropomyosin gene is alternatively spliced to produce different forms of α -tropomyosin mRNA in different cell types (Figure Q6-3). All forms of the mRNA contain the protein sequences encoded by exon 1 and exon 10. Exons 2, 3, 7, and 8 are alternative exons used in different mRNAs. Which one of the following statements about exons 2 and 3 is the most accurate? Is that statement also the most accurate one for exons 7 and 8? Explain your answers.

A. Exons 2 and 3 must have the same number of nucleotides.

B. Exons 2 and 3 must each contain an integral number of codons (that is, the number of nucleotides divided by 3 must be an integer).

C. Exons 2 and 3 must each contain a number of nucleotides that when divided by 3 leaves the same remainder (that is, 0, 1, or 2).

(A) HUMAN α -TROPOMYOSIN GENE



(B) FOUR DIFFERENT SPLICE VARIANTS

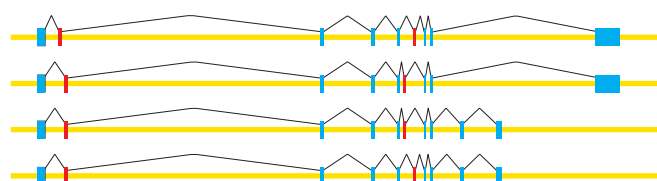


Figure Q6-3 Alternatively spliced mRNAs from the human α -tropomyosin gene (Problem 6-8). (A) Exons in the human α -tropomyosin gene. The locations and relative sizes of exons are shown by the blue and red rectangles, with alternative exons in red. (B) Splicing patterns for four α -tropomyosin mRNAs. Splicing is indicated by lines connecting the exons present in each mRNA.

6-9 After treating cells with a chemical mutagen, you isolate two mutants. One carries alanine and the other carries methionine at a site in a protein that normally contains valine (Figure Q6-4). After treating these two mutants again with the mutagen, you isolate mutants from each that now carry threonine at the site of the original valine (Figure Q6-4). Assuming that all mutations involve single-nucleotide changes, which codons are used for valine, methionine, threonine, and alanine at this site in the protein? Would you expect to be able to isolate valine-to-threonine mutants in one step?

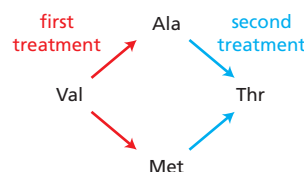


Figure Q6-4 Two rounds of mutagenesis and the altered amino acids at a single position in a protein (Problem 6-9).

6–10 Which one of the following mutational changes would you predict to be the most deleterious to gene function? Explain your answers.

1. Insertion of a single nucleotide near the end of the coding sequence.
2. Removal of a single nucleotide near the beginning of the coding sequence.
3. Deletion of three consecutive nucleotides in the middle of the coding sequence.
4. Substitution of one nucleotide for another in the middle of the coding sequence.

6–11 Prokaryotes and eukaryotes both protect against the dangers of translating broken mRNAs. What dangers do partial mRNAs pose for the cell?

6–12 Both hsp60-like and hsp70 molecular chaperones share an affinity for exposed hydrophobic patches on proteins, using them as indicators of incomplete folding. Why do you suppose hydrophobic patches serve as critical signals for the folding status of a protein?

6–13 Most proteins require molecular chaperones to assist in their correct folding. How do you suppose the chaperones themselves manage to fold correctly?

6–14 If an RNA molecule could form a hairpin with a symmetric internal loop, as shown in **Figure Q6–5**, could the complement of this RNA form a similar structure? If so, would there be any regions of the two structures that are identical? Which ones?

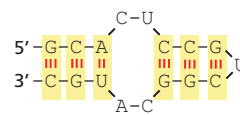


Figure Q6–5 An RNA hairpin with a symmetric internal loop (Problem 6–14).

6–15 What is so special about RNA that it is hypothesized to be an evolutionary precursor to DNA and protein? What is it about DNA that makes it a better material than RNA for storage of genetic information?

6–16 Imagine a warm pond on primordial Earth. Chance processes have just assembled a single copy of an RNA molecule with a catalytic site that can carry out RNA replication. This RNA molecule folds into a structure that is capable of linking nucleotides according to instructions in an RNA template. Given an adequate supply of nucleotides, will this single RNA molecule be able to use itself as a template to catalyze its own replication? Why or why not?

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